CREATED BY





SIGNALS & SYSTEMS

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Introduction:

بسم الله الرحمن الرحيم، الحمد لله والصلاة والسلام على سيدنا محمد وعلى آله وصحبه اجمعين وبعد:

نضع بين يديكم دفتر مادة أنظمة وإشارات الشامل لمادتنا في الجامعة، وضعت بطريقة مناسبة جدا للدراسة، وتحتوي على الكثير من الأسئلة بأجوبتها وبتفصيل للنقاط المهمة (المصادر في آخر صفحة).

نرجوا نشر هذا الدفتر حتى تعم الفائدة لكل طلاب الهندسة الكهربائية (شريطة عدم وضع اي شعارات لاتحادات أخرى)، شاكرين لكم، متمنين لكم التوفيق.

احمد فيصل محمد ابو عليان

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1 Representing signals

1.1 Classification of signals

1.1.1 Introduction:

- **Signals and systems** are very important in almost every area of electrical engineering, and also in many other science and engineering fields.
- Signals carry information (a message) about something in the real world. Like **voice, light intensity, video**, etc.
- **We represent signals using mathematics** as functions that depend on one or more variables. Like:
  - ∴ Speech can be represented as **sound pressure vs time**
  - ∴ A picture can be represented as **brightness vs two spatial directions (x and y)**
    In this course, we mainly focus on signals that depend on **one variable only** (usually time).
- **Electrical signals are very useful** because they are easy to measure, send (transfer), and use to represent real physical values.
- In this chapter, we learn how to describe signals and systems using math, how to represent them, and how to classify them.

1.1.2 Continuous & Discrete Signals:

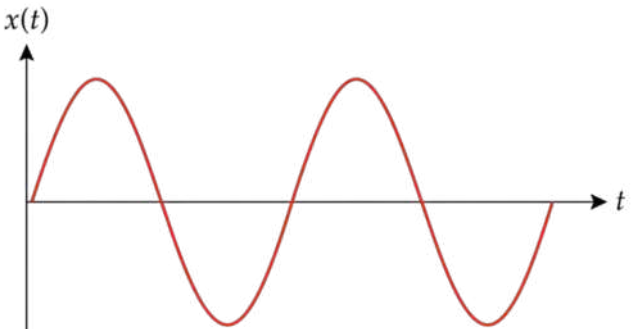
1 . Continuous Signal:

- A time signal in which all its values of $(t, F(t))$ are continuous on the interval (T_0, T_1).

◆ **Examples:**

$$x(t) = \sin(t)\,, t > 0$$

- If $x(t)$ is continuous, the values on $y - axis$ will also be continuous as a direct result of the continuity of $x(t)$.



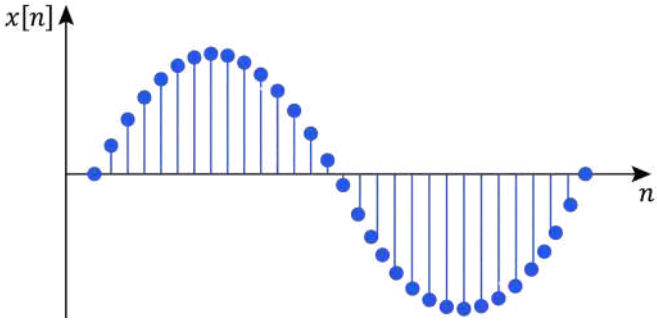
2 . Discrete Signal:

- A time signal in which all its values of $(t, x(t))$ are discontinuous or sampled from a continuous function and often denoted by $x[n]$ or a sequence x_n.

◆ **Example:**

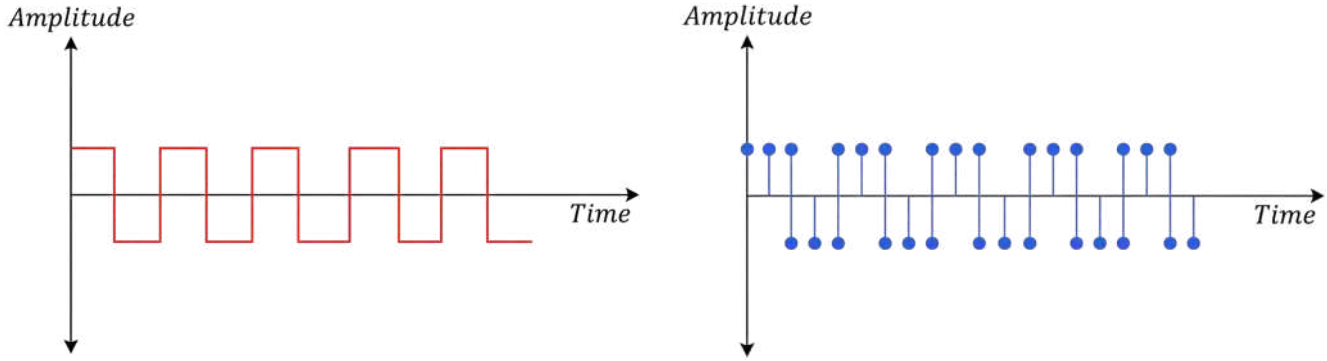
$$x_n = x[n] = \sin[n]\,, n \geq 0$$

- If $x[n]$ is defined for discrete time, the values on $y - axis$ will be discrete and correspond to the defined amplitudes at each sample point n.



Digital Signal

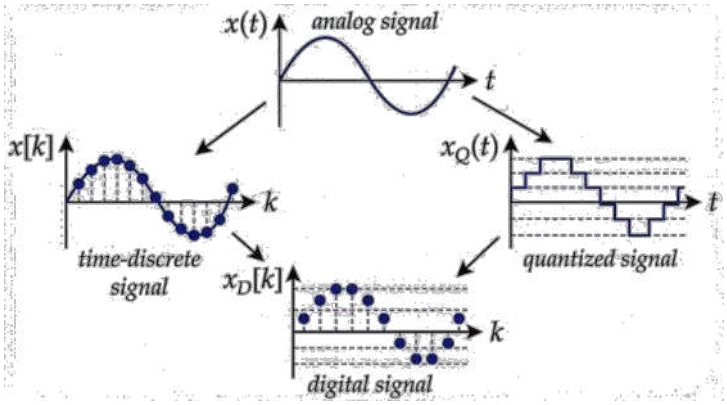
- The $x - axis$ values appear continuous in digital time signals due to the plotting method and high sampling rates, which can create the illusion of continuity.
- However, it's important to remember that the actual data points are discrete and represent the sampled values of the continuous signal.



- Analog to digital conversion (ADC): is the process of converting a continuous (analog) signal into a discrete (digital) signal that can be processed by digital systems, like computers or digital electronics.

The process involves 3 main steps:

1. Sampling
2. Quantization
3. Encoding



- A signal that is both discrete on the $x, y\ axis$, that can also be encoded into numbers like 0 or 1.

1. Sampling → **Discrete For $x - axis$.**
2. Quantization → **Discrete For $y - axis$.**

1.1.3 Complex Signal:

- A complex signal $x(t)$ can be expressed (general form):

$$x(t) = x_{real}(t) + jx_{imag}(t)$$

$\therefore$ $x_{real}(t)$ is the real part of the signal.

$\therefore$ $x_{imag}(t)$ is the imaginary part of the signal.

- j is the imaginary unit, and $j = \sqrt{-1}$ or on the complex plane as 90°.

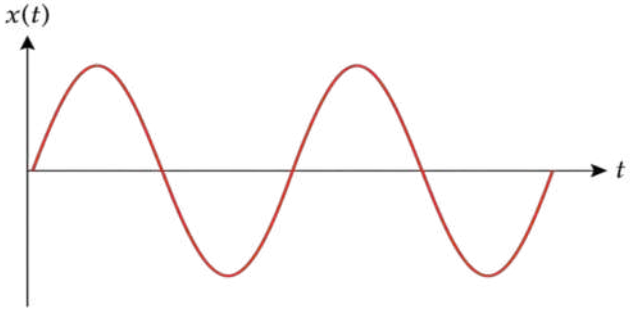
1.1.4 Deterministic & Random Signals:

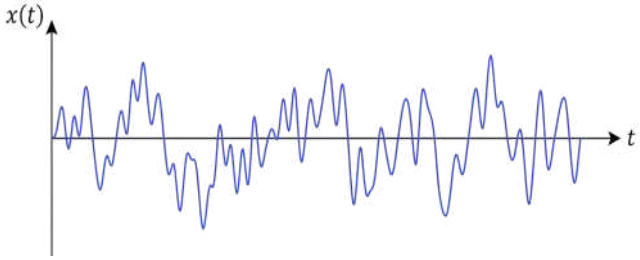
1. Deterministic Signals:

- are those signals whose values are completely specified for any given time (known for the entire interval of t). So, a deterministic signal can be written as a function of t; $F(t)$.

2. Random Signals:

- are those signals that take random values at any given time and must be characterized statistically (difficult to predict).





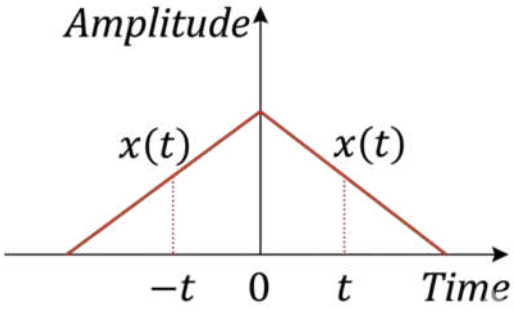
1.1.5 Even & Odd Signals:

1 . Even Signals:

- if $x(t)$ is symmetric about the $y-axis$, then it's even.

◆ **Examples:**

$x(t) = \cos t\,,\qquad x(t) = x^2$



- Mathematically:

$$x(-t) = x(t)$$

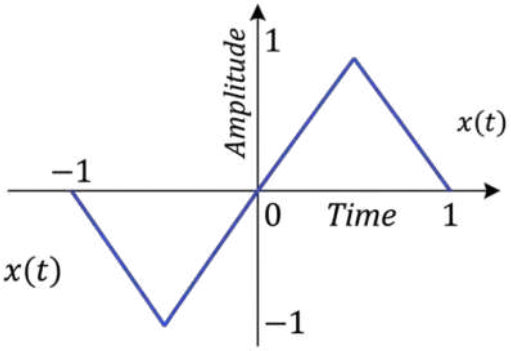
- Reflect about the $(y-axis)$.

2 . Odd Signals:

- if $x(t)$ is symmetric about the origin (0,0) then it's odd.

◆ **Examples:**

$x(t) = \sin t\,,\qquad x(t) = x^3$



- Mathematically:

$$x(-t) = -x(t)$$

- Reflect about the $(x\ \&\ y-axis)$.

Some Signals Contain Even and Odd Parts:

- The general form of any signal is:

$$x(t) = x_e(t) + x_o(t), \qquad x[n] = x_e[n] + x_o[n]$$

- To find even and odd part:

$$x_e(t) = \frac{1}{2}\big(x(t) + x(-t)\big), \qquad x_o(t) = \frac{1}{2}\big(x(t) - x(-t)\big)$$

Q1: Find the even part and odd part for:

$$x(t) = 2\cos(t) - 7\sin(t)$$

Finding the even part $\rightarrow x_e(t) = \frac{1}{2}\big(x(t) + x(-t)\big)$

$$x_e(t) = 2\cos(t)$$

Finding the odd part $\rightarrow x_o(t) = \frac{1}{2}\big(x(t) - x(-t)\big)$

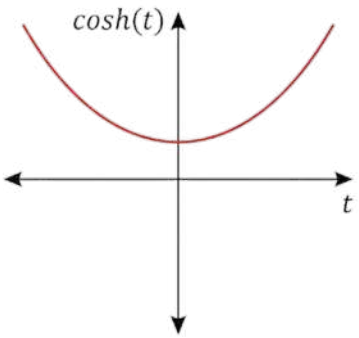
$$x_o(t) = -7\sin(t)$$

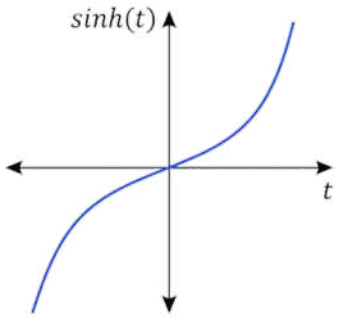
Q2: Write and draw $x_e(t)$ and $x_o(t)$ for:

$$x(t) = e^t, (-\infty < t < \infty)$$

$$x_e(t) = \frac{(e^t + e^{-t})}{2} = \cosh(t)$$

$$x_o(t) = \frac{(e^t - e^{-t})}{2} = \sinh(t)$$





1.1.6 Periodic & Aperiodic Signals:

1 . Periodic Signal:

- The signal is **periodic** if it repeats itself at regular intervals over time. Formally, a signal $x(t)$ is periodic if there exists a positive constant T, $(T > 0)$ such that:

$$x(t+T) = x(t), \qquad For\ all\ t$$

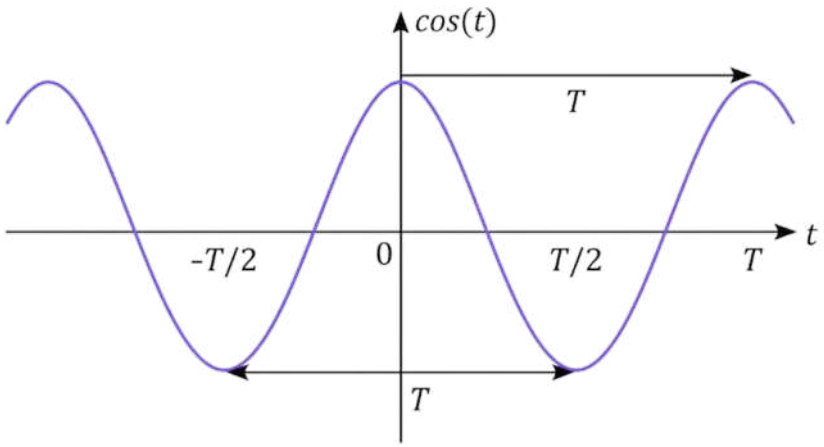
$$or$$

$$x(t+nT), \qquad n = 1,2,3 \ldots etc$$

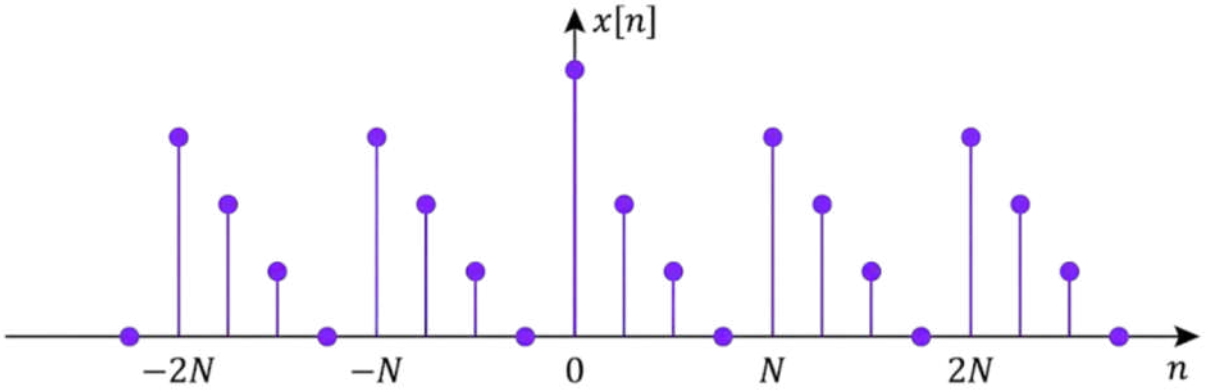
- **Periodic time T:** The time required for a signal to repeat)
- **Frequency f:** The number of times the signal repeats itself per second
- **Angular frequency ω:** How quickly a signal oscillates in radians per second

$$f = \frac{1}{T}(Hz), \omega = 2\pi f \left(rad/sec\right)$$

- There is also the **fundamental period** To which is the minimum value of the period T



• The **discrete signal** $x[n]$ is defined as **periodic** if it repeats itself after a fixed number of samples.



• Formally, a discrete signal is periodic if there exists a positive integer N such that:

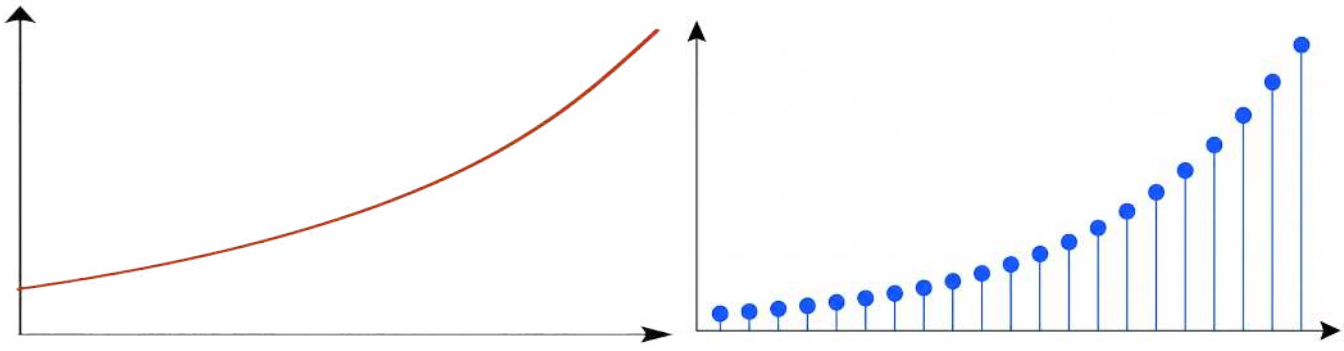
$$x[n+N] = x[n] \; for \; all \; integers \; n$$

2 . Aperiodic Signal:

• It is a signal that does not repeat itself at regular intervals. In both **continuous** and **discrete** domains, an aperiodic signal lacks a fundamental period $T_{\text{for continuous signals}}$ or $N_{\text{for discrete signals}}$ such that:

∴ **Continuous:** $f(t+T) \neq f(t)$ for all $T > 0$.

∴ **Discrete** : $x[n+N] \neq x[n]$ for any integer $N > 0$.



Summation of Two Periodic Signals:

- The sum of two periodic signals is **periodic only if** the ratio of their periods is a **rational number**:

$$\frac{T_1}{T_2} = \frac{a}{b}$$

- Examples of rational ratios: $4/2$, $5/4$, $3/1$
- If the ratio is **irrational**, the sum is **not periodic.**
- Examples of irrational numbers: π, e, $\sqrt{2}$, $\sqrt{3}$, $\sqrt{7}$

Finding the Period of the Sum

- The new period is equal to the lowest common denominator (LCD) of the two periods.
- To find LCD:
  1. When $T_1 = \frac{A}{B}$ and $T_2 = \frac{C}{D} \rightarrow \frac{T_1}{T_2} = \frac{A/B}{C/D} = \frac{AD}{BC}$
  2. Reduce $\frac{AD}{BC}$ to simplest form then say that $\frac{AD}{BC} = \frac{m}{n}$
  3. Then $T_{sum} = \text{LCD} = mT_2$ or nT_1
- Changing from non-linear operation to a linear operation by using trigonometric identities then LCD.

Remember that:

$$\therefore \sin A \cos B = \frac{\sin(A-B)+\sin(A+B)}{2}$$

$$\therefore \cos A \cos B = \frac{\cos(A-B)+\cos(A+B)}{2}$$

$$\therefore \sin A \sin B = \frac{\cos(A-B)-\cos(A+B)}{2}$$

Q3: For these data:

$$x_1(t) = \sin(10\pi t), x_2 = \cos(25\pi t), x_3(t) = \cos(20t)$$

Find the periodic time of the given signals:

1. $x_1(t) + x_2(t)$
2. $x_1(t) + x_3(t)$
3. $x_1(t) \cdot x_2(t)$

$$\omega_1 = \frac{2\pi}{T_1} = 10\pi \rightarrow T_1 = \frac{2\pi}{10\pi} = \frac{1}{5}\ sec$$

$$\omega_2 = \frac{2\pi}{T_2} = 25\pi \rightarrow T_2 = \frac{2\pi}{25\pi} = \frac{2}{25}\ sec$$

$$\omega_3 = \frac{2\pi}{T_3} = 20 \rightarrow T_3 = \frac{2\pi}{20} = \frac{\pi}{10}\ sec$$

1. $x_1(t) + x_2(t)$

Check ratio:

$$T_1 = \frac{1}{5}, T_2 = \frac{2}{25} \rightarrow \frac{T_1}{T_2} = \frac{1/5}{2/25} = \frac{5}{2} \rightarrow \text{Rational}$$

Find LCD:

$$\frac{T_1}{T_2} = \frac{5}{2} = \frac{m}{n}, \qquad m = 5, \qquad n = 2$$

$$T_{sum} = T_1 n = \frac{1}{5} * 2 = \frac{2}{5} = 0.4 \text{ sec } or$$

$$T_{sum} = T_2 m = \frac{2}{25} * 5 = \frac{2}{5} = 0.4 \text{ sec}$$

2. $x_1(t) + x_3(t)$

Check ratio:

$$\frac{T_1}{T_3} = \frac{1/5}{\pi/10} = \frac{2}{\pi} \rightarrow \text{Irrational}$$

So, the sum is **Aperiodic**.

3. $x_1(t) \cdot x_2(t)$

Use identity:

$$\sin A \cos B = \frac{1}{2}[\sin(A+B) + \sin(A-B)]$$

$$x_1(t) \cdot x_2(t) = \frac{1}{2}[\sin(35\pi t) + \sin(15\pi t)]$$

$$T_A = \frac{2\pi}{35\pi} = \frac{2}{35}, \qquad T_B = \frac{2\pi}{15\pi} = \frac{2}{15}$$

Check ratio:

$$\frac{T_A}{T_B} = \frac{30}{70} = \frac{3}{7} \rightarrow \text{Rational}$$

Find LCD:

$$\frac{T_A}{T_B} = \frac{3}{7} = \frac{m}{n}, \qquad m = 3, \qquad n = 7$$

$$T_{sum} = T_A n = \frac{2}{35} * 7 = \frac{2}{5} = 0.4 \text{ sec}$$

Euler's identity:

- Assume the continuous-time signal is:

$$x(t) = e^{j\omega_0 t}$$

- A continuous-time signal is **periodic** if there exists a **positive** value T_0 such that:

$$x(t) = x(t + T_0)$$

- So:

$$e^{j\omega_0 t} = e^{j\omega_0 (t+T_0)} = e^{j\omega_0 t}\ e^{j\omega_0 T_0}$$

- Divide both sides by $e^{j\omega_0 t}$:

$$e^{j\omega_0 T_0} = 1$$

$$e^{j\omega_0 T_0} = \cos(\omega_0 T_0) + j\sin(\omega_0 T_0)$$

- For this to equal $1 + 0j$, we need:

$$\cos(\omega_0 T_0) = 1, \qquad \sin(\omega_0 T_0) = 0$$

- This happens when:

$$\omega_0 T_0 = 2\pi m, \qquad m = 0,1,2,3,\ldots \rightarrow T_0 = \frac{2\pi m}{\omega_0}$$

- The **fundamental period** is the **smallest positive** period, so we take $m = 1$:

$$T_0 = \frac{2\pi}{\omega_0}$$

- Assume we sample the continuous-time signal:

$$x(t) = e^{j\omega_0 t}$$

- Using sampling time T_s, the discrete-time signal becomes:

$$x[n] = x(nT_s) = e^{j\omega_0 nT_s}$$

- A discrete-time signal is periodic if there exists an integer N_0 such that:

$$x[n] = x[n + N_0]$$

- So:

$$e^{j\omega_0 nT_s} = e^{j\omega_0 (n+N_0)T_s}$$

- Cancel the common term $e^{j\omega_0 nT_s}$:

$$e^{j\omega_0 N_0 T_s} = 1$$

$$e^{j\omega_0 N_0 T_s} = \cos(\omega_0 N_0 T_s) + j\sin(\omega_0 N_0 T_s)$$

- For this to equal $1 + 0j$, we need:

$$\omega_0 N_0 T_s = 2\pi m, m = 0,1,2,3,\ldots$$

- Now solve for T_s (and $T_s > 0$):

$$T_s = \frac{2\pi m}{\omega_0 N_0} = \frac{2\pi m}{\left(\frac{2\pi}{T_0}\right) N_0} = \frac{mT_0}{N_0}$$

$$\frac{T_s}{T_0} = \frac{m}{N_0}$$

- When:

  ∴ T_s: Sampling period.

  ∴ T_0: Fundamental period of main signal $x(t)$.

  ∴ N_0: Fundamental period of sampled signal $x(nT_s)$.

  ∴ m: the number that makes N_0 integer.

Q4: Consider the signal $x(t) = \cos 15t$.

Find the fundamental period of $x(nT_s)$ if $T_s = 0.1\pi$ s

$$T_0 = \frac{2\pi}{\omega_0} = \frac{2\pi}{15}$$

$$\frac{m}{N_o} = \frac{T_s}{T_0} = \frac{0.1\pi}{2\pi/15} = \frac{15}{20} = \frac{3}{4} \rightarrow m = 3, \qquad N_0 = 4$$

Q5: Find whether the signals given are periodic or Aperiodic:

A) $x(t) = t\sin(t)$

t is an aperiodic function

, so: $x(t)$ is aperiodic.

B) $x(t) = e^{\sin(t)}$

The exponential function depends on its exponent, and $\sin t$ is periodic, so:

$x(t)$ is periodic.

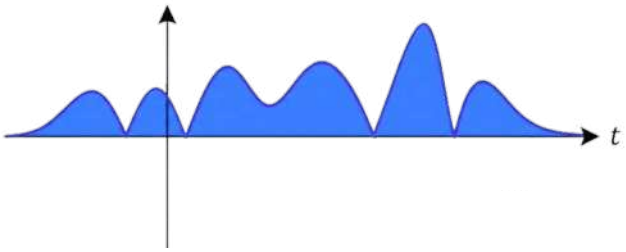
1.1.7 Energy & Power Signal:

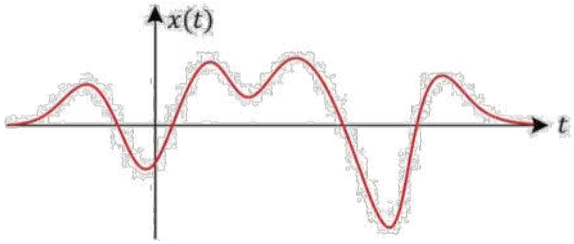
1 . Energy Signal:

- An **energy signal** is a signal that has a **finite (limited) amount of energy**.
- Since a signal changes with time, we need a way to measure how "strong" the signal is.
- A natural idea is to measure the area under the curve.
- **But there is a problem:**
- A signal can **have** positive and negative values, so the positive and negative areas may cancel each other, even though the signal is still strong.
- To avoid this, we measure the strength by using:

∴ the **absolute value**,

∴ then the **square of the signal**





- In signals and systems, we define the **energy of a signal** as the area under the **squared magnitude** of the signal.

$$E_{continuous} = \int_{-\infty}^{\infty} |x(t)|^2 \, dt$$

$$E_{discrete} = \sum_{n=-\infty}^{\infty} |x[n]|^2$$

2. Power Signal:

- A signal that has finite power.

▶ Why do we need power?

- The energy definition is good, but it fails for signals that **never decay** or **never end** (like a sine wave).
- For example, a 60 Hz sine wave continues forever, so its energy becomes:

$$E = \int_{-\infty}^{\infty} |x(t)|^2 \, dt = \infty$$

- But that does **not** mean the signal is "infinitely strong". So instead of total energy, we measure **average power**, which means:

$$Power \; = \; Energy \; per \; time \; unit$$

- Average Power of a Continuous Signal:

$$P = \lim_{T\to\infty} \frac{1}{T} \int_{-T/2}^{T/2} |x(t)|^2 \, dt = \lim_{T\to\infty} \frac{1}{T} \int_{0}^{T} |x(t)|^2 \, dt$$

- The interval length is: $T - (-T) = 2T$
- So we divide by $2T$ to get the **average.**
- This is just the same average power idea, but using a centered interval $(-T/2, T/2)$.

- Average Power of a Discrete Signal:

$$P = \lim_{N\to\infty} \frac{1}{2N+1} \sum_{n=-N}^{N} |x[n]|^2$$

- The number of samples from $-N$ to N is: $(2N + 1)$ because we include $n = 0$.

Energy vs Power Classification

Signal Type	Energy	Power	Classification
Aperiodic (finite duration)	$finite$	$(P = 0)$	Energy signal
Periodic (sine wave)	$(E = \infty)$	$finite$	Power signal
Aperiodic (grows without decaying)	$(E = \infty)$	$(P = \infty)$	Neither

Important Notes

- **Periodic signals** are usually **power signals**, because they repeat forever:

$$E = \infty, \qquad P \neq 0$$

- For a **periodic signal**, you don't need the limit.
You can compute the power over **one period.**
- **Aperiodic signals** can be **energy signals** if they exist for a limited time (from $t_{initial}$ to t_{final}).

Q6: Let $x(t) = Ae^{-kt}$, Find its energy and power.

$$E = \int_0^{\infty} (Ae^{-kt})^2 dt = \int_0^{\infty} A^2 e^{-2t} dt = -\frac{A^2}{2k}\left[e^{-2kt}\right]_0^{\infty}$$

$$= -\frac{A^2}{2k}[0 - 1] = \frac{A^2}{2k}$$

$$P = \lim_{T\to\infty} \frac{1}{2T}\int_0^{T} (Ae^{-kt})^2 dt = \lim_{T\to\infty} \frac{1}{2T}\int_0^{T} A^2 e^{-2kt} dt$$

$$= \lim_{T\to\infty} \frac{1}{2T}\left(-\frac{A^2}{2k}\left[e^{-2kt}\right]_0^{T}\right)$$

$$= \lim_{T\to\infty} \frac{1}{2T}\left(-\frac{A^2\left(e^{-2kT} - 1\right)}{2k}\right) = \lim_{T\to\infty} A^2 \frac{1 - e^{-2T}}{4T}$$

$$= A^2 \frac{1}{\infty} = 0$$

Q7: Let $x(t) = A \sin \omega t$, Find its power.

$$P = \lim_{T\to\infty} \frac{1}{T}\int_{-T/2}^{T/2} x^2(t) dt$$

$$P = \lim_{T\to\infty} \frac{1}{T}\int_{-\frac{T}{2}}^{\frac{T}{2}} (A \sin \omega t)^2 (t) dt =$$

$$\lim_{T\to\infty} \frac{1}{T}\int_{-\frac{T}{2}}^{\frac{T}{2}} A^2 \sin^2 \omega t\,(t) dt = \lim_{T\to\infty} \frac{A^2}{T}\int_{-\frac{T}{2}}^{\frac{T}{2}} \frac{(1-\cos 2\omega t)}{2} dt =$$

$$\lim_{T\to\infty} \frac{A^2}{T}\int_{-\frac{T}{2}}^{\frac{T}{2}} \left(\frac{1}{2} - \frac{\cos 2\omega t}{2}\right) dt = \lim_{T\to\infty} \frac{A^2}{T}\left(\frac{1}{2}[t]_{-T/2}^{T/2} - 0\right) =$$

$$\lim_{T\to\infty} \frac{A^2}{T}\left(\frac{1}{2}T\right) = \lim_{T\to\infty} \frac{A^2}{2} = \frac{A^2}{2}$$

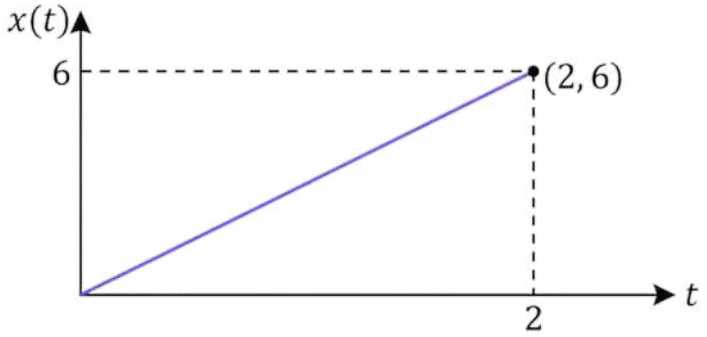
Q8: Given a square-wave periodic signal as defined by the equation:

$$y = \begin{cases} -A, & -T/2 < t < 0 \\ A, & 0 < t < T/2 \end{cases}$$

Find its power.

$$P = \frac{1}{T}\int_{-\frac{T}{2}}^{\frac{T}{2}} |y|^2 dt = \frac{1}{T}\int_{-\frac{T}{2}}^{\frac{T}{2}} |A|^2 dt = \frac{A^2}{T} t|_{-T/2}^{T/2} = A^2$$

Q9: Consider $x(t)$ as shown in the Figure:



Determine if the signal is Energy or Power Signal.

The signal $x(t)$ is aperiodic

$$E = \int_{-\infty}^{\infty} |x(t)|^2 dt$$

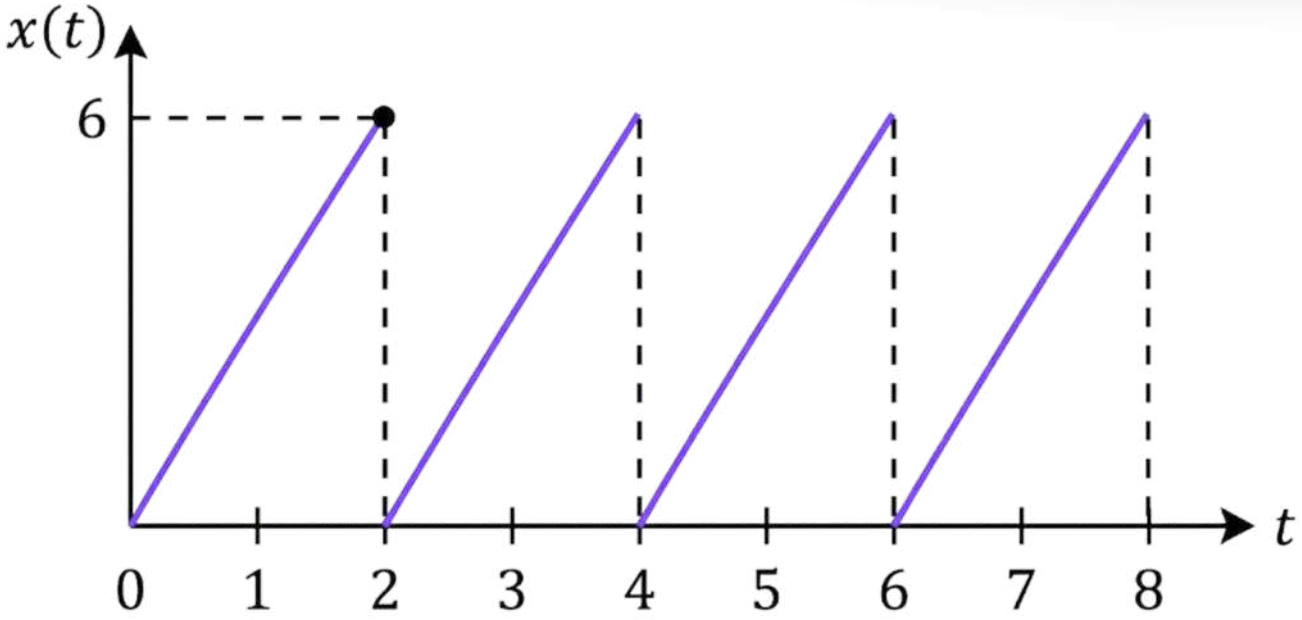
Since $0 < t < 2$, we integrate from $[0,2]$:

$$E = \int_0^2 \big(x(t)\big)^2 dt = \int_0^2 (3t)^2 dt = 9\int_0^2 t^2 dt$$

$$E = 9\left[\frac{t^3}{3}\right]_0^2 = 24$$

Since E is finite and $x(t)$ is not periodic, so; it is an **Energy Signal.**

Consider $x(t)$ shown below:



Determine if the signal is Energy or power Signal.

The signal $x(t)$ is periodic, with a repeating triangular waveform.

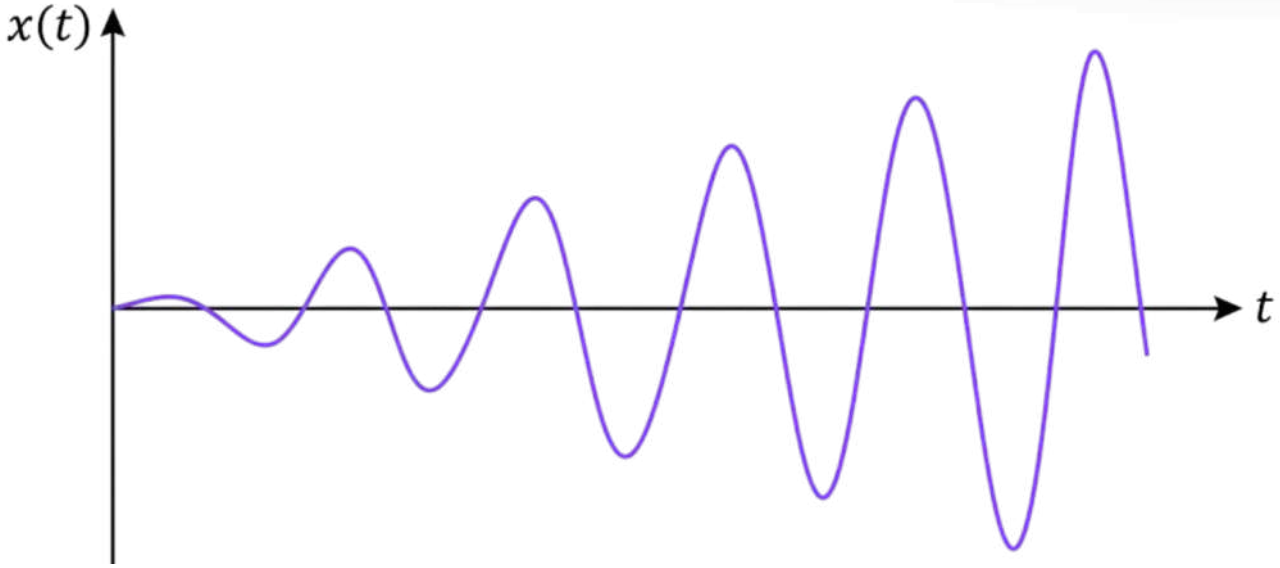
$$P = \frac{1}{T}\int_{-\frac{T}{2}}^{\frac{T}{2}} |x(t)|^2 dt$$

The signal is periodic with period $T = 4$, and we can compute the power by integrating over one period.

$$P = \frac{1}{4}\int_{-2}^{2} (3t)^2\, dt = \frac{1}{4}\int_{-2}^{2} 9t^2\, dt = \frac{1}{4}\left[3t^3\right]_{-2}^{2} = 12$$

Since P is finite, the signal is a **Power Signal**.

Q10: For $x(t) = t \sin \omega t$:



Check if the signal is:

A) Even or odd.

$x(-t) = -t * \sin(-t) = t \sin t = x(t) \rightarrow Even.$

B) Discrete or continuous.

$x(t)$ is defined for all real values of t →Continuous.

C) Periodic or Aperiodic.

Because the amplitude to grow linearly over time, the signal does not repeat →Aperiodic.

D) Power or Energy Signal.

The amplitude grows indefinitely as $t \rightarrow \infty$, causing both the energy and average power to diverge.

Thus → Neither Power nor Energy Signal.

Q11: Is $x(t) = 3\cos(3\pi t) + \sin(4\pi t) + 5$ periodic? If yes, what is the Period of this signal? Then compute the power.

$$T_1 = \frac{3}{2}s, \qquad T_2 = 2s, \qquad T_{by\ LCD} = 2 \rightarrow \text{periodic},$$

$$P = \lim_{T\to\infty} \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} (3\cos(3\pi t) + \sin(4\pi t) + 5)^2 dt =$$

$$\lim_{T\to\infty} \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} A^2 \sin^2 \omega t\,(t) dt = \lim_{T\to\infty} \frac{A^2}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} \frac{(1-\cos 2\omega t)}{2} dt =$$

$$\lim_{T\to\infty} \frac{A^2}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} \left(\frac{1}{2} - \frac{\cos 2\omega t}{2}\right) dt = \lim_{T\to\infty} \frac{A^2}{T} \left(\frac{1}{2}[t]_{-T/2}^{T/2} - 0\right) =$$

$$\lim_{T\to\infty} \frac{A^2}{T} \left(\frac{1}{2}T\right) = \lim_{T\to\infty} \frac{A^2}{2} = \frac{A^2}{2}.$$

1.3 Transformations of continuous Signals

1.3.1 Time-Shifting:

- **Time-shifting:** the process that moves the signal to right or to the left on the diagram.
- **Delay process:** the shifting the signal to the right only.
- To perform time-shifting on $x(t)$ by a time t_o the:

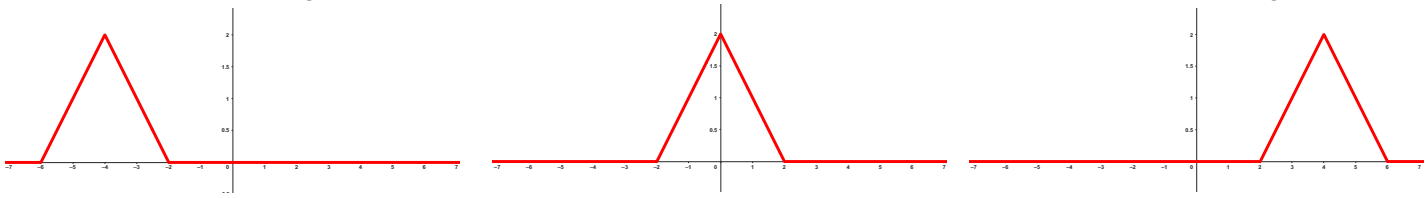
A) Right: by subtracting a constant positive quantity "t_o" in $x(t)$:

$x(t - t_o) \rightarrow$ shift to the right; $t_o > 0$

B) Left: by adding a constant positive quantity "t_o" in $x(t)$:

$x(t + t_o) \rightarrow$ shift to the left; $t_o > 0$

$x(t + t_o)$ $x(t)$ $x(t - t_o)$



Q1: If $x(t) = t^2 - 5$, write down the new version of $x(t)$ after performing time shifting to:

1. $x(t)$ to the right by 2 units:

$$x(t-2) = (t-2)^2 - 5 = t^2 - 4t - 1$$

2. $x(t)$ to the left by 4 units.

$$x(t+4) = (t+4)^2 - 5 = t^2 + 8t + 11$$

Q2: If $x(t) = \sin(\pi t - 3)$, write the new version of $x(t)$ if:

A) Shift t to the right by 3 units.

$$x(t-3) = \sin(\pi(t-3) - 3) = \sin(\pi t - 3\pi - 3)$$

$$x_A(t) = \sin(\pi t - 3(\pi + 1))$$

A) Shift t to the left by $3/\pi$ units.

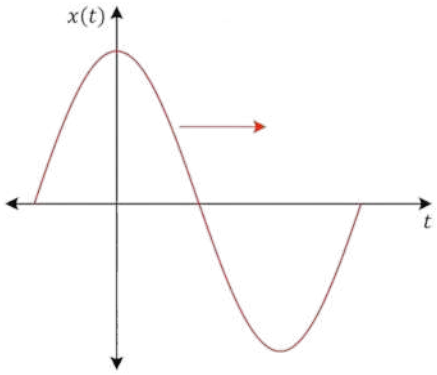
$$x(t + \pi 3) = sin(\pi(t + \pi 3) - 3) = sin(\pi t + 3 - 3)$$

$$x_B(t) = \sin(\pi t)$$

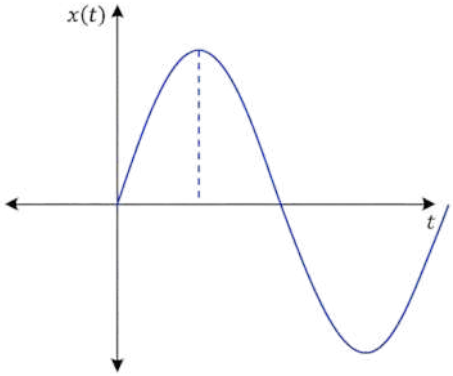
Q3: Shift $x(t) = \cos t$ to the right by $\pi/2$.

$$x(t + \pi/2) = \cos(t + \pi/2) = \sin(t)$$

$\cos(t)$



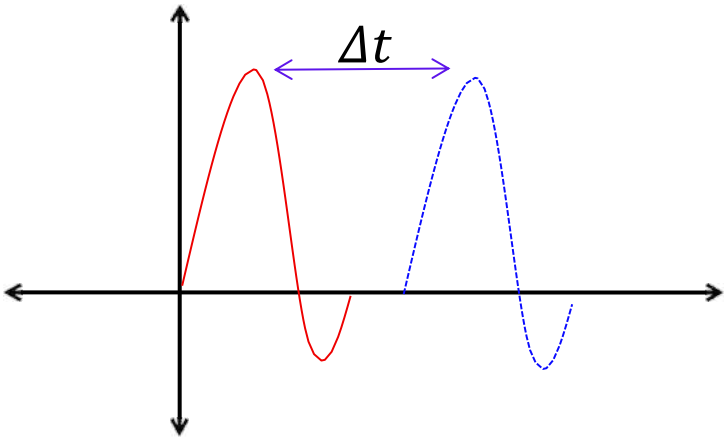
$\sin(t)$



- The same result can be obtained graphically if we shift the $\cos t$ to the right by $\pi/2$.

Applications on time Shifting:

- **RADAR:** is a system that uses pulses of electromagnetic waves to determine the position of a distant object.
- RADAR operates by sending out a signal and then listening for echoes.
- If there is an object in range, the signal will bounce off that object and return to the RADAR station.
- The echo signal is a shifted version of the sent signal, by measuring the delayed time (the shifted time) as shown in the figure below, then multiply it by the speed of the light, you get the distance as follows:



$$2D = C \times \Delta t$$

- Where D is the distance and C is the light speed.

1.3.2 Time-Reflection:

- **Reflection:** the process that rotate the signal around the $y-axis$.
- $x(-t)$ is the reflected version of $x(t)$.

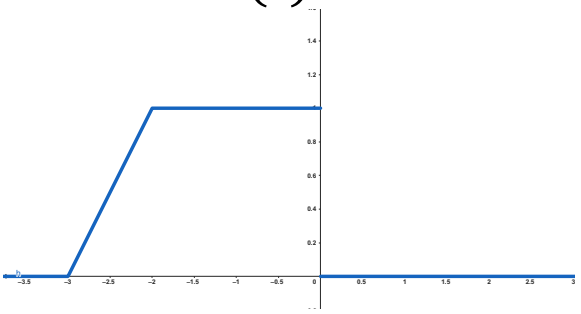
Notes:

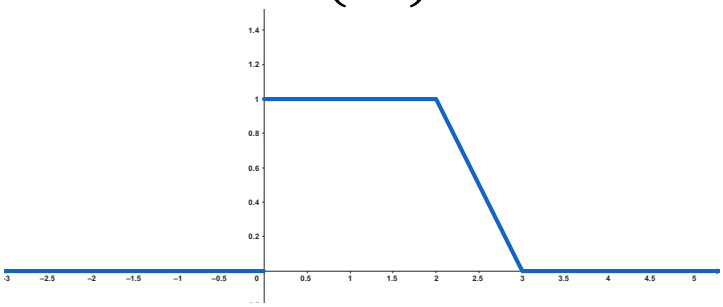
∴ Reflection can be symmetrical or not symmetrical.
If symmetrical, then we may have two types:

∴ Even Symmetry, when $x(t) = x(-t)$.

∴ Odd Symmetry, when $x(t) = -x(-t)$.

$x(t)$ $\qquad$ $x(-t)$

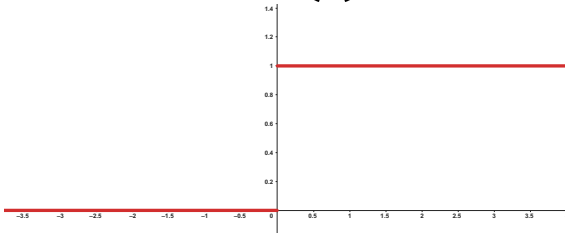


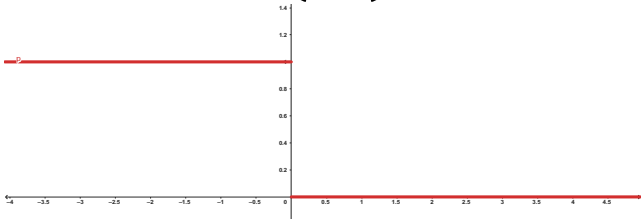


Q4: Reflect the following function:

$$u(t) = \begin{cases} 1, & t \geq 0 \\ 0, & t < 0 \end{cases}$$

$u(t)$ $\qquad$ $u(-t)$



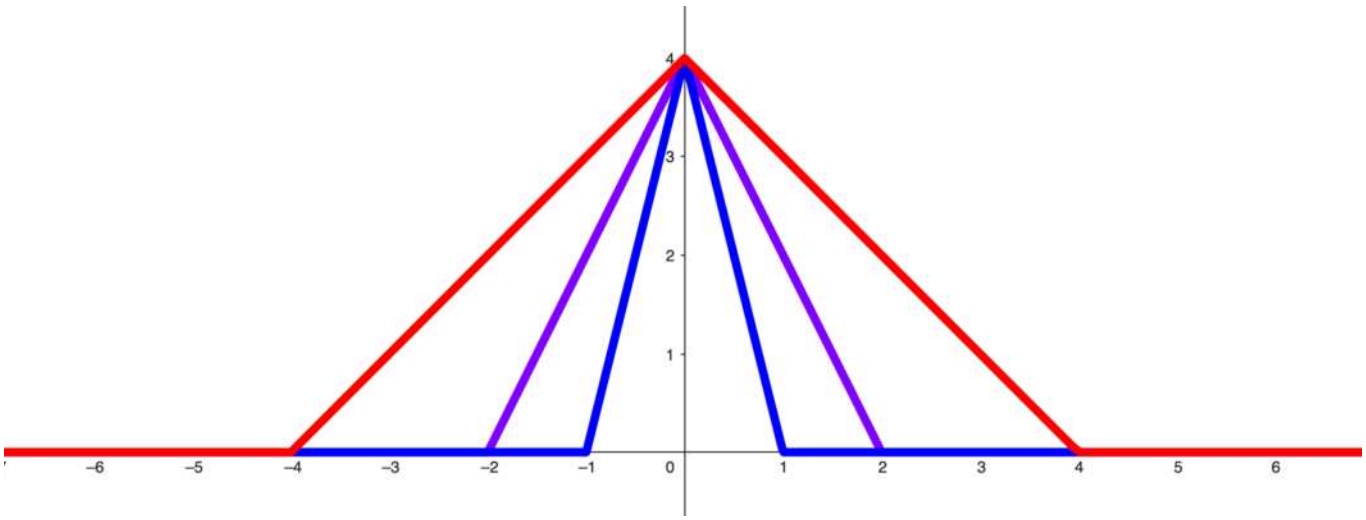


$$u(-t) = \begin{cases} 1, -t \geq 0 \\ 0, -t < 0 \end{cases} = \begin{cases} 1, t \leq 0 \\ 0, t > 0 \end{cases}$$

1.3.3 Time-Scaling:

- **Time scaling:** the process that expands or compresses the signal in time.
- Time scaling of a signal $x(t)$ implies $x(\alpha t)$ leading to:
  - $\therefore$ If $\alpha > 1$, then $x(t)$ is compressed by factor α.
  - $\therefore$ If $\alpha < 1$, then $x(t)$ is expanded by factor α.
- Where α is constant.

$x(2t)$	$x(t)$	$x(t/2)$
$\alpha > 1$	$\alpha = 1$	$\alpha < 1$



1.3.4 Combination of Operators:

- From the previous operators we conclude that $x(-t)$ is the reflected version of $x(t)$, $x(2t)$ for example is the scaled version of $x(t)$, and $x(t-3)$ is the shifted version.
- But something like $x(2t-6)$is a scaled and reflected version of $x(t)$at the same time. That means this function has two operators.
- And something like $x(-3t+6)$ has three operators: scaling, shifting, and reflection.
- The question is how to do such an operation?

▶ **To do so, use one of the following procedures:**

1. Do the shift first, then scaling, then reflection.

or

2. Do the scale first, then reflection, then shifting.

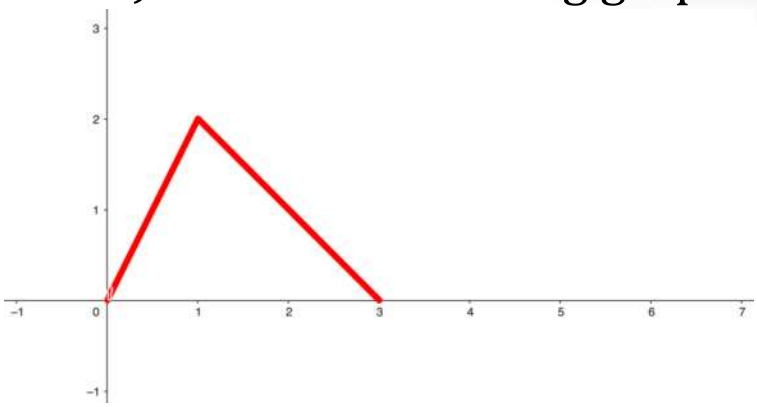
Q5: Find $x(3t-6)$ from:

$$x(t)=\begin{cases} t+3, & -3\le t<0 \\ 3-t, & 0\le t<3 \end{cases}$$

$$x(3t-6)=\begin{cases} (3t-6)+3, & -3\le 3t-6<0 \\ 3-(3t-6), & 0\le 3t-6<3 \end{cases}$$

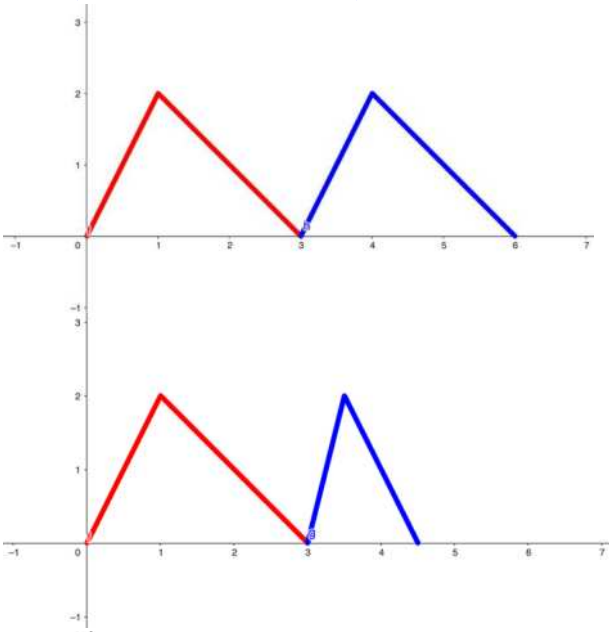
$$=\begin{cases} (3t-3), & 1\le t<2 \\ (-3t+3), & 2\le t<3 \end{cases}$$

Q6: Draw $x(2t-6)$ from the following graph:

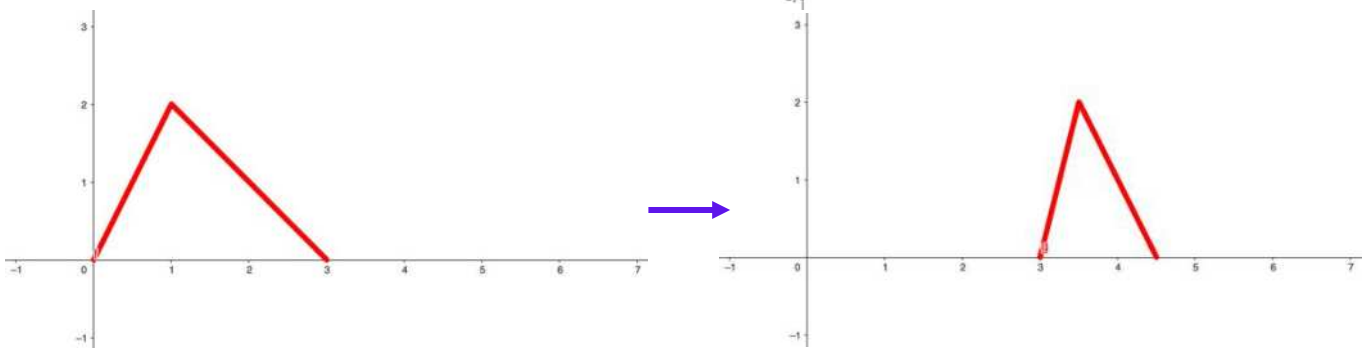


First, from $x(2t-6) = x\big(2(t-3)\big)$ we conclude that $x(t)$ is shifted (to right) by 3 then scaled (compressed) by 2.

Shifting to right by 3 first:
$(0 \to 3) \to (3 \to 6)$



Then scaling by 2:
$(3 \to 6) \to (3 \to 4.5)$



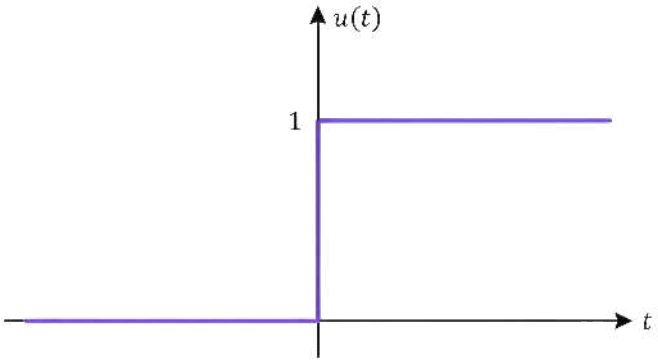
1.4 Elementary Signals

1.4.1 Unit Step Function:

- The unit step Signal $u(t)$is defined as:

$$u(t) = \begin{cases} 1, & t > 0 \\ 0, & otherwise \end{cases}$$

- Graphically:



- The shifted version of the unit step is:

$$u(t - t_o) = \begin{cases} 1, & t \geq t_o \\ 0, & otherwise \end{cases}$$

- The reflected unit step is:

$$u(-t) = \begin{cases} 1, & t \leq 0 \\ 0, & otherwise \end{cases}$$

- The scaled unit step remains the same as $u(t)$:

$$u(\alpha t) = \begin{cases} 1, & t \geq 0 \\ 0, & otherwise \end{cases}$$

- α doesn't effect on $u(t)$ because that it has a slope with zero value, so when you scale it by α the slope remains zero $(0\alpha = 0)$

From the step signal we can represent other types of signals, such as:

1. Rectangular Signal "$rect(t)$".
2. Signum Signal "$Sgn(t)$".

Unit Step Function Application:

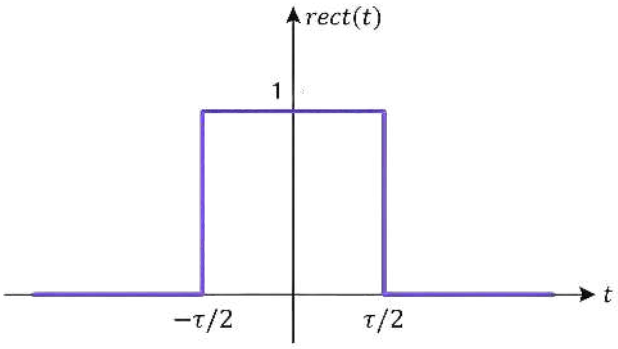
1. These signals are an important signal for analytic studies, and they also have many practical applications.
2. The function is used in the mathematics of control theory and signal processing to represent a signal that switches on at a specified time and stays switched on indefinitely.
3. It is also used in structural mechanics together with Dirac Delta function to describe different types of structural loads.

1.4.2 Rectangular Function:

- The rectangular signal $rect\left(\frac{t}{\tau}\right)$ is defined as:

$$rect\left(\frac{t}{\tau}\right) = \begin{cases} 1, & -\frac{\tau}{2} \le t \le \frac{\tau}{2} \\ 0, & otherwise \end{cases}$$

- Where τ is the width of the rectangular signal.
- Graphically:



- We can represent $rect\left(\frac{t}{\tau}\right)$ using $u(t)$:

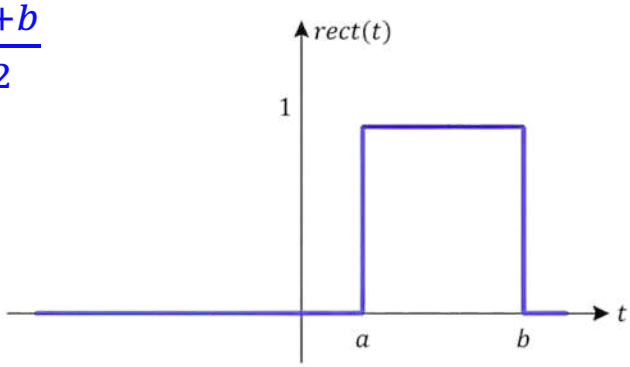
$$rect\left(\frac{t}{\tau}\right) = u\left(t + \frac{\tau}{2}\right) - u\left(t - \frac{\tau}{2}\right)$$

Q7: Represent the following graph using the:

A) *rect* signal.

$width = b - a,\ center\ point = \frac{a+b}{2}$

$$x(t) = A\ rect\left(\frac{t - \frac{a+b}{2}}{b-a}\right)$$



B) unit step $u(t)$ signal.

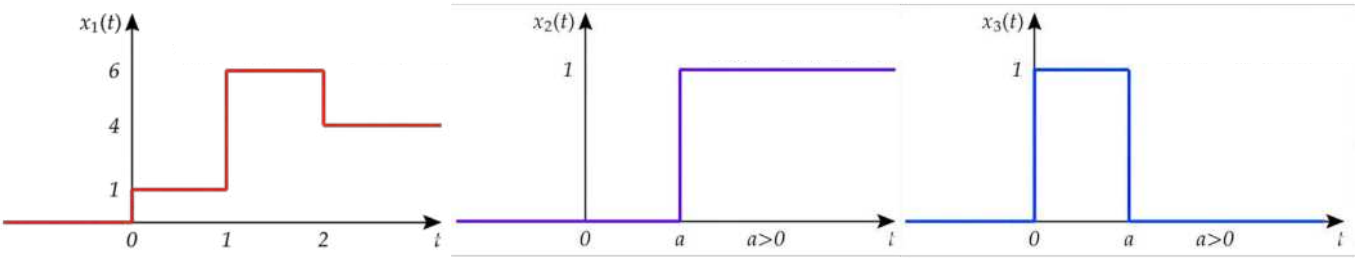
$$x(t) = A\big(u(t-a) - u(t-b)\big)$$

Q8: Sketch the following signals:

A) $x_1(t) = u(t) + 5u(t-1) - 2u(t-2).$

B) $x_2(t) = u(t) \cdot u(t-a),\ a > 0$

C) $x_3(t) = u(t) \cdot u(-t-a),\ a > 0$

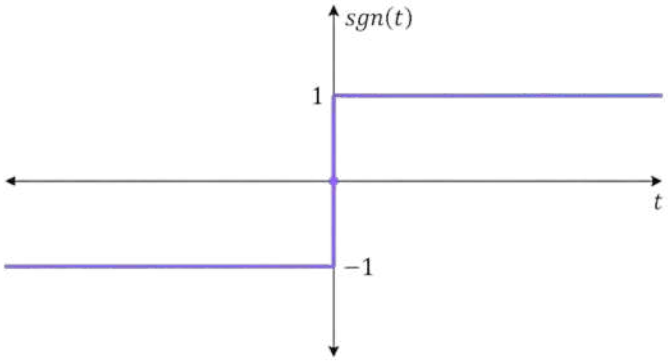


1.4.3 Signum Function:

- The signum signal $sgn(t)$ is defined as:

$$sgn(t) = \begin{cases} 1, & t > 0 \\ 0, & t = 0 \\ -1, & t < 0 \end{cases}$$

- Graphically:



- And we can represent $sgn(t)$ using $u(t)$:

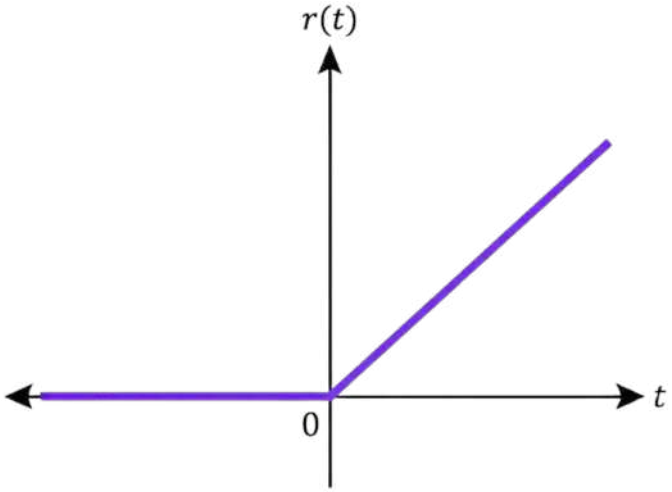
$$sgn(t) = u(t) - u(-t)$$

1.4.4 Ramp Function:

- The ramp signal $r(t)$ is defined as:

$$r(t) = \begin{cases} t, & t \leq 0 \\ 0, & otherwise \end{cases}$$

- Graphically:



- Ramp function is a unary real function, whose graph is shaped like a ramp. It can be expressed by numerous definitions, for example "0 for negative inputs, output equals input for non- negative inputs".
- The term "ramp" can also be used for other functions obtained by scaling and shifting, the unit ramp function (slope 1, starting at 0).
- In mathematics, the ramp function is also known as the positive part.

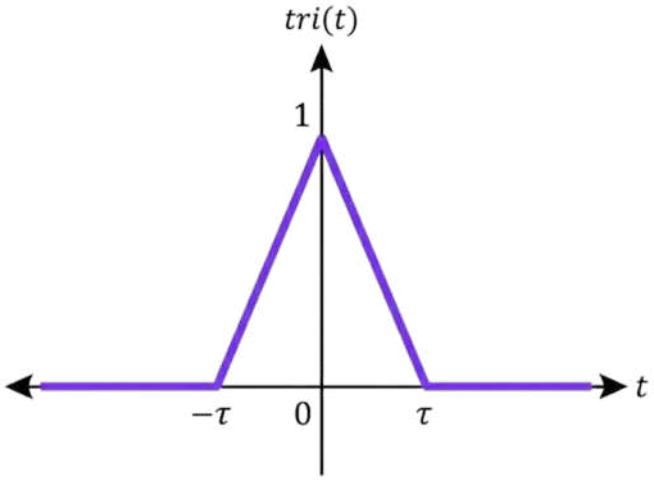
Triangular Function:

- From the ramp signal we can represent other types of signals, such as triangular function $tri(t)$ or $\Delta(t)$:

$$tri\left(\frac{t}{\tau}\right) = \frac{2}{\tau} r\left(t + \frac{\tau}{2}\right) - \frac{4}{\tau} r(t) + \frac{2}{\tau} r\left(t - \frac{\tau}{2}\right)$$

$$= \begin{cases} 1 - \dfrac{|t|}{\tau}, & |t| \le \tau \\ 0, & otherwise \end{cases}$$

- Graphically:



Ramp Function applications:

1. In machine learning, it is commonly known as a rectifier (neural networks) or ReLU (rectified linear unit) activation function or a rectifier in analogy to half-wave rectification in electrical engineering.
2. In statistics (when used as a likelihood function) it is known as a Tobit model.

Ramp Function properties:

1. The derivative of $r(t)$ is $u(t)$:

$$\frac{dr(t)}{dt} = u(t)$$

2. The integral of $u(t)$ is $r(t)$:

$$r(t) = \int_{-\infty}^{t} u(t)dt$$

- The device that accomplishes this operation is **integrator.**

Important notes for drawing:

1. If $u(t)$ and $r(t)$ shifted, they will start from that shift point:

$$u(t-a)=\begin{cases}1, & t>a\\ 0, & t<a\end{cases}, \qquad r(t-a)=\begin{cases}t, & t>a\\ 0, & t<a\end{cases}$$

2. If $u(t)$ multiplied by a constant, it represents its magnitude:

$$A\cdot u(t)=\begin{cases}A, & t>0\\ 0, & t<0\end{cases}$$

but in $r(t)$ it represents its slope:

$$A\cdot r(t)=\begin{cases}At, & t>0\\ 0, & t<0\end{cases}$$

in addition, amplitudes and slopes can be added as a linear operation:

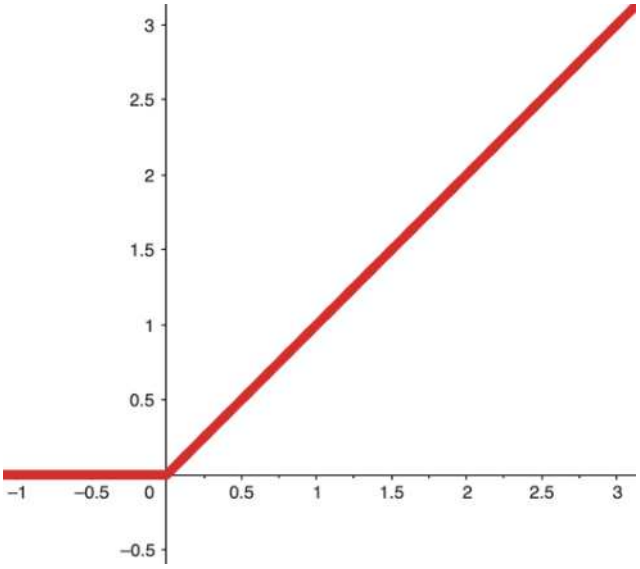
- **Example:**

$$2u(t)+3u(t)=5u(t), \qquad 3r(t)-4r(t)=-r(t)$$

- Since you have $A\,r(t)$ and it goes to ∞ with slope A, when you put to the signal $-A\,r(t-a)$ you make the slope 0 at the point a; so, you change it from $r(t)$ to $u(t)$ at that shifted point.

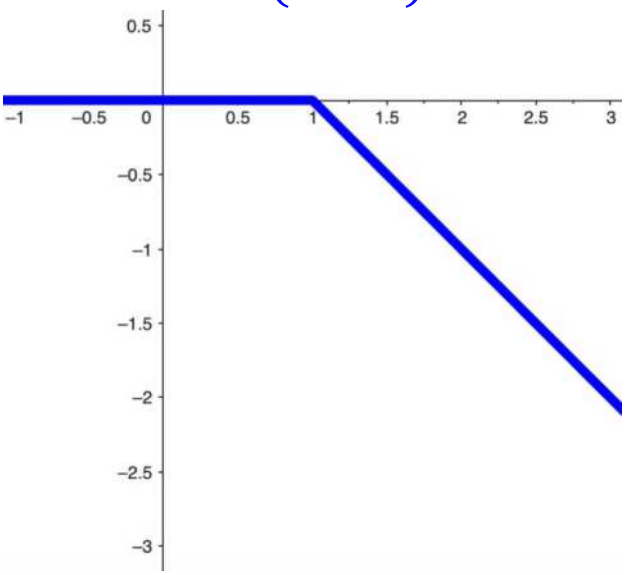
- **Example:**

$r(t)$



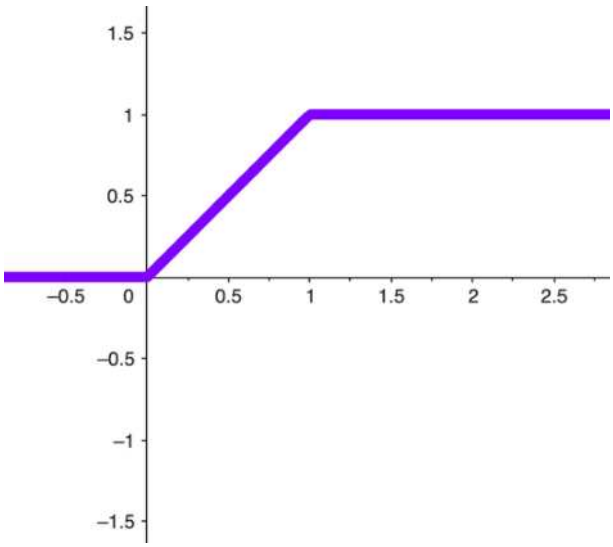
$+$

$-r(t-1)$

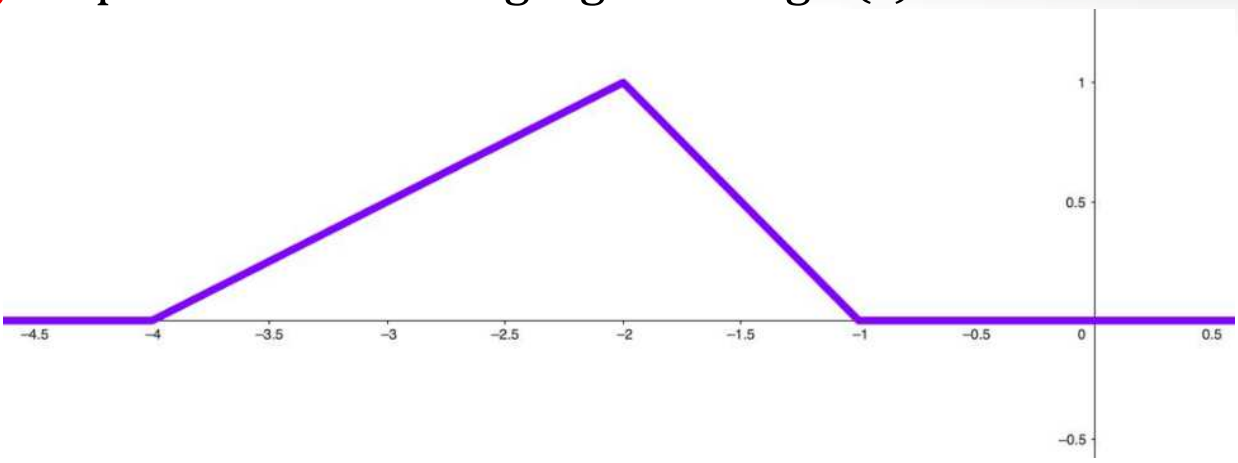


$\rightarrow$

$r(t)-r(t-1)$



Q9: Express the following signal using $r(t)$:



At starting point (-4), we conclude that the slope at that ramp is 2: $\frac{\Delta y}{\Delta x} = \frac{0-1}{-4--2} = \frac{1}{2}$:

$$x_1(t) = \frac{1}{2}r(t+4)$$

Then, at the point (-2), the ramp changes its slope from $\frac{1}{2}$ to $-1 \rightarrow \left(\frac{\Delta y}{\Delta x} = \frac{1-0}{-2--1}\right)$, then we say that we add ramp function with slope $-\frac{3}{2}$ at the point -2 $\left(-\frac{1}{2}\right.$ to make it 0 and -1 as final slope$\left.\right)$; so:

$$x_2(t) = -\frac{3}{2}r(t+2)$$

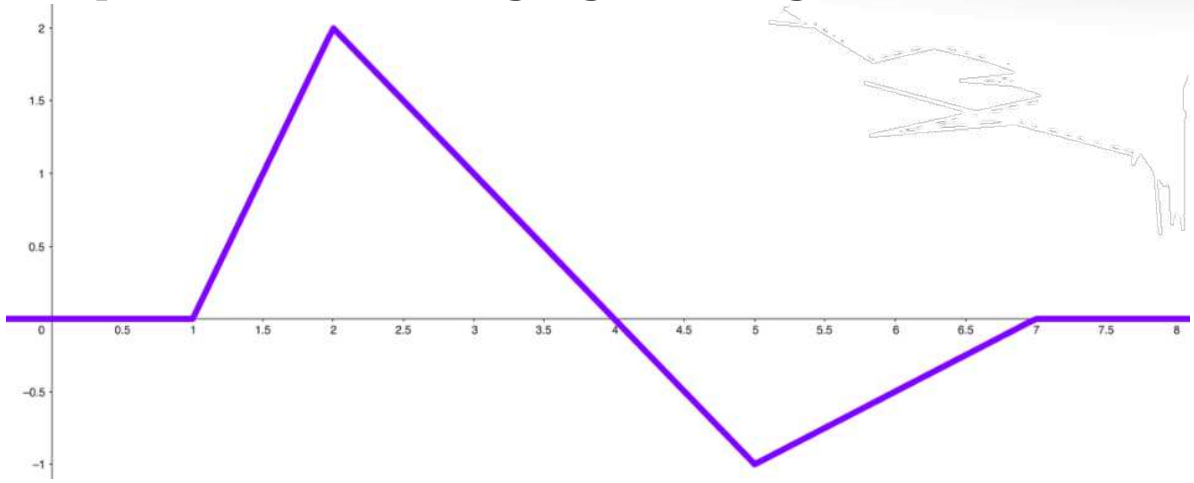
Then, at the point (-1), the ramp changes its slope from -1 to 0, then we say that we add ramp function with slope 1 at the point -1; so:

$$x_3(t) = r(t+1)$$

Finally, combine $x_1(t) + x_2(t) + x_3(t) = x(t)$:

$$x(t) = \frac{1}{2}r(t+4) - \frac{3}{2}r(t+2) + r(t+1)$$

Q10: Express the following signal using $r(t)$:



At starting point (1), we conclude that the slope at that ramp is 2:

$$x_1(t) = 2r(t-1)$$

Then, at the point (2), the ramp changes its slope from 2 to -1, so:

$$x_2(t) = -2r(t-2) - r(t-2) = -3r(t-2)$$

Then, at the point (5), the ramp changes its slope from -3 to $\frac{1}{2}$, so:

$$x_3(t) = 3r(t-5) + \frac{1}{2}r(t-5) = \frac{7}{2}r(t-5)$$

Then, at the point (7), the ramp changes its slope from $\frac{1}{2}$ to 0 , so:

$$x_4(t) = -\frac{1}{2}r(t-7)$$

After combining:

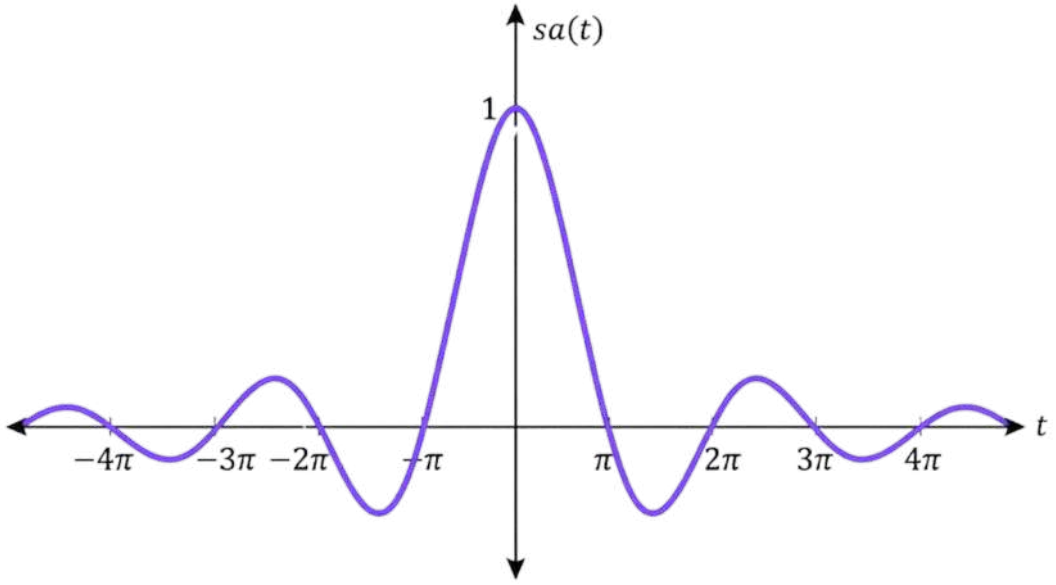
$$x(t) = 2r(t-1) - 3r(t-2) + \frac{7}{2}r(t-5) - \frac{1}{2}r(t-7)$$

1.4.5 Sampling Function:

- The Sampling Signal $\mathrm{sa}(t)$ is defined as:

$$sa(t) = \begin{cases} 1, & t = 0 \\ \dfrac{\sin t}{t}, & otherwise \end{cases}$$

- Graphically:



Sampling Function properties:

1. The zero crossing $(\mathrm{sa}(t) = 0)$ occurs at:

$$t = \pm k\pi, \qquad where\ k = 1,2,3, \ldots$$

2. The area under the curve of $sa(t)$ is π, That means:

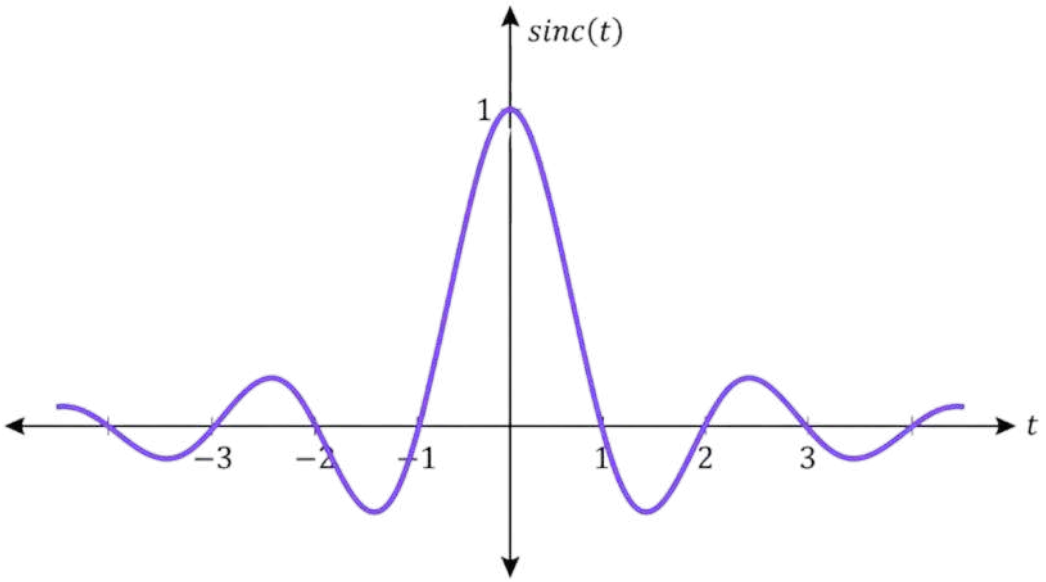
$$\int_{-\infty}^{\infty} \mathrm{sa}(t)\, dt = \pi$$

1.4.6 Sinc Function:

- The Sinc Signal $\text{sinc}(t)$ is defined as:

$$sinc(t) = \begin{cases} 1 , & t = 0 \\ \dfrac{\sin \pi t}{\pi t}, & otherwise \end{cases}$$

- Graphically:



Sampling Function properties:

1. The zero crossing ($\text{sinc}(t) = 0$) occurs at:

$$t = \pm k, \qquad where\ k = 1,2,3, \ldots$$

2. The area under the curve of $\text{sinc}(t)$ is 1, That means:

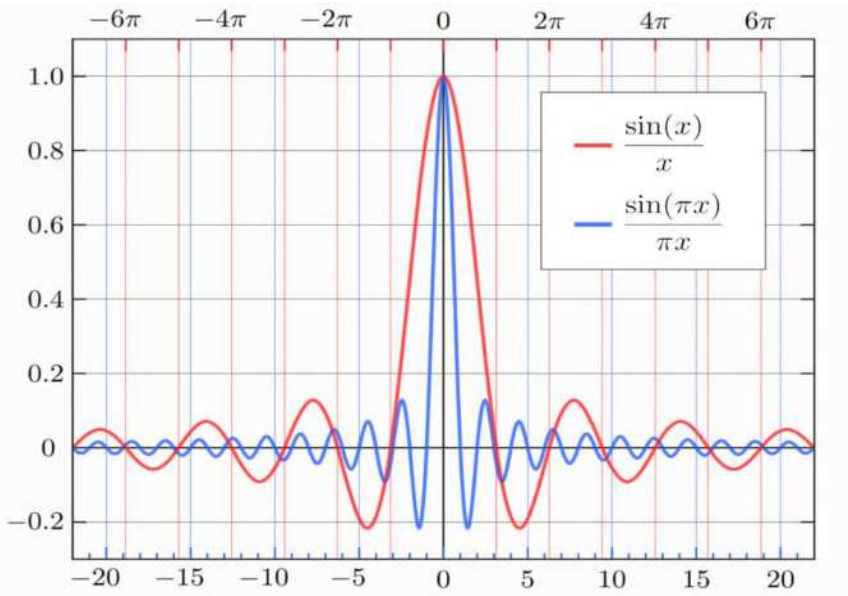
$$\int_{-\infty}^{\infty} \text{sinc}(t)\, dt = 1$$

The differences between Sa(t) & Sinc(t):

1. Note that:

$$\text{sinc}(t) = \text{sa}(\pi t), \qquad \text{sa}(t) = \text{sinc}\left(\frac{t}{\pi}\right)$$

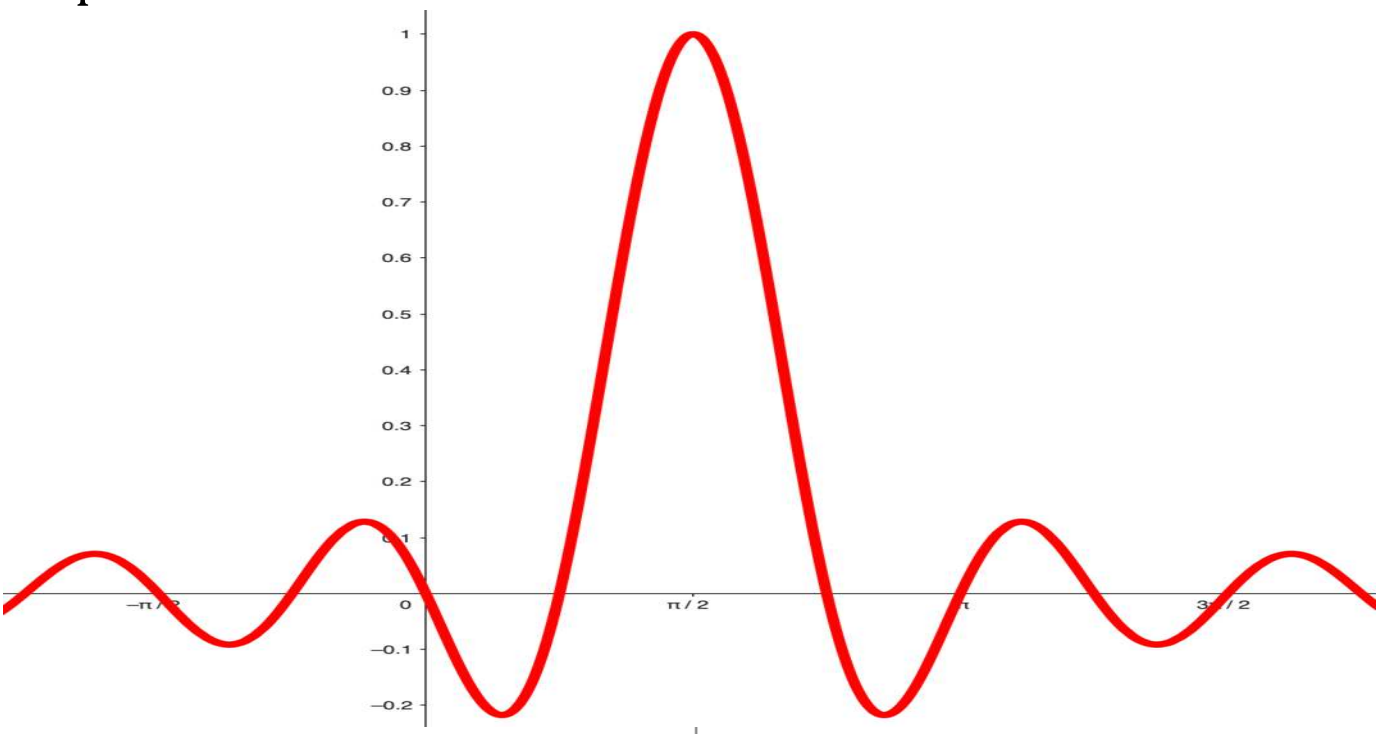
2. The normalized Sinc(t), is defined such that the definite integral of the function over all $\mathbb{R}$ is 1.
In contrast, the unnormalized sinc function $\text{sa}(t) = \frac{\sin t}{t}$, has a definite integral over all $\mathbb{R}$ is π.

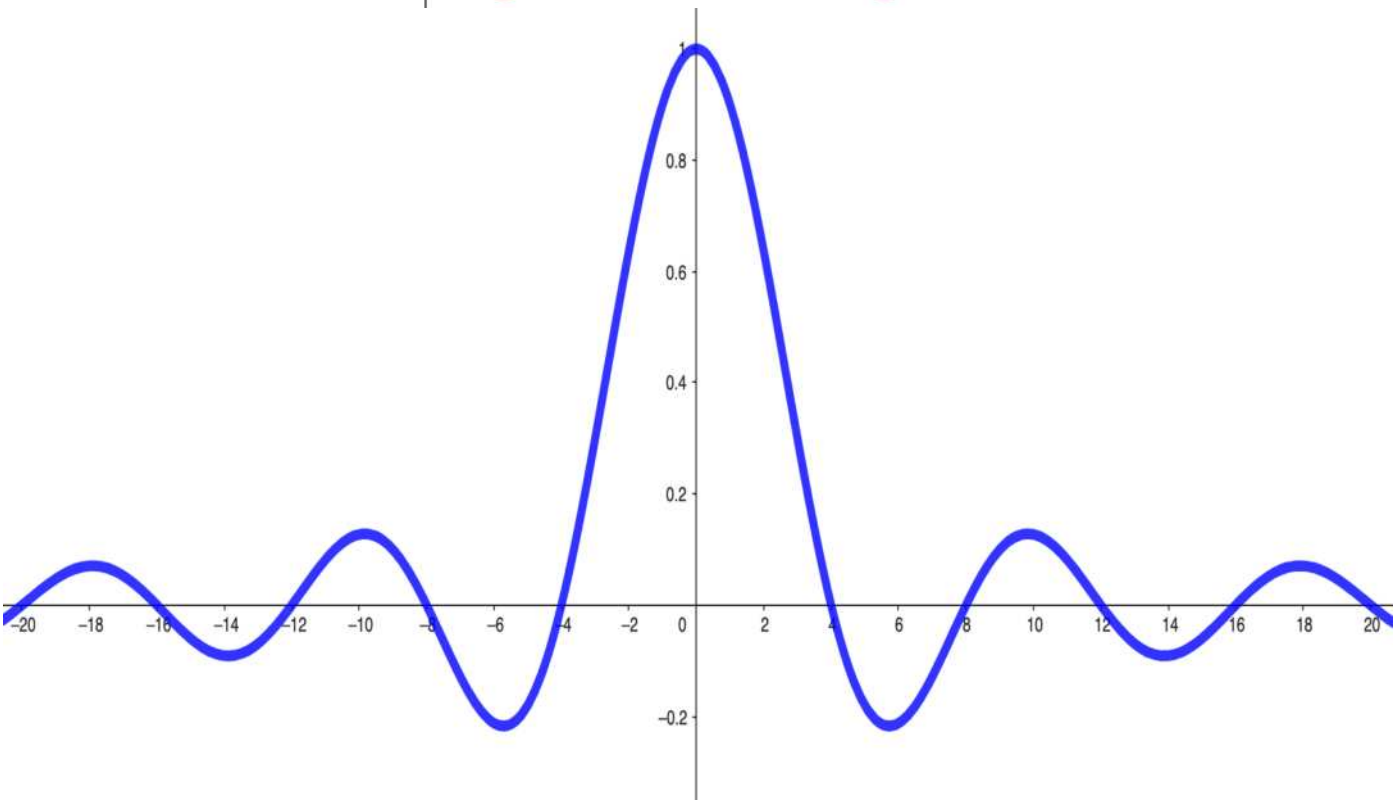


Sinc Function applications:

1. sinc(t) is the Fourier transform (F.T) of the rectangular function which has many interests in the telecommunication world.
2. sinc(t) is fundamental in the concept of reconstructing the original continuous bandlimited signal from uniformly spaced samples of that signal.
3. sinc(t) is commonly used in DSP applications.

Q11: Draw $\text{Sinc}\left(\frac{t}{4}\right)$ & $\text{sa}(4t - 2\pi)$ and determine the first five zero crossing points of the negative and the positive side.



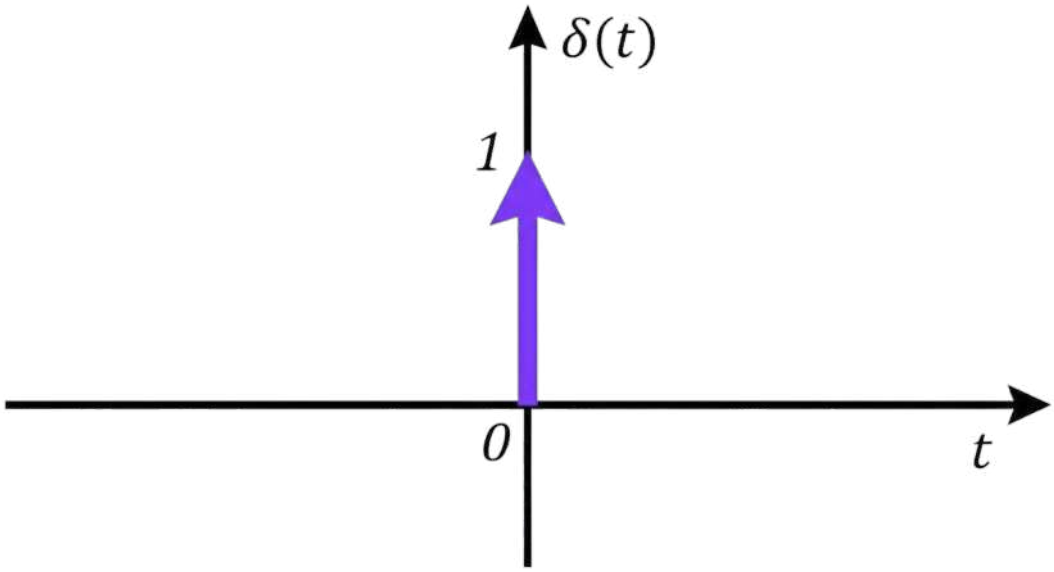


1.4.7 Unit Impulse Function:

- The Unit Impulse (Dirac or Dirac Delta) Signal $\delta(t)$ is defined as:

$$\delta(t) = \begin{cases} 1, & t = 0 \\ 0, & otherwise \end{cases}$$

- Graphically:



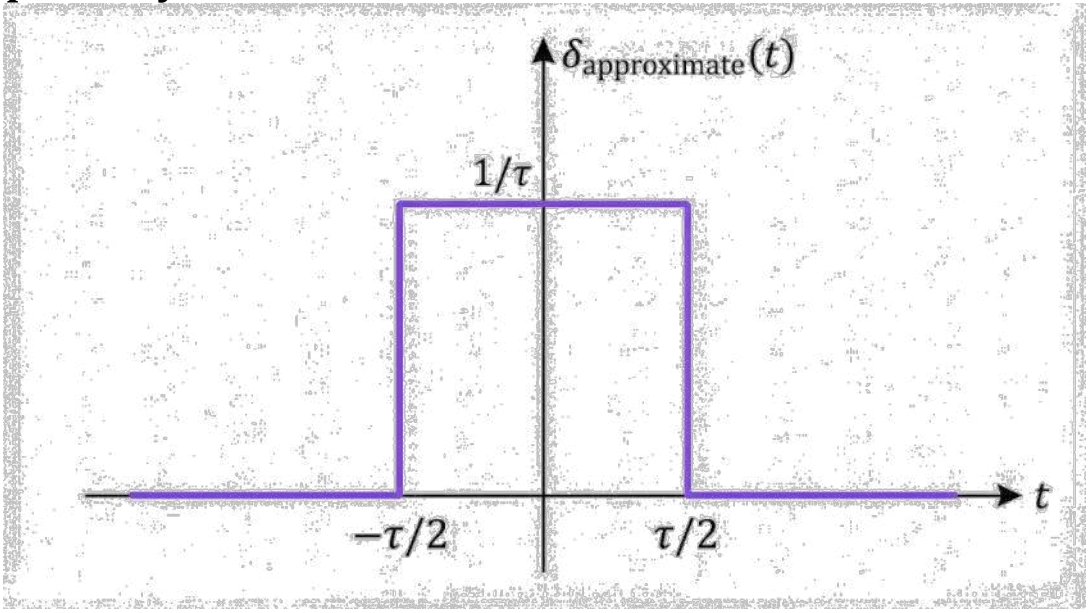
▶ **$\delta(t)$ satsfies:**

1. The peak occurs at $t = 0$.
2. Their duration is short, so as $\tau \to 0$. Where τ is the width of the signal.
3. The total area under each function equal to 1.

- We can approximate $\delta(t)$ using rectangular pulse:

$$\delta_a(t) = \begin{cases} \frac{1}{\tau}, & |\tau| <= \frac{\tau}{2} \\ 0, & otherwise \end{cases}$$

- Graphically



- $\delta(t)$ is even function.

Delta Function applications:

1. It occupies a central place in signal analysis.
2. Many physical phenomena can be modeled as Delta signal such as:

A) point sources.
B) point charges.
C) concentrated loads on structures,
D) voltage or current sources acting for very short times.

Delta Function properties:

1. When you multiply a function by $delta$:

$$x(t-A)\delta(t-B) = x(B-A)\delta(t-B) \rightarrow (1)$$

2. When you integrate a function that it multiplied by $delta$:

$$\int_a^b x(t-A)\delta(t-B)dt = x(B-A) \rightarrow (2)$$

$$When \;\; a \leq t-B \leq b$$

3. When you integrate scaled & shifted $delta$:

$$\int_a^b \delta\big(At-B\big)dt = \frac{1}{|A|}\int_a^b \delta\left(t-\frac{B}{A}\right)dt \rightarrow (3)$$

4. When you integrate a derived function that multiplied by $delta$:

$$\int_a^b x(t-A)\delta^{(n)}(t-B)dt = (-1)^n x^{(n)}(B-A) \rightarrow (4)$$

$$When \;\; a \leq t-B \leq b$$

Q12: Evaluate:

A) $\int_0^3 e^{t-2} \cdot \delta(2t-4)dt$

$\int_0^3 e^{t-2} \cdot \delta\big(2(t-2)\big)dt$ *2 as a C.F.

$= \frac{1}{2}\int_0^3 e^{t-2} \cdot \delta(t-2)dt$ *Using (3) Rule

$= \frac{1}{2}e^{-2-(-2)} = \frac{1}{2}$ *Using (2) Rule

B) $\int_{-4}^{4}(t-2)^2 \cdot \delta'\left(-\frac{1}{3}t+\frac{1}{2}\right)dt$

$\int_0^3 (t-2)^2 \cdot \delta'\left(\frac{-1}{3}\left(t-\frac{3}{2}\right)\right)dt$ *$\frac{-1}{3}$ as a C.F

$= 3\int_{-4}^{4}(t-2)^2 \cdot \delta'\left(t-\frac{3}{2}\right)dt$ *Using (3) Rule

$= -3\frac{d}{dt}\left((t-2)^2\right)|_{t=\left(\frac{3}{2}-2\right)}$ *Using (4) Rule

$= -3\left(2\left(\frac{3}{2}-2\right)\right) = 3$ *Substitution

C) $\int_{-\infty}^{\infty} t \cdot u(t+1) \cdot u(2-t)dt$

$u(t+1), u(2-t)$ *Limiting the domain

$\rightarrow t+1>0,\ 2-t>0$

$\rightarrow -1<t<2$

$\rightarrow \int_{-1}^{2} t\,dt = \frac{3}{2}$

D) $\int_{-\infty}^{\infty} e^{-t} \cdot \delta\left(3 - \frac{5}{2}t\right) dt$

$\int_{-\infty}^{\infty} e^{-t} \cdot \delta\left(\frac{-5}{2}\left(t - \frac{6}{5}\right)\right) dt$ *$-5/2$ as a C.F.

$= \frac{2}{5}\int_{-\infty}^{\infty} e^{-t} \cdot \delta\left(\left(t - \frac{6}{5}\right)\right) dt$ *Using (3) Rule

$= \frac{2}{5} e^{-\left(\frac{6}{5}\right)} = \frac{2}{5} e^{-\frac{6}{5}}$ *Using (2) Rule

E) $\int_{0}^{\infty} \sqrt{t+1} \cdot \delta(t+2) dt$

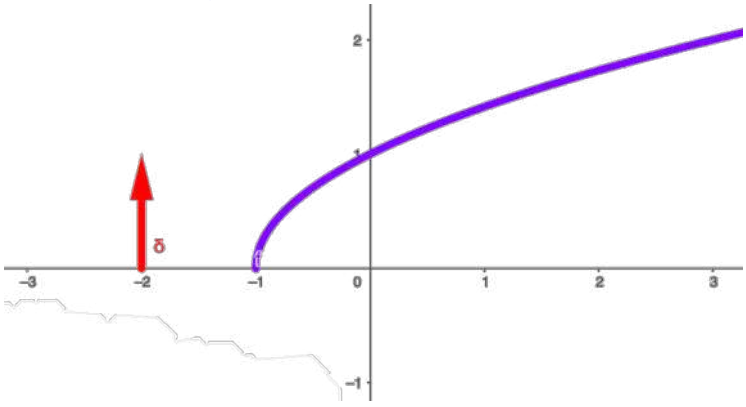
$\int_{0}^{\infty} \sqrt{(t+1)} \cdot \delta\big(t - (-2)\big) dt$

$= \sqrt{-2 - (-1)} =$ undefind *Using (2) Rule

We must deal with real numbers

The domain is $(-1, \infty)$and the delta function is out of it

"Shifted from 0 to left by 2" so, it is 0.



F) $\int_{-\infty}^{\infty} t^3 \cdot \delta''(t+2) dt$

$\int_{-\infty}^{\infty} t^3 \cdot \delta''\big(t - (-2)\big) dt$

$= \frac{d^2}{dt^2}\left(t^3\right)|_{t=-2}$ *Using (4) Rule

$= 6(-2) = -12$ *Simplifying

G) $\sin\left(\frac{t^2-2}{\pi}\right)\cdot\delta\left(\frac{2}{\pi}t+2\right)$

$\sin\left(\frac{t^2-2}{\pi}\right)\cdot\delta\left(\frac{2}{\pi}(t-(-\pi))\right)$ *$\frac{2}{\pi}$ as a C.F

$\delta(t+\pi)\sin\left(\frac{(-\pi)^2-2}{\pi}\right)$ *Using (4) Rule

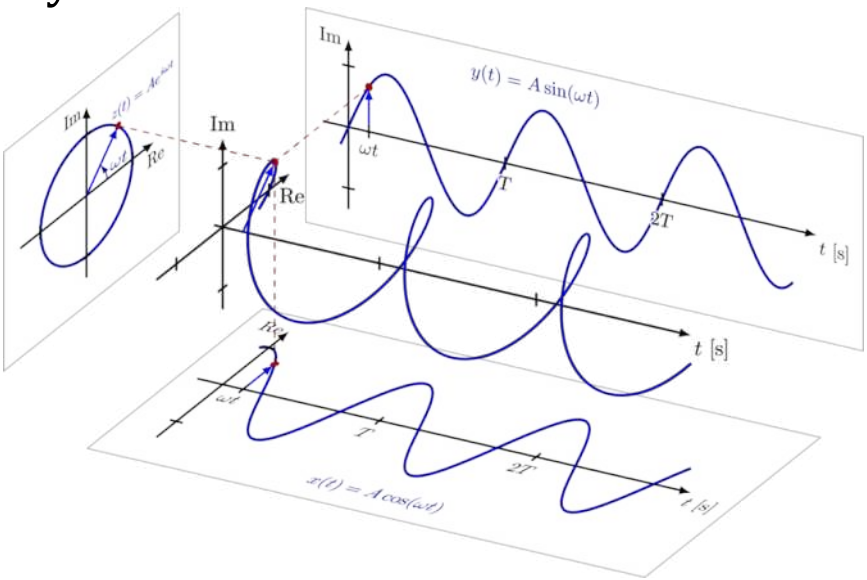
$=\delta(t+\pi)\sin\left(\pi-\frac{2}{\pi}\right)$ *Simplifying

Remember that:

- The Complex Signal is defined as:

$$x(t)=e^{j\omega_0 t}=\cos\omega_0 t+j\sin\omega_0 t$$

- Graphically:



- The fundamental period T_0 of $x(t)$ is given by:

$$T_0=\frac{2\pi}{\omega_0},\qquad T_0\omega_0=2\pi$$

- This relation is very important when we deal with Fourier Series & Fourier Transformation (Chapters 3&4)

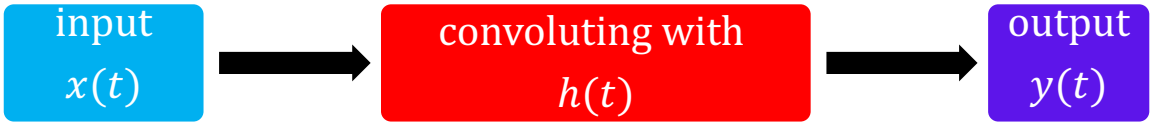
2 Continuous-Time Systems

2.1 Continuous systems classification

2.1.1 Introduction:

- **Systems can be described by one of the following:**

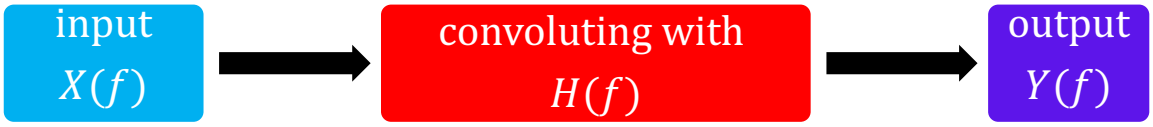
H) Continuous-Time Systems



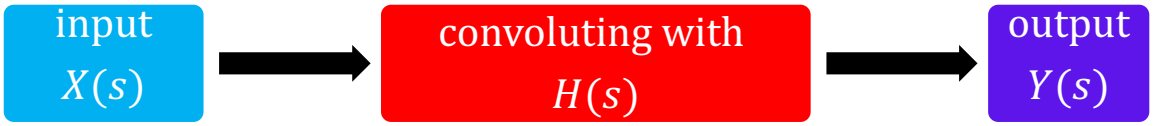
I) Discrete-Time Systems



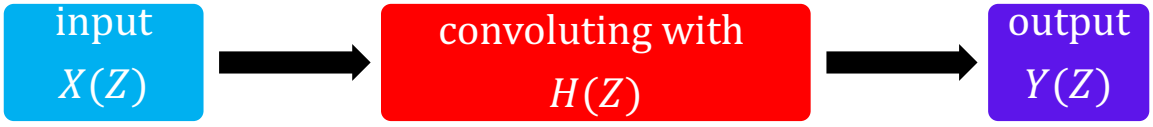
J) f-Domain Systems



K) s-Domain systems



L) Z-Domain systems



▶ **You can classify continuous-time systems with the following classes:**

1. Linear & Zant & Time-Variant Systems.
2. Causal & Anti-Causal Systems.
3. Memory & Memoryless Systems.
4. Stable & Unstable Systems.

2.1.2 Linear & Non-Linear Systems:

- To make the distinction between linearity and nonlinearity a bit more precise, recall that a mathematical equation can be thought of as a function.
- The equation $y = x$, for instance, is equivalent to a function that takes as its input a value for x and produces as its output a value for y.
- In other words, in a Linear system the superposition principle is applicable.

▶ **To check the linearity of a system:**

1. Superposition Method:

∴ Let $x_1(t)$ as an input to get $y_1(t)$ as an output.

∴ Let $x_2(t)$ as an input to get $y_2(t)$ as an output.

∴ Then apply $x_1(t) + x_2(t)$ as an input signal to the system.

∴ If the output is $y_1(t) + y_2(t)$, Then the system is **linear system.**

2. Scaling Method:

∴ Let $x(t)$ as an input to get $y(t)$ as an output.

∴ Let $y(x(t))$ as another output, If you can write it as $y_1(t)$,Then the system is **linear system.**

3. Fast Method:

- If the system is written in this formula:

$$y(t) = Ax(t-a) + Bx(t-b) + \cdots + Nx(t-n) \rightarrow (2.1)$$

- The system is Linear directly when:

∴ A, B, N, a, b, n are constants

∴ Without constants and with degree 1 (to power of 1).

Q13: Check if these systems are linear or not:

A) $y(t) = 3x(t-1) + x(t-3)$

This system is written as the formula (2.1), so this system is **linear.**

Proving in superposition method:

$$y_1(t) = 3x_1(t-1) + x_1(t-3)$$
$$y_2(t) = 3x_2(t-1) + x_2(t-3),$$
$$y(x_1(t) + x_2(t))$$
$$= 3x_1(t-1) + x_1(t-3) + 3x_2(t-1) + x_2(t-3)$$
$$y(x_1(t) + x_2(t)) = y_1(t) + y_2(t)$$

B) $y(t) = x(t-2) + 3$

This system is not written as formula (2.1), so this system is **non-linear.**

Proving in scaling method:

$$Ay(t) = A[x(t-2) + 3] = Ax(t-2) + 3A$$

$$y\big(Ax(t)\big) = Ax(t-2) + 3$$

$$y\big(Ax(t)\big) \neq Ay(t)$$

A) $y(t) = t \cdot x(t-1)$

Proving in scaling method:

$$Ay(t) = A[t \cdot x(t-1)] = At \cdot x(t-1)$$

$$y\big(Ax(t)\big) = At \cdot x(t-1)$$

$$y\big(Ax(t)\big) = Ay(t)$$

so this system is **linear.**

B) $y(t) = x^2(t)$

This system is not written as formula (2.1), so this system is **non-linear.**

Proving in scaling method:

$$Ay(t) = A\big[x^2(t)\big] = Ax^2(t)$$

$$y\big(Ax(t)\big) = (Ax)^2(t) = A^2x^2(t)$$

$$y\big(Ax(t)\big) \neq Ay(t)$$

2.1.3 Time-Invariant & Time-Variant Systems:

- Time Invariant means that a shift in time of the input signal causes an identical shift in time in the output signal.
- In other words, A time-invariant system is one whose output does not depend explicitly on time.

▶ **To check time-Invariance of a system:**

1. Shifting Method:

∴ Let $x_1(t)$ as an input to get $y_1(t)$ as an output.

∴ Shift $y_1(t)$ by a to get $y_1(t-a)$

∴ Let $x_2(t)$ as an input to get $y_2(t)$ as an output.

∴ Now replace $x_2(t)$ by $x_1(t-a)$ in $y_2(t)$.

∴ If $y_2(t) = y_1(t-a)$, then the system is **time-invariant system.**

2. Fast Method:

- If the system is written in this formula:

$$y(t) = Ax(t-a) + Bx(t-b) + \cdots + C \rightarrow (2.2)$$

- The system is Time-Invariant directly when:

∴ A, B, N, a, b, n are constants

∴ It doesn't depend on time t.

Q14: Check if these systems are time invariant or not:

A) $y(t) = x(t) + 5$

This system is written as the formula (2.2), so this system is **time-invariant.**

Proving:

$$y_1(t) = x_1(t) + 5 \rightarrow y_1(t-a) = x_1(t-a) + 5$$
$$y_2(t) = x_2(t) + 5$$
$$y_2(t) = x_1(t-a) + 5$$
$$y_2(t) = y_1(t-a)$$

B) $y(t) = tx(t)$

This system is not written as the formula (2.2), and it depends on variable t, so this system is **time-variant.**

Proving:

$$y_1(t) = tx_1(t) \rightarrow y_1(t-a) = (t-a)x_1(t-a)$$
$$y_2(t) = tx_2(t)$$
$$y_2(t) = tx_1(t-a)$$
$$y_2(t) = y_1(t-a)$$

2.1.4 Causal & Anti-Causal Systems:

- A causal system is a system where the output depends on past and current inputs but not future inputs i.e. the output $y(t_0)$ only depends on the input $x(t)$ for values of $t \leq t_0$.
- For a causal system, the impulse response of the system must be 0 for all $t < 0$. That is the sole necessary as well as sufficient condition for causality of a system, linear or non-linear

▶ **To check causality of a system:**

- If the system is written in this formula:

$$y(t) = Ax(t-a) + Bx(t-b) + \cdots + C$$

∴ When $a \geq 0$ *and* $b \geq 0$, the system is causal because it depends on past and current inputs.

Q15: Check if these systems are causal or not:

A) $y(t) = x(t) + 2x(t-1)$

This system is written as the above formula, only depends on current value ($t_0 = 0$), and past value ($t_0 = -1$), so it is **causal system.**

B) $y(t) = x(t+2)$

This system is not written as the above formula, and it depends on future value ($t_0 = +1$), so it is **anti-causal system.**

2.1.5 Memory & Memoryless Systems:

- A system's output generally depends on past inputs, but for memoryless systems, the output at any instant t
- depends only on the input at that same instant.
- Memoryless systems have no stored information about past inputs, making them static or instantaneous systems.
- Systems that rely on past inputs are dynamic (or systems with memory).
- All memoryless systems are causal, but not all causal systems are memoryless.

▶ **To check on a memory of a system:**

- If the system is written in this formula:

$$y(t) = Ax(t-a) + Bx(t-b) + \cdots + C$$

∴ if $a \neq 0$ *or* $b \neq 0$ at least, the system has a memory because it depends on past and future inputs.

Q16: Check if these systems are memory or memoryless:

A) $y(t) = 3x(t)$

This system only depends on current value ($t_0 = 0$), so it is **memoryless system.**

B) $y(t) = x(t+2) + 2x(t) + 2$

This system depends on future value ($t_0 = +2$), so it is **memory system.**

2.1.6 Stable & Unstable Systems:

- A system is stable if a bounded input signal yields a bounded output signal.
- A signal is said to be bounded if its absolute value is less than some finite value for all time:

$$|x(t)| < \infty, \qquad -\infty < t < \infty.$$

- A system for which the output signal is bounded when the input signal is
- bounded is called bounded-input-bounded-output (BIBO) stable system.
- A system is BIBO stable if every bounded input results in a bounded output.

∴ Bounded input : $|x(t)| \leq M$, where M is finite.

∴ Bounded output: $|y(t)| \leq N$, where N is finite.

- Shifting and adding by constant don't effect on the stability of a system.

▶ **To check stability of a system:**

∴ Write the mathematical relationship between input and output,

∴ Then assume the input is bounded and analyze if the output is also bounded.

∴ Confirm that the output does not grow unbounded under the given input.

Q17: Check if these systems are stable or not:

A) $y(t) = 2x(t+2)$

The bounded input: $|x(t)| \leq M$,

The assumed bounded output:

$$|y(t)| = 2|x(t+2)|$$

Since $x(t)$ is bounded ($|\ x(t)\ | \leq M$, $x(t-2)$ is also bounded by M (shifting does not change the bound).

$$|y(t)| \leq 3M$$

Thus, this system is **stable**

B) $y(t) = \sin(t)\, x(t)$

The bounded input: $|x(t)| \leq M$,

The assumed bounded output:

$$|y(t)| = |\sin(t)| \cdot |x(t)|$$

Since we have $x(t)$ is bounded and $\sin(t)$ is also bounded (it's value always between $[-1,1]$, thus.

$$-M \leq |y(t)| \leq M$$

Thus, this system is **stable**

C) $y(t) = t \cdot x(t)$

The bounded input: $|x(t)| \leq M$,

The assumed bounded output:

$$|y(t)| = t \cdot |x(t)|$$

For large t, $|y(t)| \to \infty$

so, the system is **unstable**

Q18: Classify these systems in:
(linearity, time invariance, causality, memory, stability)

A) $y(t) = 3x(t) + 4$:

Non-linear, time-invariant, causal, memoryless, stable.

B) $y(t) = x^2(t) - 2$:

Non-linear, time-invariant, causal, memoryless, stable.

C) $y(t) = t \cdot x(t+1)$:

Linear, time-variant, anti-causal, memory, unstable.

D) $y(t) = x(t-2) + x(t+4)$:

Linear, time-invariant, anti-causal, memory, stable.

E) $y(t) = \int_{-\infty}^{t} x(\tau)\, d\tau$

Linear, time-invariant, causal, memory, unstable.

F) $y(t) = e^{-t}u(t) \cdot x(t)$

Linear, time-variant, causal, memoryless, stable.

2.2 LTI-Systems Convolution Integral

2.2.1 Introduction:

- Every physical system is broadly characterized by its ability to accept an input such as Voltage, Current, pressure, force, displacement...etc. and to produce an output in response to this input.

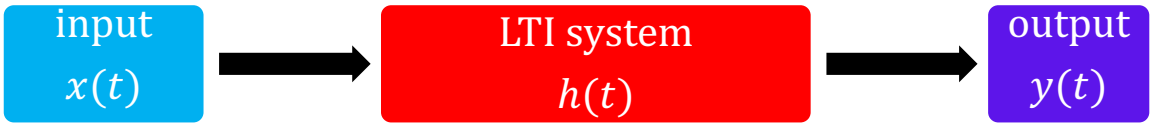
- **For Example:**

1. Electrical Filter whose input signal corrupted by noise and interference and whose output signal is the desired signal.
2. A Robot is a system whose input is an electric control signal and whose output is a motion or action.
3. Radar receiver is an electronic system whose input is the reflection of an electromagnetic signal from the target and whose output is a Video signal displayed on the radar screen.

- Linear Time- Invariant systems (LTI), play a fundamental role in signal and system analysis because of the many physical phenomena that can be modeled by Linear time-Invariant systems and because a mathematical analysis of the behavior of such systems can be carried out in a straightforward manner.

2.2.2 Convolution Integral:

- If $x(t)$ an input to LTI system $h(t)$, the output signal $y(t)$ can be evaluated by the convolution process.



- The operation of the convolution is represented symbolically as:

$$y(t) = x(t)^{*}h(t)$$

- The convolution integral is the dot product of the two functions $h(\tau), x(t-\tau)$:

$$\int_{-\infty}^{\infty} h(\tau) \cdot x(t-\tau)\, d\tau$$

- Where $h(t)$ assumed known, is the response of the system to a unit impulse input $\delta(t)$.
- The term $x(t-\tau)$ is the reflected shifted version of $x(\tau)$
- The integration with respect to τ is performed to result $y(t)$.
- So, t is treated as a constant in the integration, and it is representing the shifted value of the $x(\tau)$.

2.2.3 Convolution Graphically:

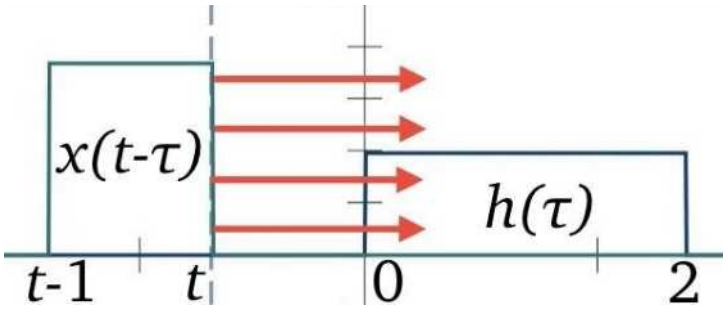
To perform the convolution integral graphically:

1. Change t to τ, and draw $x(\tau)$ and $h(\tau)$.
2. Reflect $x(\tau)$ to produce $x(-\tau)$.
3. Shift $x(-\tau)$ to the left by t to produce $x(t-\tau)$.
4. Integrate $h(\tau)x(t-\tau)$as a function of τ.

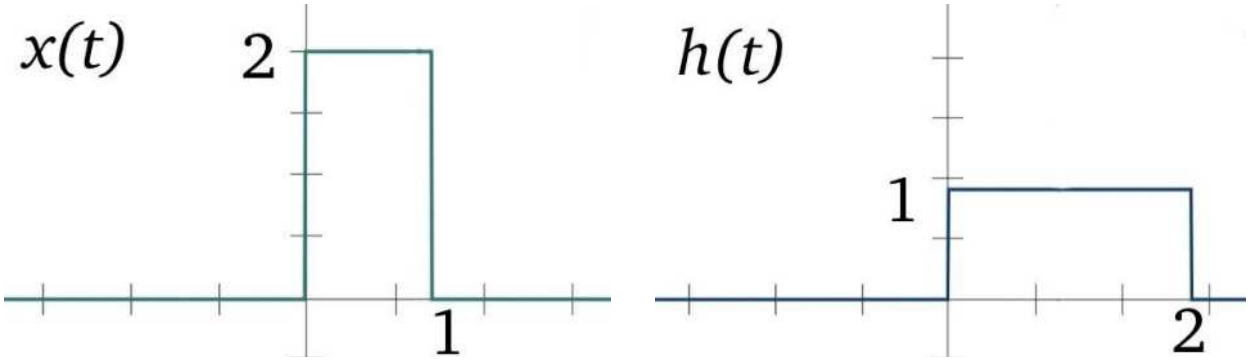
Notes:

∴ After integration with respect to τ is performed, result will be $y(t)$.

∴ The overall result represents the overlapping area of $h(t)$ and $x(t)$ when moving the reflected version of $x(t)$ from $-\infty \rightarrow \infty$.

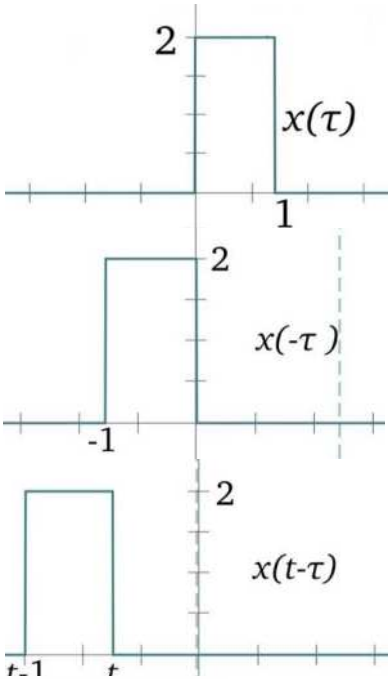


Q19: From the input $x(t)$ and the impulse response $h(t)$ as shown below, find the output $y(t)$.



Converting from function of to function of τ:

$$x(t) \rightarrow x(\tau)$$



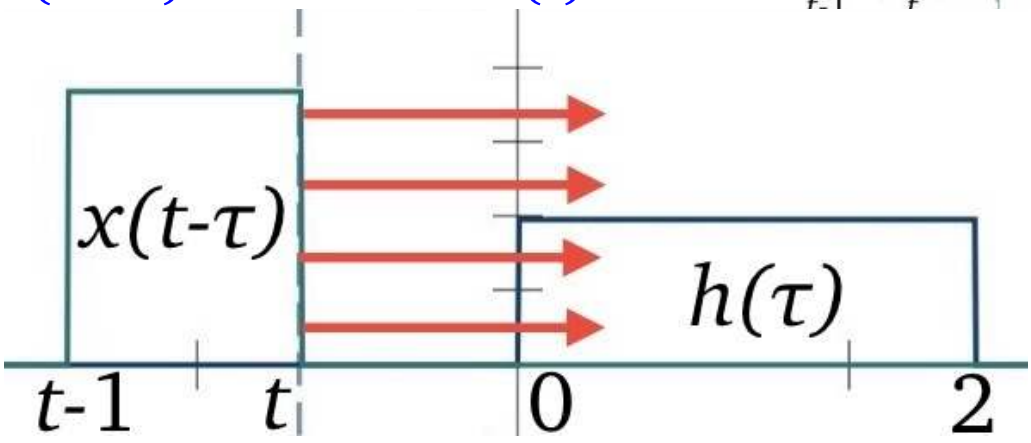
Reflecting about the $y-axis$:

$$x(\tau) \rightarrow x(-\tau)$$

Shifting by a magntiude of t:

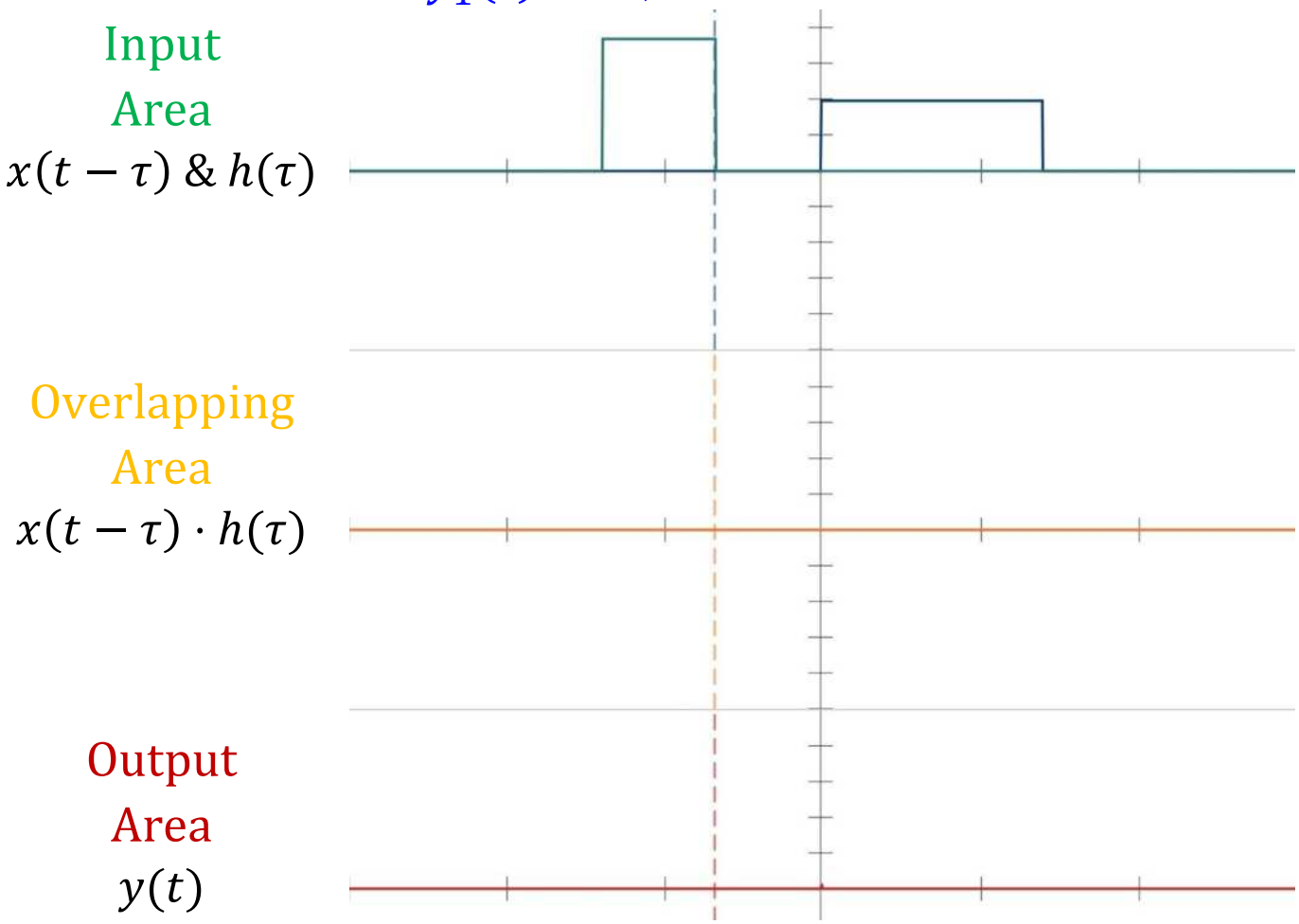
$$x(-\tau) \rightarrow x(t-\tau)$$

Making $x(t-\tau)$ tend toward $h(t)$:



When $t < 0, x(t-\tau)$ don't touch $h(t)$, there will be no overlapping area; so:

$$y_1(t) = 0, \qquad t < 0$$

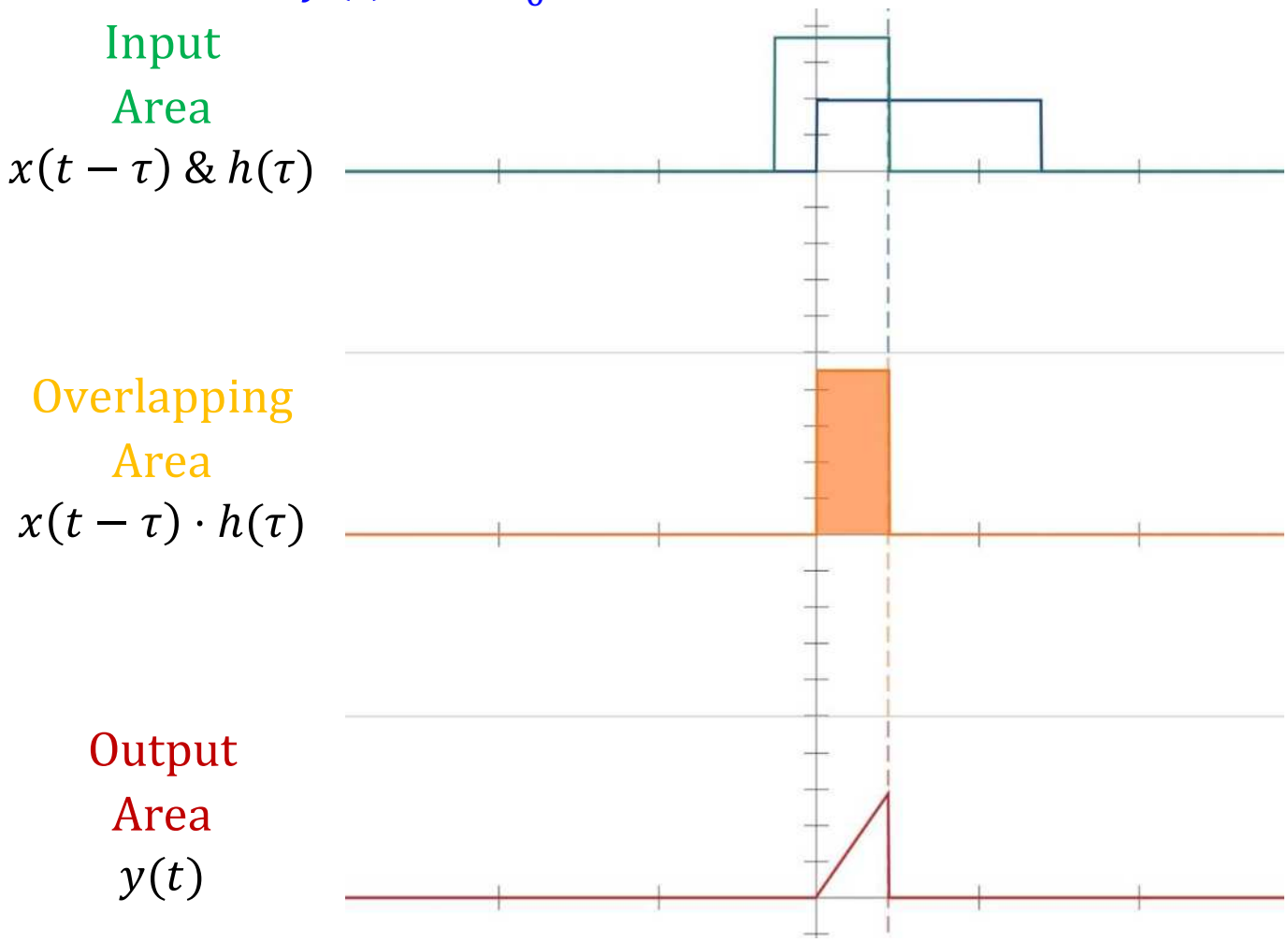


Overlapping $x(t-\tau)$ and $h(t)$ on the interval $(0,t) \rightarrow$ Integrating $x(t-\tau)\cdot h(t)$ from 0 to t.

This is called **Initial Overlap:**

$$y_2(t) = \int_0^t x(t-\tau).h(t)d\tau = \int_0^t (2)(1)d\tau$$

$$y(t) = 2\tau|_0^t = 2t\,, \qquad 0 \le t < 1$$



Overlapping $x(t-\tau)$ and $h(t)$ on the interval $(t-1, t)$ because we know that the difference between x_i, x_o of $x(t)$ is one unit → Integrating $x(t-\tau)\cdot h(t)$ from $t-1$ to t.

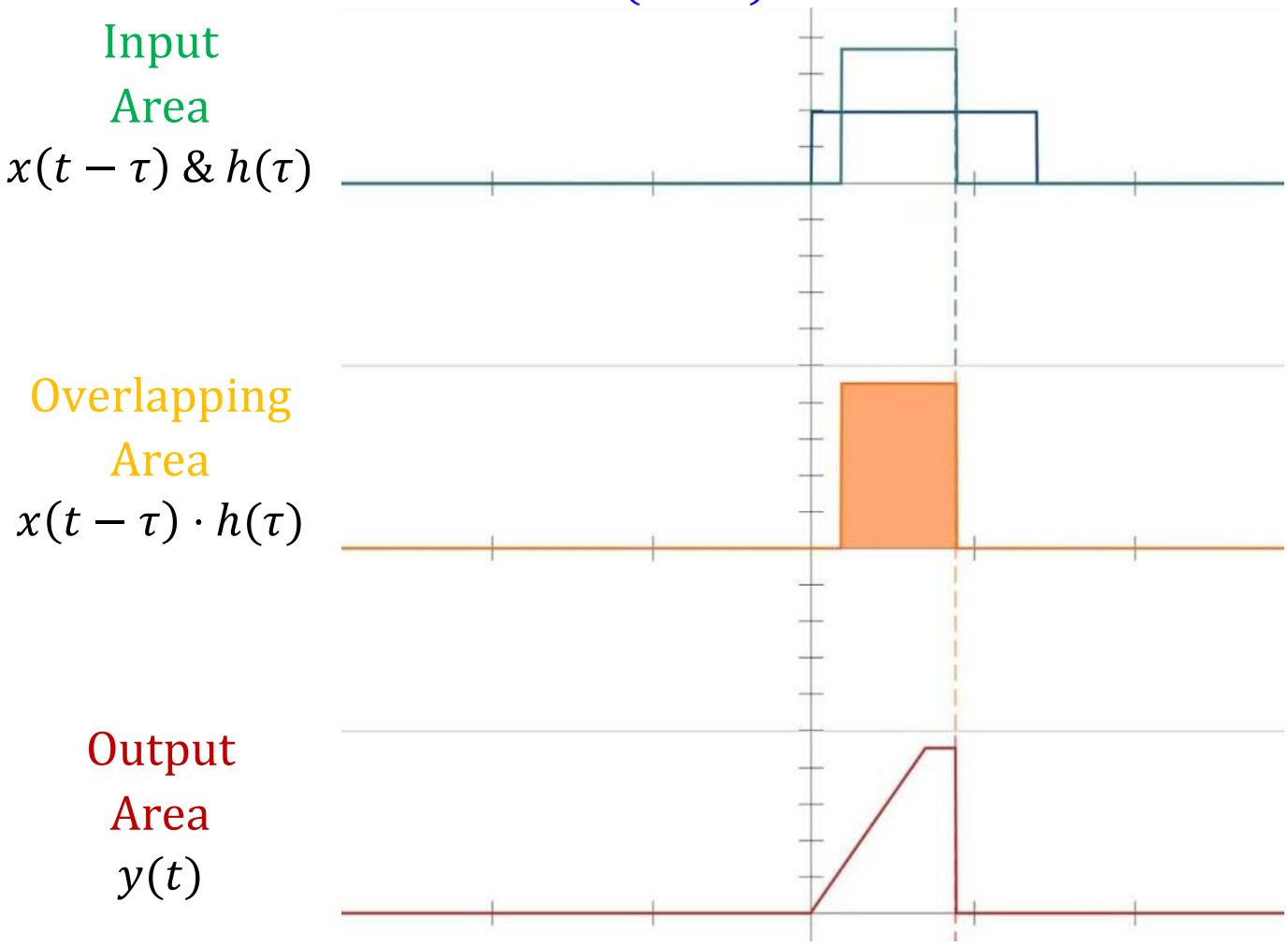
This is called **Completed Overlap:**

$$y_3(t) = \int_{t-1}^{t} h(\tau)\cdot x(t-\tau)\,d\tau = \int_{t-1}^{t} (1)(2)\,d\tau = 2\tau\Big|_{t-1}^{t}$$

$$= 2\big(t-(t-1)\big) = 2, \qquad 1 \le t < 2$$

$$\uparrow$$

$$(t-1) > 0 \cup t < 2$$



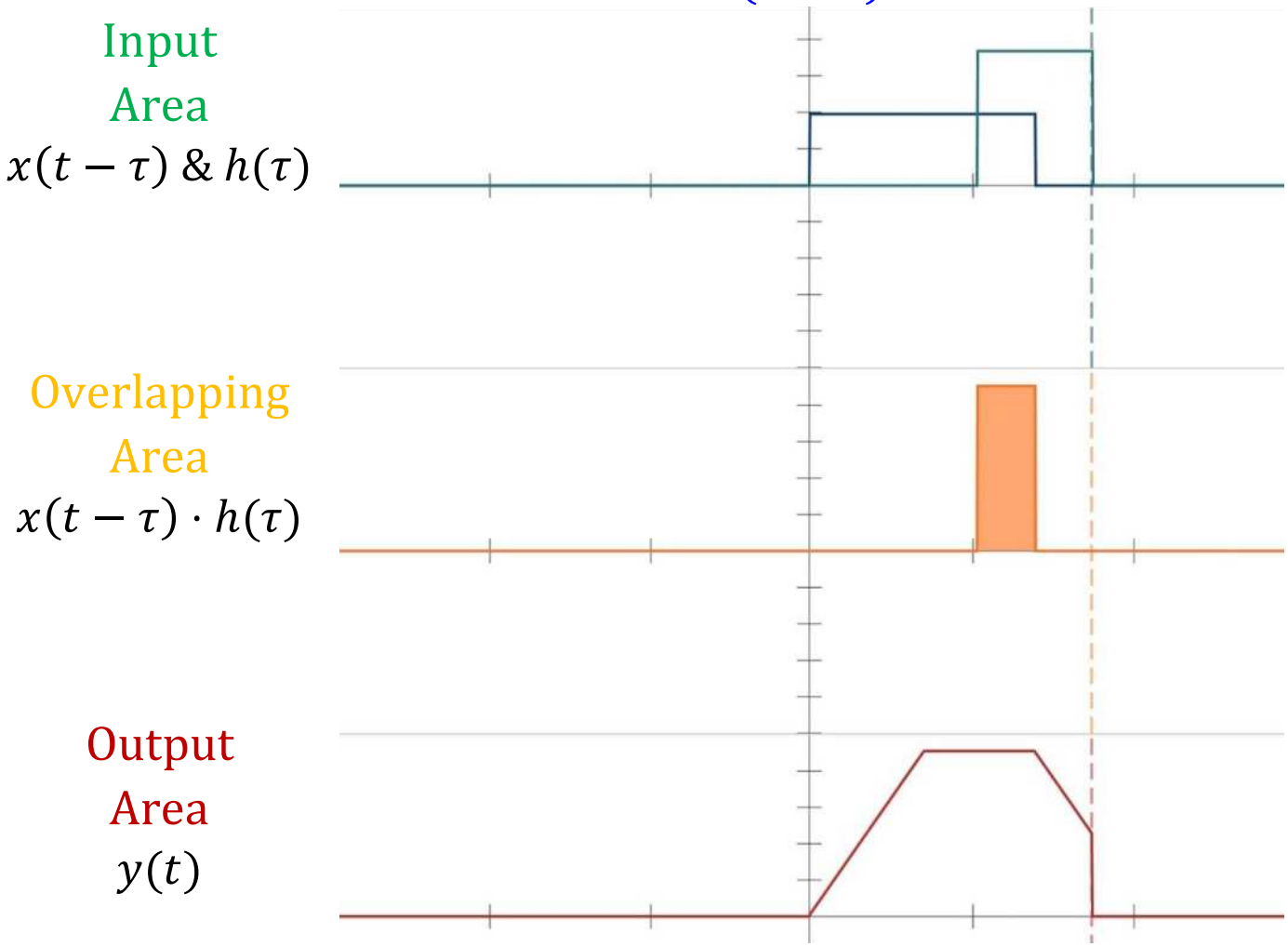
Overlapping $x(t-\tau)$ and $h(t)$ on the interval
$(t-1, 2) \rightarrow$ Integrating $x(t-\tau) \cdot h(t)$ from $t-1$ to 2.
This is called **Final Overlap:**

$$y_4(t) = \int_{t-1}^{2} h(\tau) \cdot x(t-\tau)\, d\tau = \int_{t-1}^{2} (1)(2)\, d\tau = 2\tau\Big|_{t-1}^{2}$$

$$= 2\big(2-(t-1)\big) = 6-2t, \qquad 2 \le t \le 3$$

$$\uparrow$$

$$(t-1) < 2 \cup t > 2$$



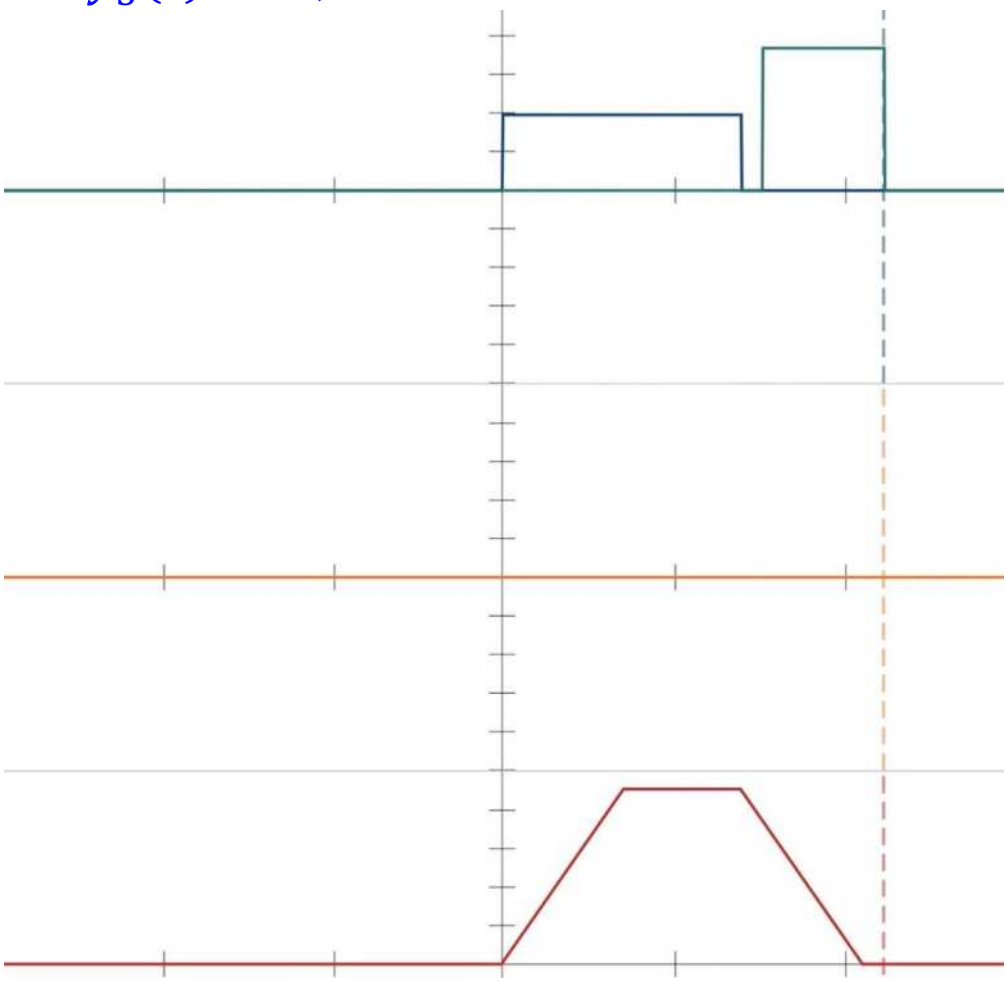
When $t < 3, x(t-\tau)$ don't touch $h(t)$, there will be no overlaping area; so:

$$y_5(t) = 0, \qquad t > 3$$

Input Area
$x(t-\tau)$ & $h(\tau)$

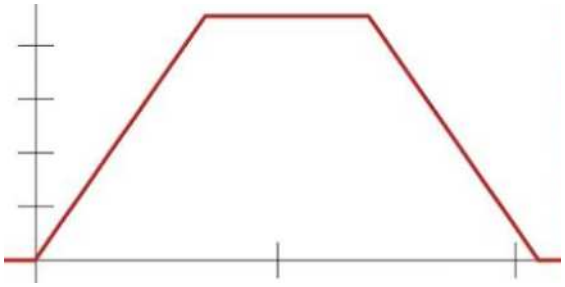
Overlapping Area
$x(t-\tau) \cdot h(\tau)$

Output Area
$y(t)$



Collect $y_1(t), y_2(t), y_3(t), y_4(t)$ and $y_5(t)$ in one piecewise function, then we find:

$$y(t) = \begin{cases} 0, & t < 0 \\ 2t, & 0 \le t < 1 \\ 2, & 1 \le t < 2 \\ 6 - 2t, & 2 \le t \le 3 \\ 0, & 3 > t \end{cases}$$



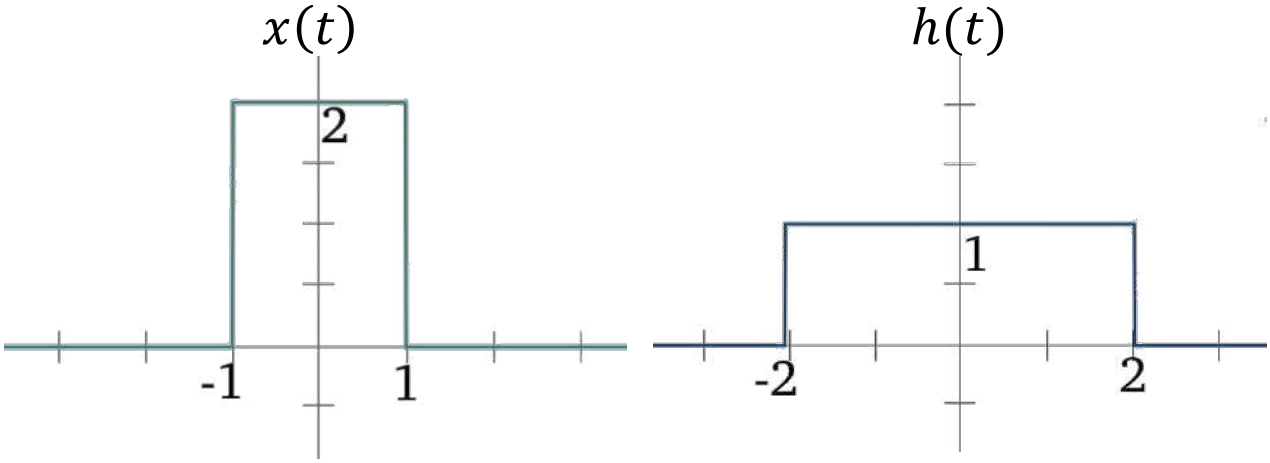
Visit this website and make convolutions by yourself!

↓↓↓↓↓↓↓↓↓↓↓↓↓↓

See the simulation for the last example on this video!

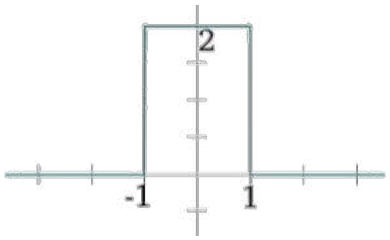
↓↓↓↓↓↓↓↓↓↓↓↓↓↓

Q2o: From the input $x(t)$ and the impulse response $h(t)$ as shown below, find the output $y(t)$.



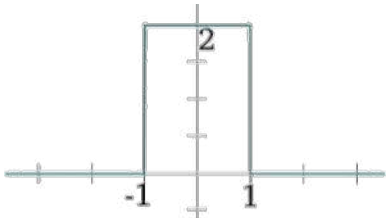
Converting from function of to function of τ:

$$x(t) \rightarrow x(\tau)$$



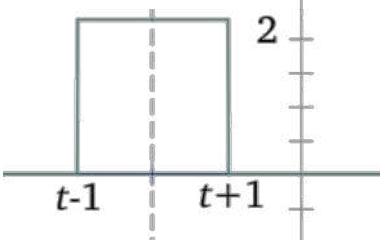
Reflecting about the $y-axis$:

$$x(\tau) \rightarrow x(-\tau)$$

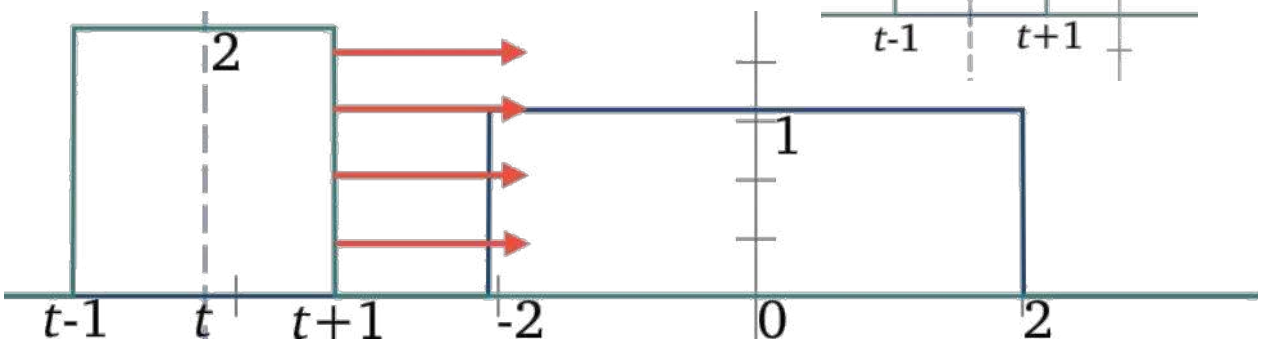


Shifting by a magntiude of t:

$$x(-\tau) \rightarrow x(t-\tau)$$



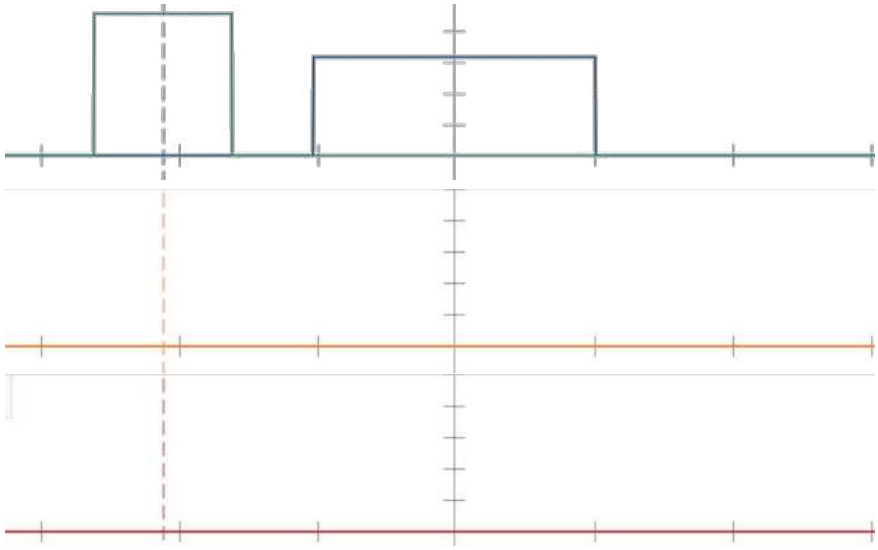
Making $x(t-\tau)$ tend toward $h(t)$:



When $t+1<-2$, no overlap area; so:

$$y_1(t) = 0, \qquad t+1<-2 \rightarrow t<-3$$

Input
Area
$x(t-\tau)\ \&\ h(\tau)$
Overlapping
Area
$x(t-\tau)\cdot h(\tau)$
Output
Area
$y(t)$

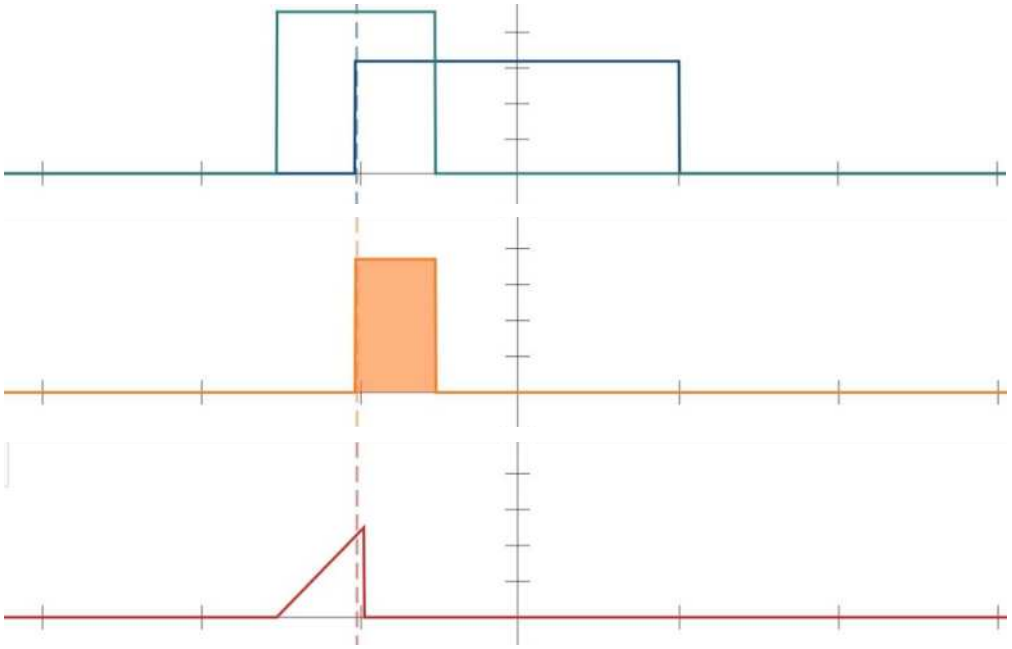


Initial Overlap on $(-2, t+1)$:

$$y_2(t) = \int_{-2}^{t+1} h(\tau)\cdot x(t-\tau)\,d\tau = \int_{-2}^{t+1} (1)(2)\,d\tau = 2\tau\Big|_{-2}^{t+1}$$

$$= 2t+6, \qquad -2<t+1<2 \rightarrow -3<t<1$$

Input
Area
$x(t-\tau)\ \&\ h(\tau)$
Overlap
Area
$x(t-\tau)\cdot h(\tau)$
Output
Area
$y(t)$



Completed Overlap on $(t-1, t+1)$:

$$y_3(t) = \int_{t-1}^{t+1} x(t-\tau).h(t)d\tau = \int_{t-1}^{t+1} (2)(1)d\tau = 2\tau|_{t-1}^{t+1}$$

$$= 4, \quad -2 < t+1 < 2 \rightarrow -3 < t < 1$$

Input

Area

$x(t-\tau)$ & $h(\tau)$

Overlap

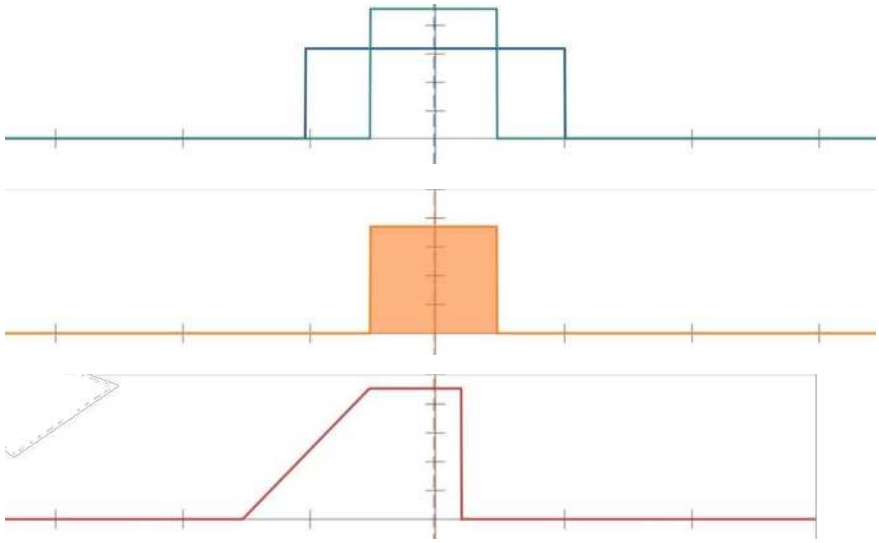
Area

$x(t-\tau) \cdot h(\tau)$

Output

Area

$y(t)$



Final Overlap on $(t-1,2)$:

$$y_4(t) = \int_{t-1}^{2} h(\tau) \cdot x(t-\tau)\, d\tau = \int_{t-1}^{2} (1)(2)\, d\tau = 2\tau|_{t-1}^{2}$$

$$= 6 - 2t, \quad -2 < t-1 < 2 \rightarrow -3 < t < 1$$

Input

Area

$x(t-\tau)$ & $h(\tau)$

Overlapping

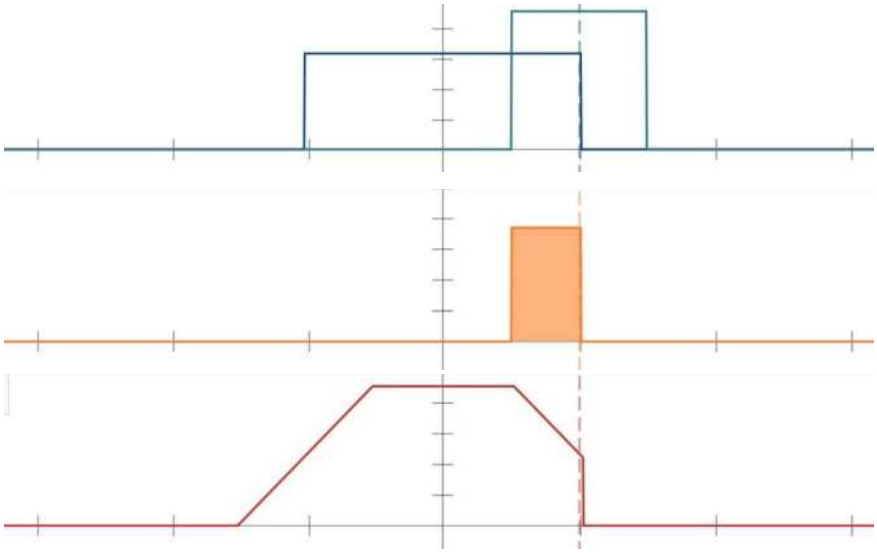
Area

$x(t-\tau) \cdot h(\tau)$

Output

Area

$y(t)$



When $t-1>2$, no overlaping area; so:

$$y_5(t)=0, \qquad t-1>2 \rightarrow t>3$$

Input

Area

$x(t-\tau)\ \&\ h(\tau)$

Overlapping

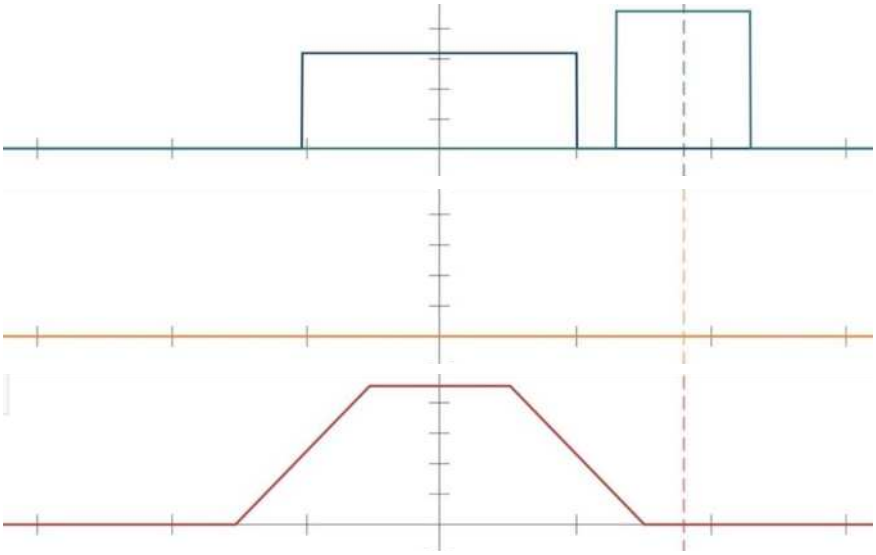
Area

$x(t-\tau)\cdot h(\tau)$

Output

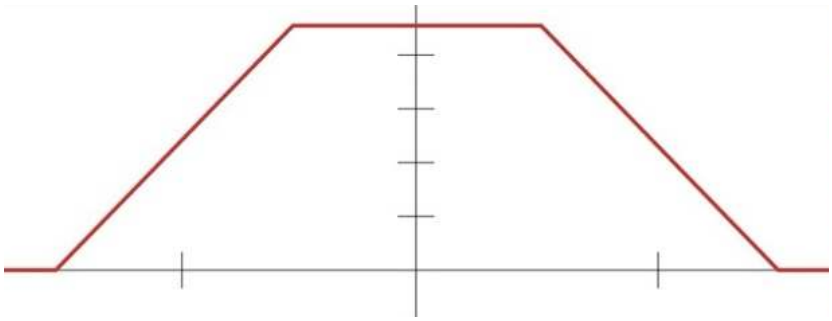
Area

$y(t)$



Collect $y_1(t), y_2(t), y_3(t), y_4(t)$ and $y_5(t)$:

$$y(t)=\begin{cases}0, & t<-3\\ 2t, & -3\le t<-1\\ 4, & -1\le t<1\\ 6-2t, & 1\le t\le 3\\ 0, & 3>t\end{cases}$$

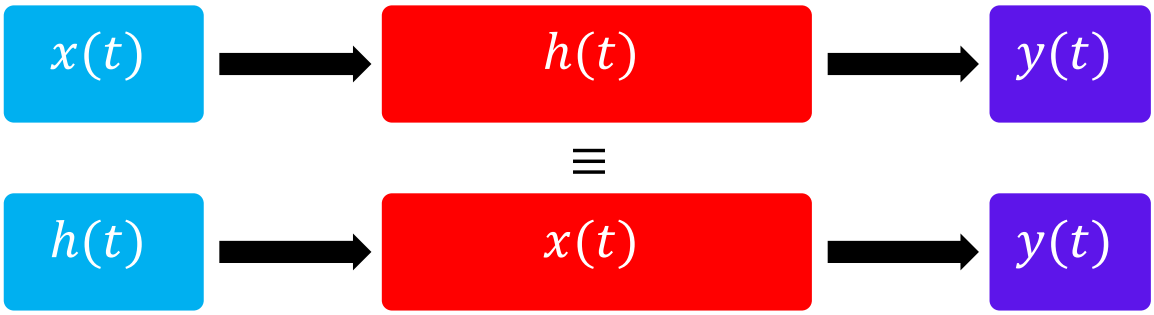


2.2.4 Properties of Convolution Integral:

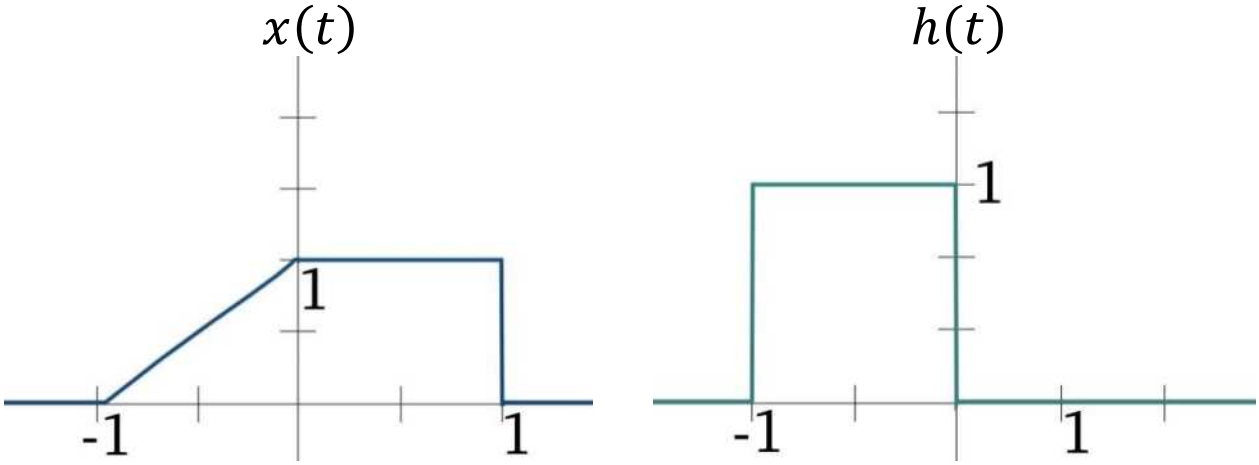
1. Commutative:

$$x(t)^{*}h(t) = h(t)^{*}x(t)$$

- This property is proved by substitution of variables.
- The property implies that the roles of the input signal and the impulse response are interchangeable.



Q21: From the input $x(t)$ and the impulse response $h(t)$ as shown below, find the output $y(t)$.



Based on Commutative property we can solve this example by making $h(t)$ the input signal instead of $x(t)$ which will result in a simpler way of solution.

When $t + 1 < -1$, no overlap; so:

$$y_1(t) = 0, \qquad t + 1 < -1 \rightarrow t < -2$$

Input

Area

$h(t-\tau)$ & $x(\tau)$

Overlapping

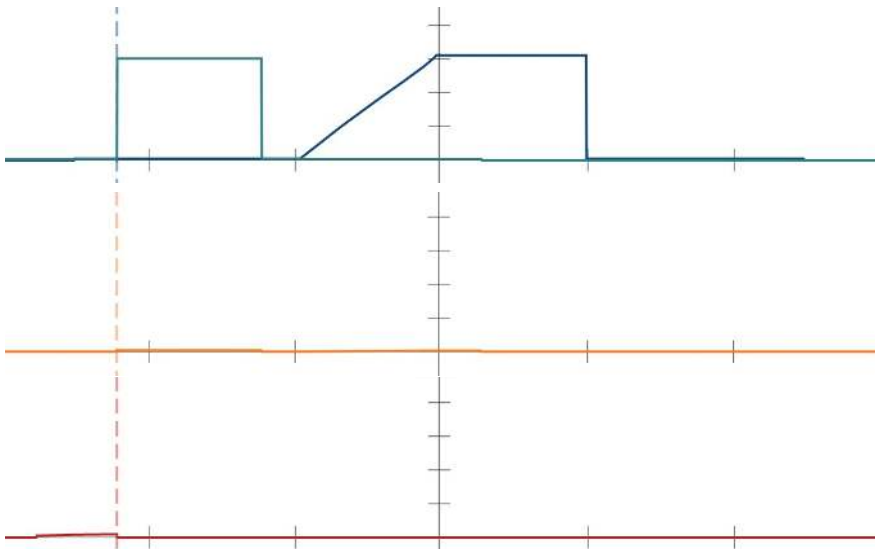
Area

$h(t-\tau) \cdot x(\tau)$

Output

Area

$y(t)$



Initial overlap on $(-1, t+1)$:

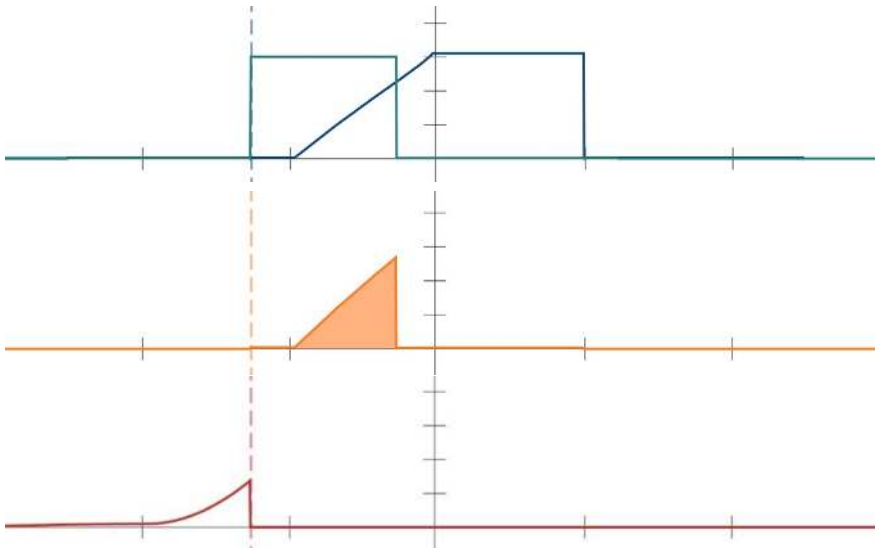
$$y_2(t) = \int_{-1}^{t+1} x(\tau) \cdot h(t-\tau)\, d\tau = \int_{-1}^{t+1} (\tau+1)(1)\, d\tau$$

Also Note here that we substitute $x(\tau)$ as $\tau + 1$ since it represents a ramp function on $(-1, t+1)$ shifted on the left by one unit.

$$y_2(t) = \left(\frac{\tau^2}{2} + \tau\right)\Big|_{-1}^{t+1} = \frac{t^2}{2} + 2t + 2,$$

$$t + 1 \geq -1 \;\&\; t < -1 \rightarrow -2 \leq t < -1$$

Input
Area
$h(t-\tau)$ & $x(\tau)$
Overlapping
Area
$h(t-\tau) \cdot x(\tau)$
Output
Area
$y(t)$



Complete overlap 1 on $(t, t+1)$ in this case we must split the integration to 2 different integrations because we have 2 different areas of overlap on $(t, t+1)$:

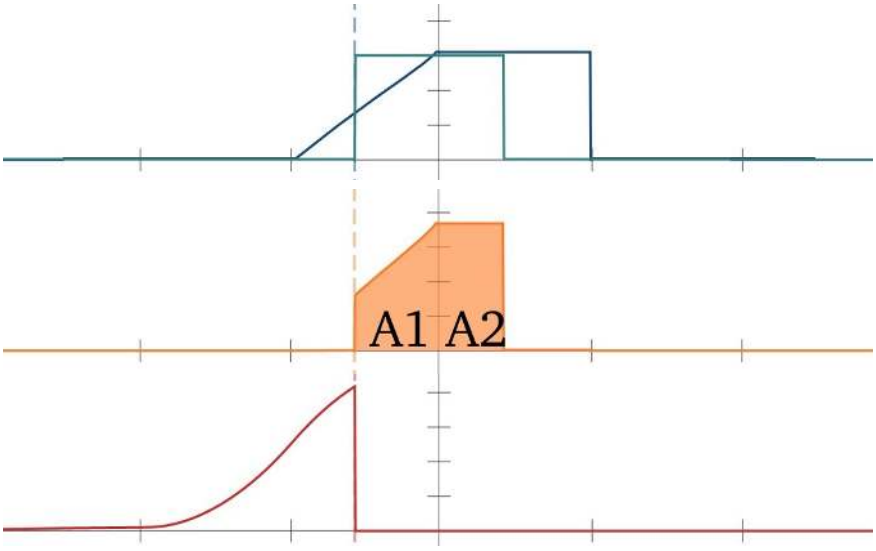
$$y_3(t) = \int_t^{t+1} x(\tau) \cdot h(t-\tau)\, d\tau$$

$$= \int_t^0 (\tau+1)(1) d\tau + \int_0^{t+1} (1)(1)\, d\tau$$

$$= \left(\frac{\tau^2}{2} + \tau\right)\Big|_t^0 + \tau|_0^{t+1} = -\frac{t^2}{2} + 1,$$

$$t+1 \geq 0 \;\&\; t < 0 \rightarrow -1 \leq t < 0$$

Input
Area
$h(t-\tau)$ & $x(\tau)$
Overlapping
Area
$h(t-\tau) \cdot x(\tau)$
Output
Area
$y(t)$

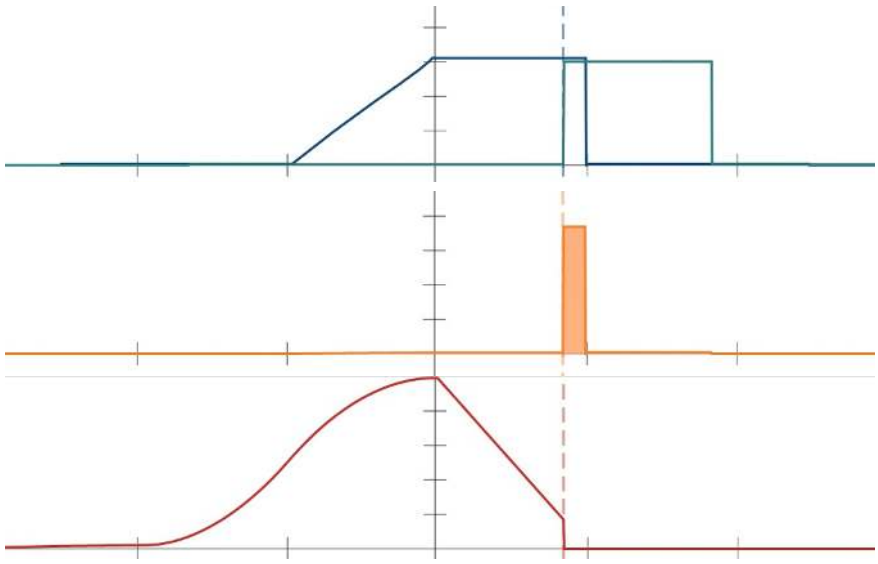


Completed overlap 2 on $(t, 1)$ Beware that this overlap is different from the previous overlap since it only has only 1 Area of overlap on $(t, 1)$

$$y_4(t) = \int_t^1 x(\tau) \cdot h(t-\tau)\, d\tau = \int_t^1 (1)(1)\, d\tau = (\tau)|_t^1 = 1 - t,$$

$$t + 1 \geq 1 \;\&\; t < 1 \rightarrow 0 \leq t \leq 1$$

Input
Area
$h(t-\tau)$ & $x(\tau)$
Overlapping
Area
$h(t-\tau) \cdot x(\tau)$
Output
Area
$y(t)$



When $t > 1$, no overlap; so:

$$y_5 = 0, \qquad t > 1$$

Input

Area

$h(t-\tau)\ \&\ x(\tau)$

Overlapping

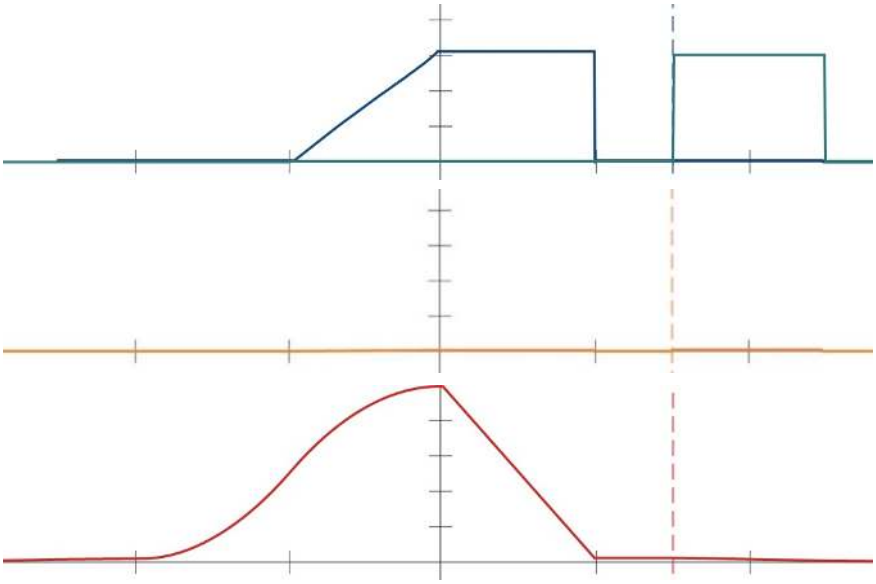
Area

$h(t-\tau) \cdot x(\tau)$

Output

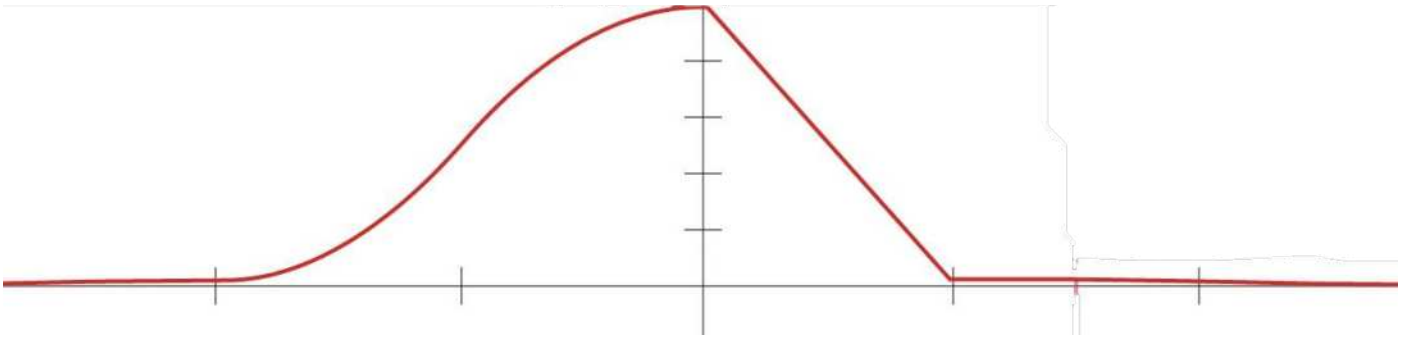
Area

$y(t)$



Combining the solutions:

$$y(t) = \begin{cases} 0, & t < -2 \\ \frac{t^2}{2} + 2t + 2, & -2 \le t < -1 \\ -\frac{t^2}{2} + 1, & -1 \le t < 0 \\ 1 - t, & 0 \le t \le 1 \\ 0, & 1 < t \end{cases}$$

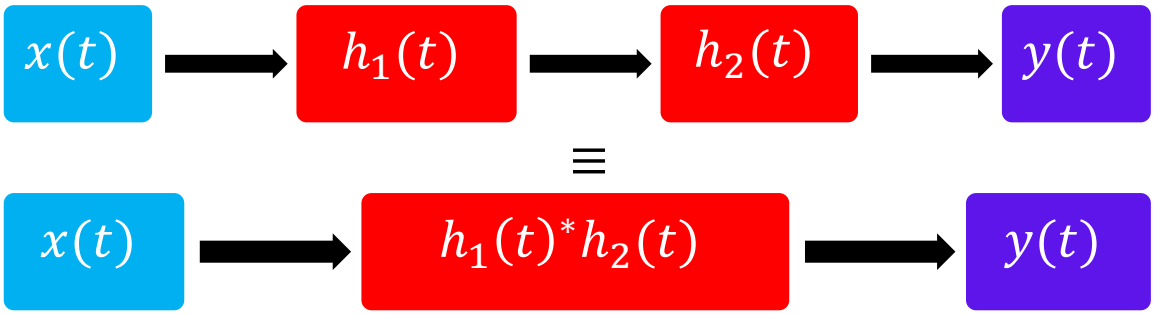


2. Associative:

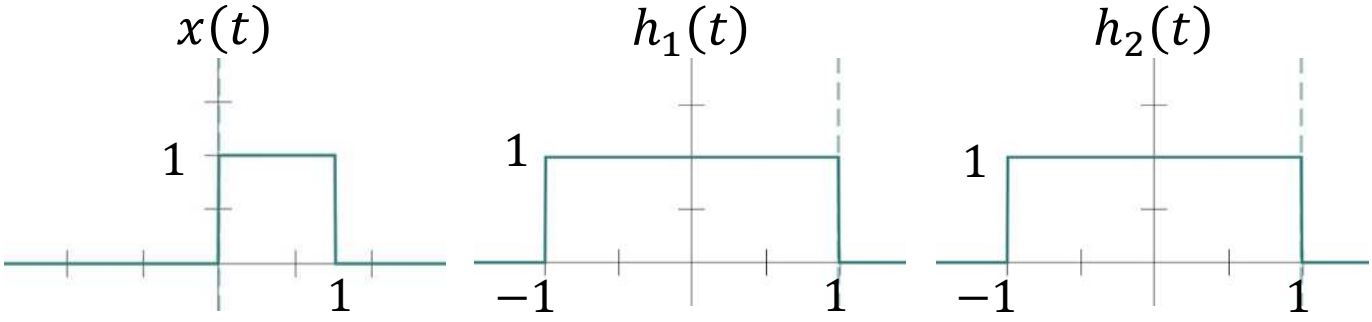
$$x(t)^{*}h_1(t)^{*}h_2(t) =$$

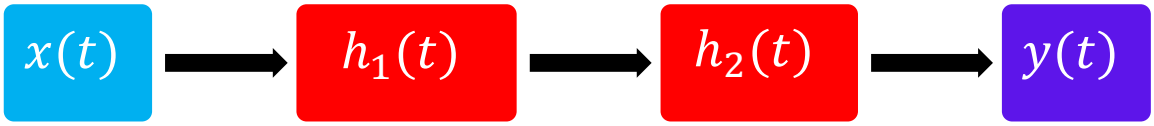
$$[x(t)^{*}h_1(t)]^{*}h_2(t) = x(t)^{*}[h_1(t)^{*}h_2(t)]$$

- This property is proved by changing the orders of integration.
- Associatively, implies that a cascade combination of LTI system can be replaced by a single system whose impulse response is the convolution of the individual impulse response.
- This property is illustrated in the following block diagram:



Q22: From the input $x(t)$ and the impulse responses $h_1(t)$ and $h_2(t)$ as shown below, find the output $y(t)$.





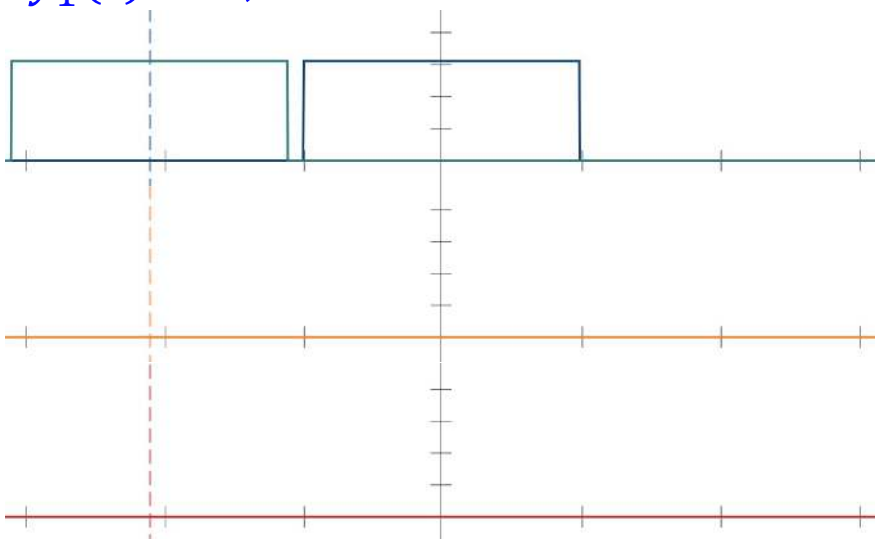
Note here that the input $x(t)$ Enters two separate functions (h_1, h_2), Based on Associative property we can convolve h_1, h_2 then convolve the result with $x(t)$.

Starting with $\hat{y}(t) = h_1(t)^* h_2(t)$:

When $t + 1 < -1$, no overlap; so:

$\hat{y}_1(t) = 0, \quad t < -2$

Input
Area
$h_1(t-\tau)$ & $h_2(\tau)$
Overlapping
Area
$h_1(t-\tau) \cdot h_2(\tau)$
Output
Area
$\hat{y}(t)$



Initial overlap on $(-1, t+1)$:

$$\hat{y}_2(t) = \int_{-1}^{t+1} h_1(t-\tau) \cdot h_2(\tau)\, d\tau = \int_{-1}^{t+1} (1)(1)\, d\tau = t\Big|_{-1}^{t+1}$$

$$= t + 2, \qquad t - 1 < -1 \;\&\; t + 1 \geq -1 \rightarrow 2 \leq t < 0$$

Input

Area

$h_1(t-\tau)$ & $h_2(\tau)$

Overlapping

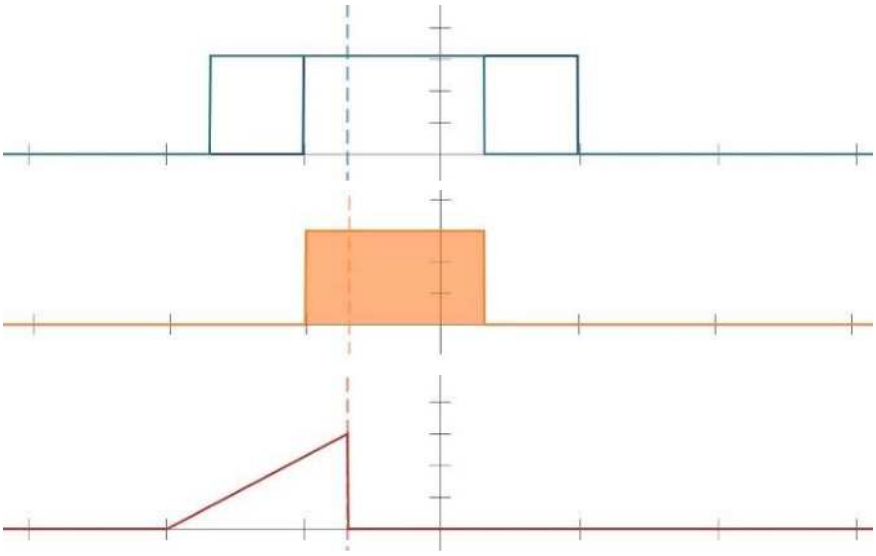
Area

$h_1(t-\tau) \cdot h_2(\tau)$

Output

Area

$\hat{y}(t)$



Final overlap on $(t-1, 1)$:

$$\hat{y}_4(t) = \int_{1}^{t+1} h_1(t-\tau).h_2(t).d\tau = \int_{1}^{t+1} (1)(1)\, t\Big|_{t-1}^{1} = 2 - t,$$

$$t - 1 \leq 1 \;\&\; t + 1 < 1 \rightarrow 0 < t \leq 2$$

Input

Area

$h_1(t-\tau)$ & $h_2(\tau)$

Overlapping

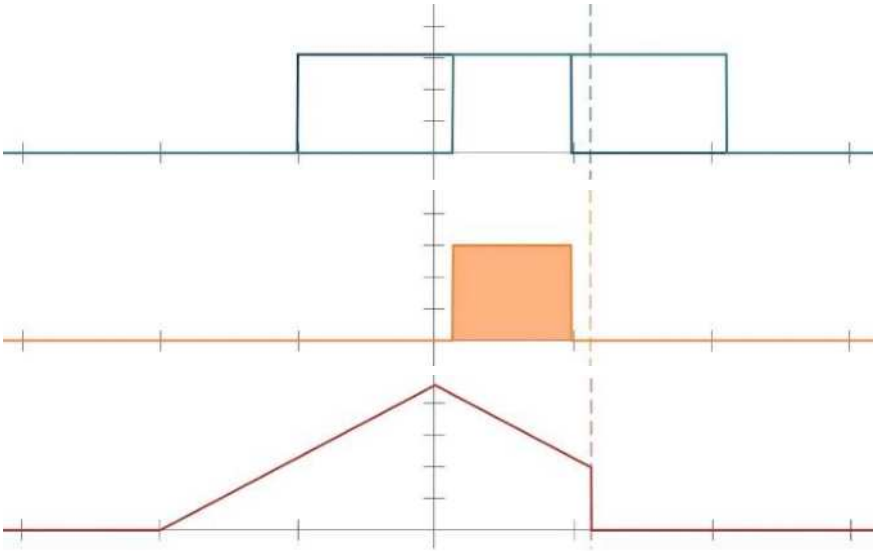
Area

$h_1(t-\tau) \cdot h_2(\tau)$

Output

Area

$\hat{y}(t)$



When $t > 2$, no overlap:

$$\hat{y}_5(t) = 0, \qquad t > 2$$

Input

Area

$h_1(t-\tau)$ & $h_2(\tau)$

Overlapping

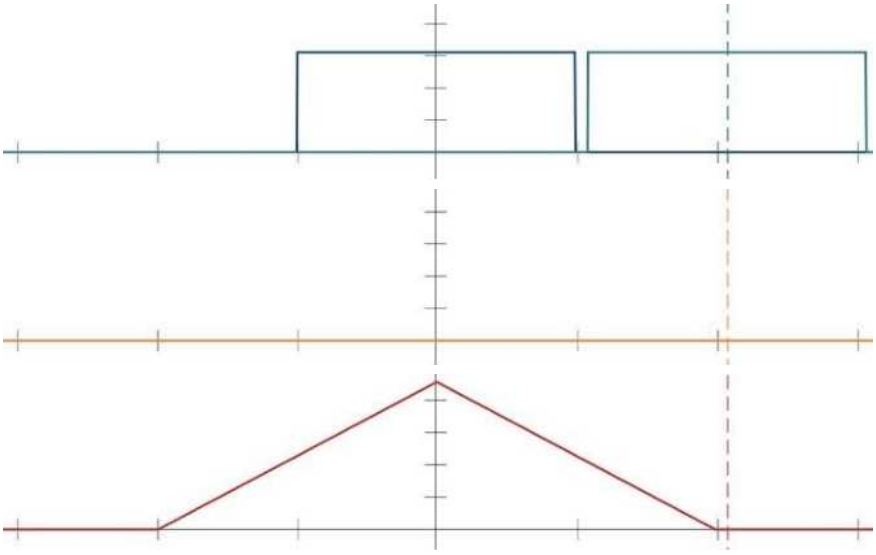
Area

$h_1(t-\tau) \cdot h_2(\tau)$

Output

Area

$\hat{y}(t)$

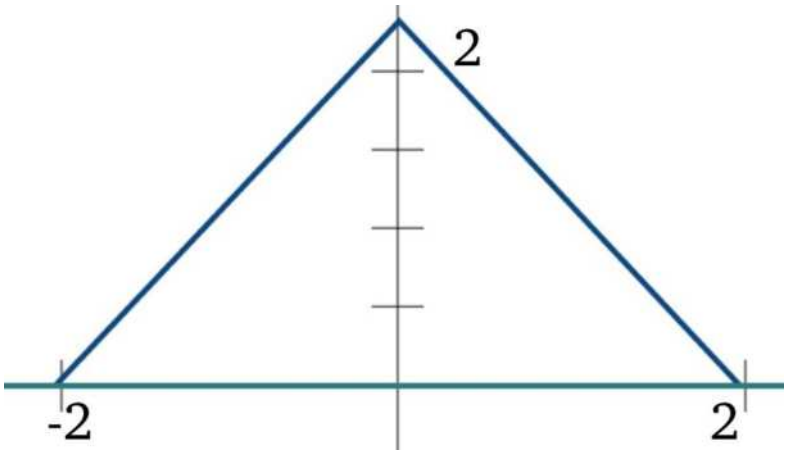


After combining:

$$\hat{y}(t) = \begin{cases} 0, & t \le -2 \\ t+2, & -2 < t < 0 \\ 2, & t = 0 \\ 2-t, & 0 \le t < 2 \\ 0, & 2 \le t \end{cases}$$

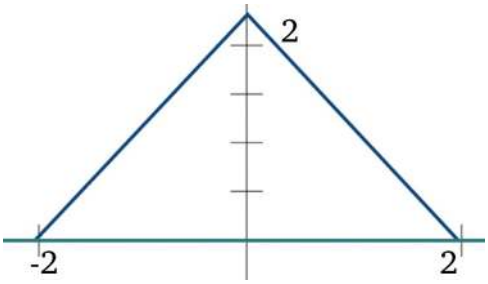
Since we must convolute once again, the graph of $h_1(t)^* h_2(t)$ is essential and we must find it to convolute it with $x(t)$.

interval	$y(t)$	point
$t < -2$	0	$(-2, 0)$
$-2 \le t < 0$	$t+2$	$(-1, 1)$
$t = 0$	2	$(0, 2)$
$0 < t \le 2$	$2-t$	$(1, 1)$
$t > 2$	0	$(2, 0)$

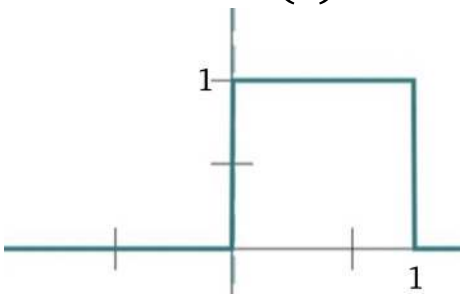


Now finding $y(t) = \hat{y}(t)^* x(t)$:

$\hat{y}(t)$



$x(t)$



When $t < -2$, no overlap:

$y_1(t) = 0, \quad t \leq -2$

Input

Area

$x(t-\tau)$ & $\hat{y}(\tau)$

Overlapping

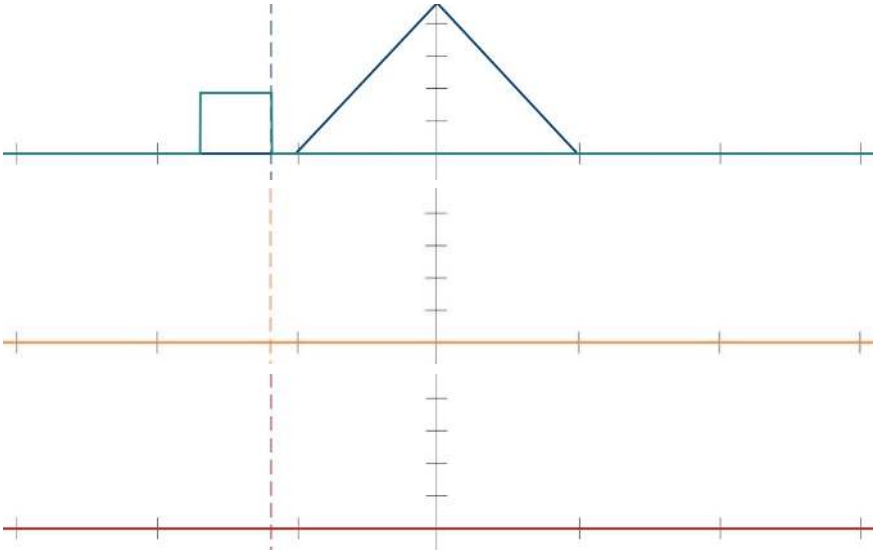
Area

$x(t-\tau) \cdot \hat{y}(\tau)$

Output

Area

$y(t)$

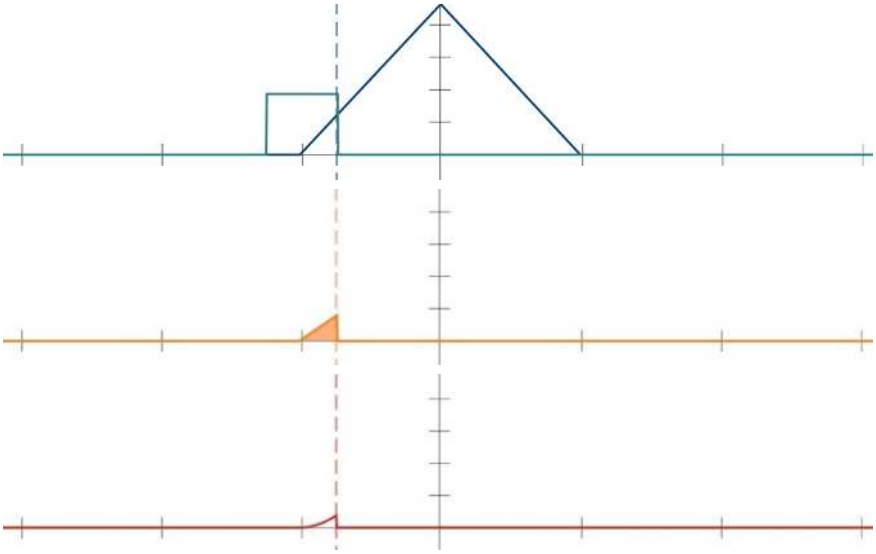


Initial overlap on $(-2, t)$:

$$y_2(t) = \int_{-2}^{t} x(t-\tau) \cdot \hat{y}(\tau)\, dt = \int_{-2}^{t} (1)(\tau+2)\, d\tau$$

$$= \frac{\tau^2}{2} + 2\tau\Big|_{-2}^{t} = \frac{t^2}{2} + 2t + 2, \qquad -2 < t < -1$$

Input
Area
$x(t-\tau)$ & $\hat{y}(\tau)$
Overlapping
Area
$x(t-\tau) \cdot \hat{y}(\tau)$
Output
Area
$y(t)$

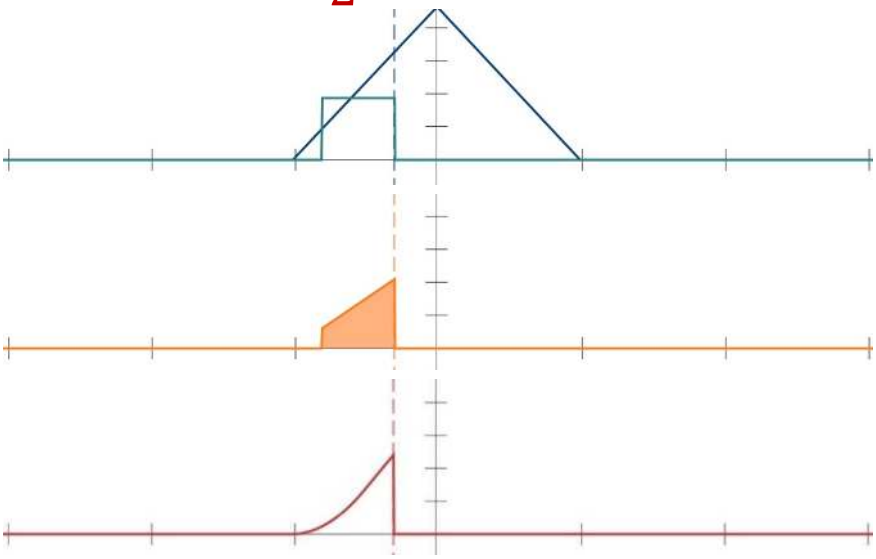


Complete overlap 1 on $(t-1, t)$:

$$y_3(t) = \int_{t-1}^{t} x(t-\tau) \cdot \hat{y}(\tau)\, dt = \int_{t-1}^{t} (1)(\tau+2)\, d\tau$$

$$= \frac{\tau^2}{2} + 2\tau\Big|_{t-1}^{t} = t + \frac{3}{2}, \qquad -1 \le t < 0$$

Input
Area
$x(t-\tau)$ & $\hat{y}(\tau)$
Overlapping
Area
$x(t-\tau) \cdot \hat{y}(\tau)$
Output
Area
$y(t)$



Complete overlap 2 on $(t-1, t)$:

$$y_4(t) = \int_{t-1}^{t} x(t-\tau) \cdot \hat{y}(\tau)\, dt$$

$$= \int_{t-1}^{0} (1)(\tau+2)\, d\tau + \int_{0}^{t} (1)(2-\tau)\, d\tau$$

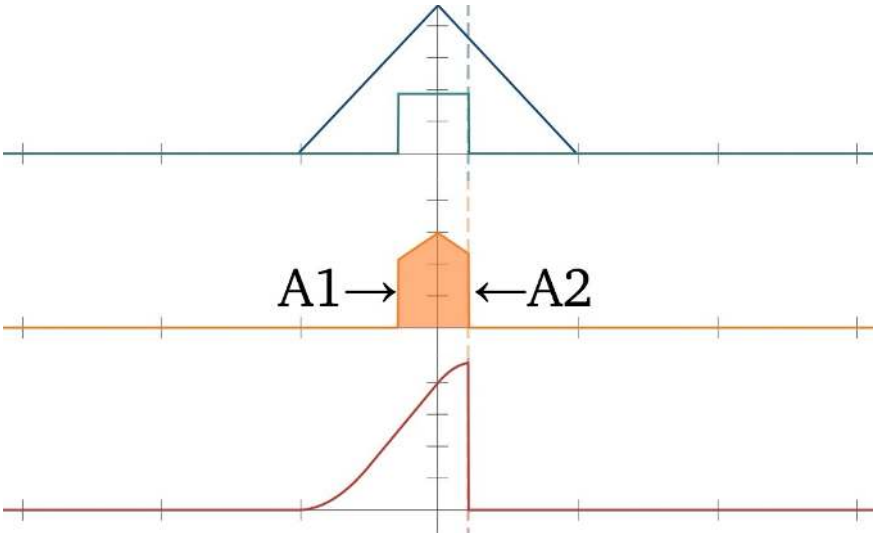
$$= \frac{\tau^2}{2} + 2\tau\Big|_{t-1}^{0} + 2\tau - \frac{\tau^2}{2}\Big|_{0}^{t} = -t^2 + t + \frac{3}{2},$$

$0 \le t < 1$

Input
Area
$x(t-\tau)$ & $\hat{y}(\tau)$

Overlapping
Area
$x(t-\tau) \cdot \hat{y}(\tau)$

Output
Area
$y(t)$

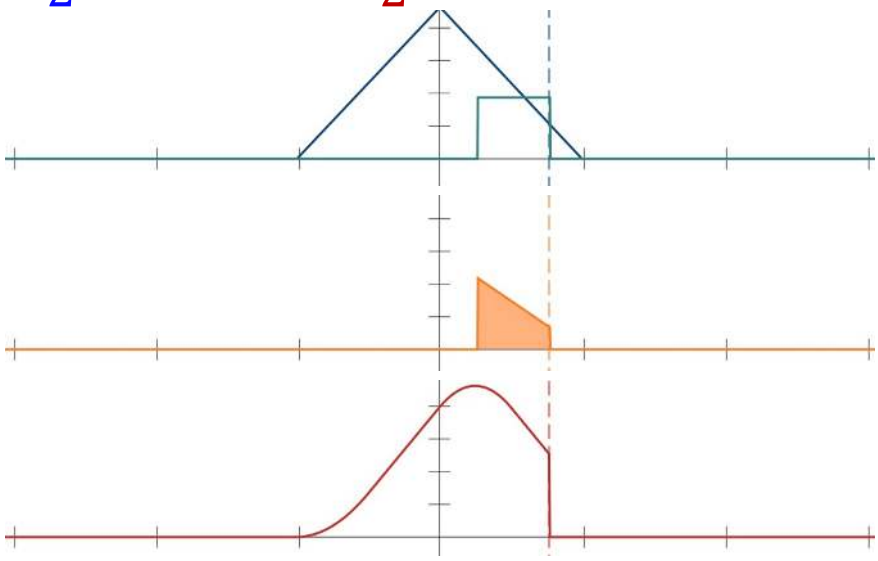


Complete overlap 3 on $(t-1, t)$:

$$y_5(t) = \int_{t-1}^{t} x(t-\tau) \cdot \hat{y}(\tau)\, dt = \int_{t-1}^{t} (1)(2-\tau)\, d\tau$$

$$= 2\tau - \frac{\tau^2}{2}\Big|_{t-1}^{t} = -t + \frac{5}{2}, \qquad 1 \le t < 2$$

Input
Area
$x(t-\tau)$ & $\hat{y}(\tau)$
Overlapping
Area
$x(t-\tau) \cdot \hat{y}(\tau)$
Output
Area
$y(t)$

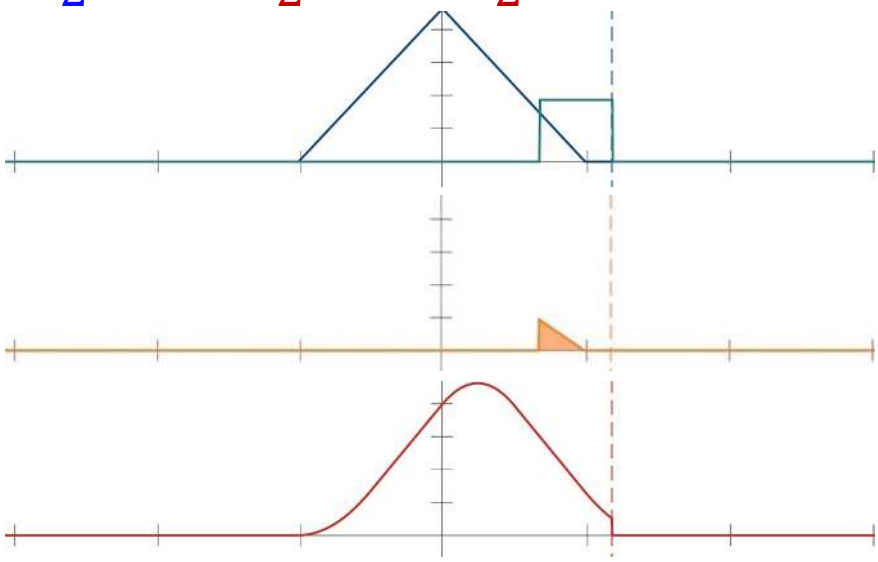


final overlap on $(-2, t)$:

$$y_6(t) = \int_{t-1}^{2} x(t-\tau) \cdot \hat{y}(\tau)\, dt = \int_{t-1}^{2} (1)(2-\tau)\, d\tau$$

$$= 2\tau - \frac{\tau^2}{2}\Big|_{t-1}^{2} = \frac{t^2}{2} - 3t + \frac{9}{2}, \qquad 2 \le t < 3$$

Input
Area
$x(t-\tau)$ & $\hat{y}(\tau)$
Overlapping
Area
$x(t-\tau) \cdot \hat{y}(\tau)$
Output
Area
$y(t)$



When $t < -2$, no overlap:

$$y_1(t) = 0, \qquad t \geq 3$$

Input

Area

$x(t-\tau)$ & $\hat{y}(\tau)$

Overlapping

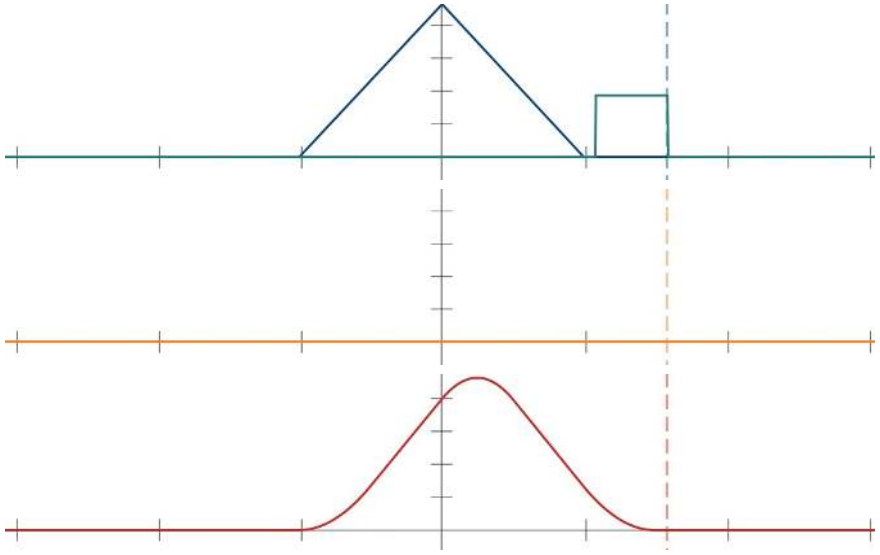
Area

$x(t-\tau) \cdot \hat{y}(\tau)$

Output

Area

$y(t)$



After combining:

$$y(t) = x(t)^* h_1(t)^* h_2(t)$$

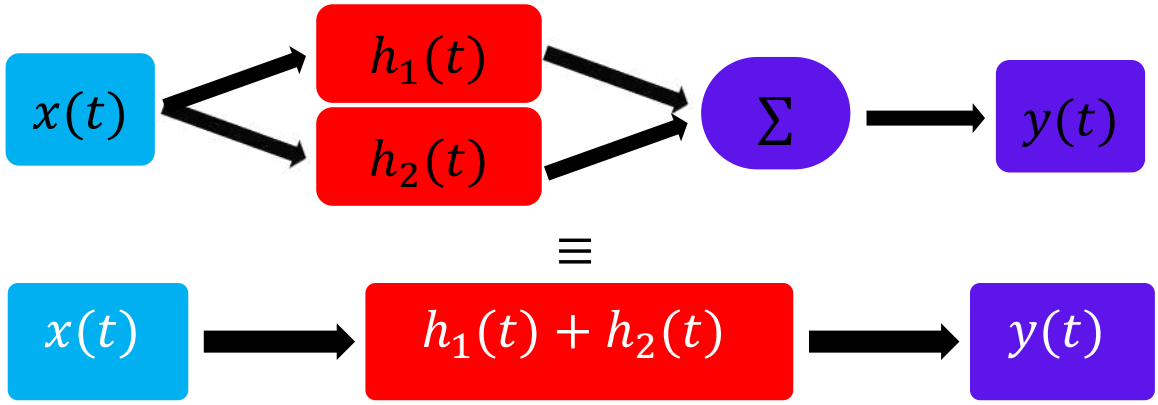
$$= \begin{cases} 0, & t \leq -2 \\ \frac{t^2}{2} + 2t + 2, & -2 < t < -1 \\ t + \frac{3}{2}, & -1 \leq t < 0 \\ -t^2 + t + \frac{3}{2}, & 0 \leq t < 1 \\ -t + \frac{5}{2}, & 1 \leq t < 2 \\ \frac{t^2}{2} - 3t + \frac{9}{2}, & 2 \leq t < 3 \\ 0, & 2 \leq t \end{cases}$$

3. Distributivity:

$$x(t)^*[h_1(t) + h_2(t)] =$$

$$[x(t)^*h_1(t)] + [x(t)^*h_2(t)]$$

- This property follows directly because of the linear property of integration.
- Distributivity states that a parallel combination of LTI systems is equivalent to a single system whose impulse response is the sum of the individual impulse responses in the parallel configuration.
- This property is illustrated in the following block diagram:



4. Convolution with $u(t)$:

- If the impulse response $h(t) = u(t)$ the convolution integral can be obtained directly using:

$$x(t)^* u(t) = \int_{-\infty}^{\infty} x(\tau)u(t-\tau)\,d\tau = \int_{-\infty}^{t} x(\tau)\,d\tau$$

- So, The LTI system with step response at $h(t) = u(t)$ is a normal integration, but it adjust the terminals of integration.

Q23: Convolve the unit step function with LTI system $h(t) = e^{-at}u(t)$.

$$h(t)^* u(t) = \int_{-\infty}^{\infty} h(\tau)\cdot u(t-\tau)\,d\tau = \int_{-\infty}^{t} e^{-a\tau}u(\tau)\,d\tau$$

$$= \int_{0}^{t} e^{-a\tau}\,d\tau = -\frac{1}{a}e^{-a\tau}\Big|_0^t = \frac{1}{a} - \frac{1}{a}e^{-at}$$

Q24: Find the output if the input to the system is:

$$h(t) = e^{-at}\cdot u(t), \qquad x(t) = e^{-bt}\cdot u(t), \qquad b \neq a$$

$$y(t) = \int_{-\infty}^{\infty} e^{-b\tau}\cdot u(\tau)\cdot e^{-a(t-\tau)}\cdot u(t-\tau)\,d\tau$$

$$= \int_{-\infty}^{t} e^{-b\tau}\cdot u(\tau)\cdot e^{-a(t-\tau)}\,d\tau$$

$$= \int_{0}^{t} e^{-at}\cdot e^{(a-b)\tau}\,d\tau = \frac{1}{a-b}\left(e^{(a-b)\tau}\cdot e^{-at}\right)\Big|_0^t$$

$$= \frac{1}{a-b}\left(e^{-bt} - e^{-at}\right), \qquad 0 \leq t$$

5 . Convolution with $\delta(t)$:

- If the impulse response $h(t) = \delta(t)$ the convolution integral result is the same input signal:

$$x(t)^*\delta(t) = \int_{-\infty}^{\infty} x(\tau)\delta(t-\tau)\,d\tau = x(t)$$

- So, The LTI system with impulse response $h(t) = \delta(t)$ is the identity system. like the previous equation leads to:

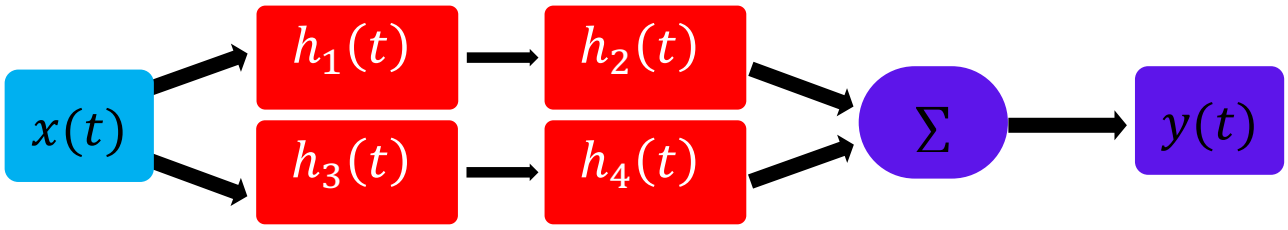
$$x(t-A)^*\delta(t-B) = x\big(t-(A+B)\big)$$

Q25: Find $x(t)^*h(t)$ if:

$$x(t) = 2e^{2t}, \qquad h(t) = \delta(t-3)$$

$$y(t) = 2e^{2(t-3)}$$

Q26: Find the impulse response of the system shown in the figure below:



$$h_1(t) = e^{-2t}u(t), \qquad h_2(t) = 2e^{-t}u(t)$$
$$h_3(t) = e^{-3t}u(t), \qquad h_4(t) = 4\delta(t)$$

$$h(t) = \left[[h_1(t)^* h_2(t)] + [h_3(t)^* h_4(t)]\right]$$
$$= \left[e^{-2t}u(t)^* 2e^{-t}u(t) + e^{-3t^*} 4\delta(t)\right]$$
$$= \int_{-\infty}^{\infty} e^{-2\tau}u(\tau)\cdot 2e^{-(t-\tau)}u(t-\tau)\,d\tau + 4e^{-3t}$$
$$= 2\int_{0}^{t} e^{-2\tau}e^{-(t-\tau)}\,d\tau + 4e^{-3t}$$
$$= 2e^{-t}\int_{0}^{t} e^{-\tau}\,d\tau + 4e^{-3t}$$
$$= 2e^{-t}(1-e^{-t}) + 4e^{-3t}$$
$$= 4e^{-3t} - 2e^{-2t} + 2e^{-t}$$

2.3 Discrete Convolution

2.3.1 Introduction:

- Discrete Signals will take values at selected time values.
- A discrete-time is a system in which all the signals are discrete-time Signals.
- A discrete-time system transforms discrete-time inputs into discrete-time output.

2.3.2 Discrete Convolution:

- Discrete Signals can be represented as follows:

$$h(n) = \{1,2,\underset{\uparrow}{1},3,1,2\}$$

- The arrow below the third element represents the current value of the signal (at $n = 0$).
- The Signal at any time between 0 and 1 is equal to zero as well as between 1 and 2 and so on. that mean, the signal will take values at selected time values. e.g: $h(1) = 3, h(2) = 1 \ldots etc.$
- Consider a Linear, Shift-invariant, discrete-time system with input $x(n)$, Any arbitrary Signal $x(n)$ can be written as the weighted sum of shifted delta functions:

$$x(n) = \sum_{k=-\infty}^{\infty} x(k)\delta(n-k)$$

- **To convolve $x(n)$ with the system $h(n)$:**

∴ Replace n by k.

∴ shift and reflect $h(k)$ by n to obtain $h(n-k)$.

∴ Then fix n, let k vary from $-\infty \rightarrow \infty$ and take the sum.

2.3.3 Discrete Convolution Fast Method:

- **The fast method is represented by:**

$h[n]$

$x[n]$	h_1	h_2	h_3	h_i
x_1	x_1h_1	x_1h_2	x_1h_3	x_1h_i
x_2	x_2h_1	x_2h_2	x_2h_3	x_2h_i
x_3	x_3h_1	x_3h_2	x_3h_3	x_3h_i
x_k	x_jh_1	x_jh_2	x_jh_3	x_jh_i

$$y[n] = \{[x_1h_1],$$
$$[x_2h_1 + x_1h_2],$$
$$[x_3h_1 + x_2h_2 + x_1h_3],$$
$$[x_jh_1 + x_3h_2 + x_2h_3 + x_1h_i],$$
$$[x_jh_2 + x_3h_3 + x_2h_i],$$
$$[x_jh_3 + x_3h_i],$$
$$[x_jh_i]\}$$

Q1: Find the convolution $y[n] = x[n]^* h[n]$ for:

$$x[n] = \{2,1,0,1,2\}, \qquad h[n] = \{1,1,-1,-1\}$$

$\therefore$ Solution:

$y[n] = \{[2],$
$[1+2],$
$[0+1-2],$
$[1+0-1-2],$
$[2+1+0-1],$
$[2-1+0],\leftarrow$
$[-2-1],$
$[-2]\}$

$= \{2, 3, -1, -2, 2, 1, -3, -2\}$

$h[n]$ ↓

$x[n]$		1	1	-1	-1
	2	2	2	-2	-2
	1	1	1	-1	-1
	0	0	0	0	0
→	1	1	1	-1	-1
	2	2	2	-2	-2

2.3.4 Properties of Discrete Convolution:

- the properties of discrete convolution are exactly like the properties of Continues LTI system:

Discrete Convolution properties:

1. Commutative:

$$x[n]^*h[n] = h[n]^*x[n]$$

2. Associative:

$$x[n]^*h_1[n]^*h_2[n] =$$

$$(x[n]^*h_1[n])^*h_2[n] = x[n]^*(h_1[n]^*h_2[n])$$

3. Distributivity:

$$x[n]^*(h_1[n] + h_2[n]) =$$

$$(x[n]^*h_1[n]) + (x[n]^*h_2[n])$$

4. Convolution with $u[n]$:

$$x[n]^*u[n] = \sum_{k=-\infty}^{\infty} x[k]u[n-k] = \sum_{k=-\infty}^{n} x[k]$$

5. Convolution with $\delta[n]$:

$$x[n]^*\delta[n] = x[n]$$

$$x[n-A]^*\delta[n-B] = x[n-(A+B)]$$



Mid-term Exam 2024-2025 1st s

3 Fourier Analysis

3.1 Fourier Series

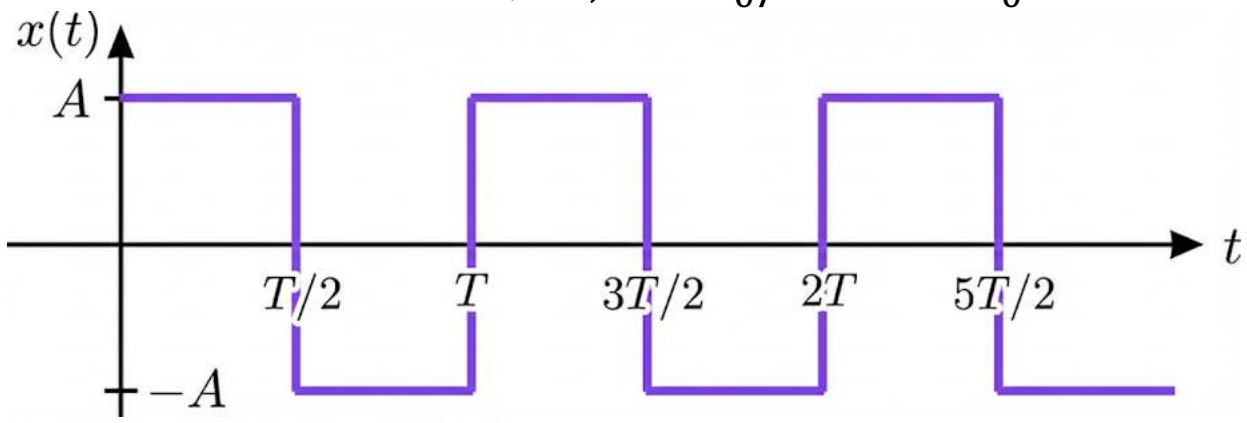
3.1.1 Introduction:

- **Fourier series** is a way to represent a periodic function as an infinite sum of sine and cosine functions.
- Any periodic signal $x(t)$ with period T_0 will have a Fourier Series (FS) representation. Aperiodic signals will have a Fourier Transform.
- When we say series, that means we are going to expand the signal to many terms.
- The expansion of the signal using FS can be exactly as $x(t)$ if we take infinity number of terms of the series.
- Finite number of terms means that we have an approximation of $x(t)$.

3.1.2 Approximating Periodic signals:

- Let's take this periodic square signal which can be represented by:

$$x(t) = \begin{cases} A, & 0 < t < T_0/2 \\ -A, & T_0/2 < t < T_0 \end{cases}$$



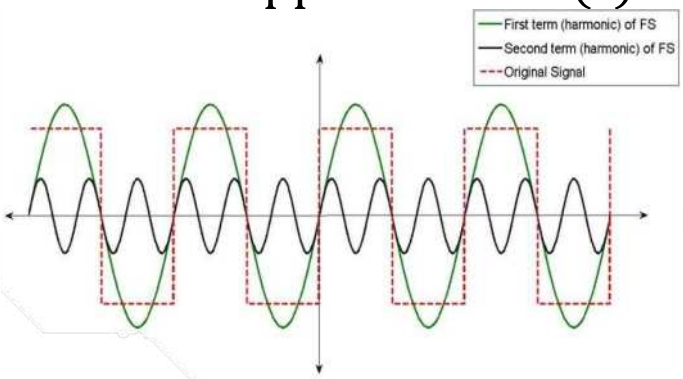
- By Fourier series, we can also represent this signal by a series complex exponential formula or sinusoids, it similar with the idea of Taylor series that transform any function to summation of polynomials.
- However, the trigonometric Fourier series for this signal (will explain later) is:

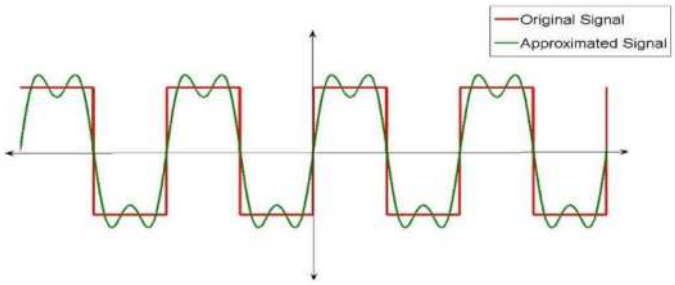
$$x(t) = \sum_{\substack{k=1, \\ k\ is\ odd}}^{\infty} \frac{4A}{k\pi} \sin(k\omega_0 t)$$

- Let's take the first two harmonics
- $(k = 1\ \&\ k = 3)$, we will find that:

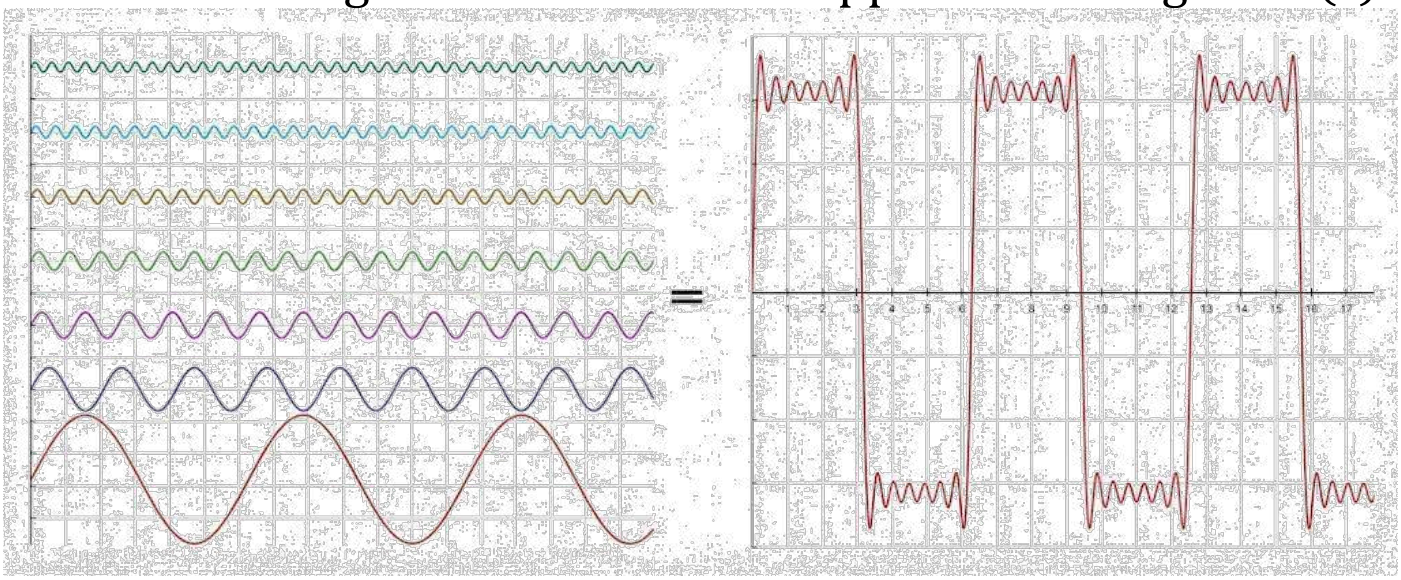
$$x(t) = \frac{4A}{\pi} \sin(\omega_0 t) + \frac{4A}{3\pi} \sin(3\omega_0 t) + \cdots$$

- as shown in the figure, we see that how these 2 harmonics approximate $x(t)$ with only sines functions.





- When taking more harmonics it approach to original $x(t)$.



Fourier Series Forms:

1. The Exponential Fourier Series.
2. The Trigonometric Fourier Series.
3. The Compact / Combined Fourier Series.

Remember that:

∴ The fundamental frequency is represented with:

$$\omega_0 = \frac{2\pi}{T_0} = 2\pi f_0$$

∴ ω_0 can be determined from the time domain signal.

∴ The kth harmonic is represented as:

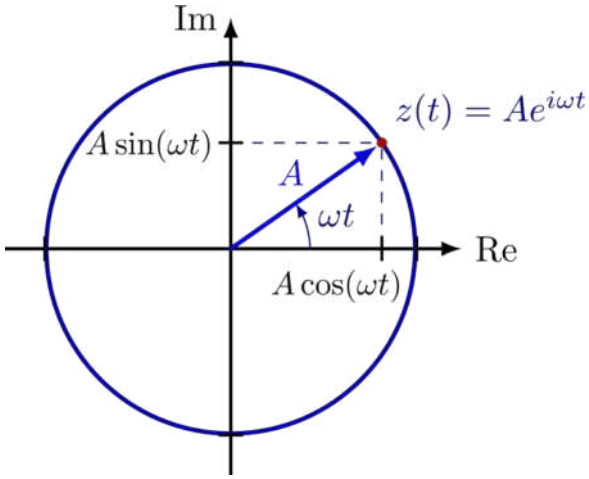
$$k = \pm 1 \rightarrow 1^{\text{st}} \text{ harmonic},$$

$$k = \pm 2 \rightarrow 2^{\text{nd}}\ harmonic, \ldots etc$$

∴ sines & cosines in unit circle:

$$\sin\left(\frac{k\pi}{2}\right) = \begin{cases} 0, & \text{even} \\ 1, & k \equiv 1(\text{mod}4) \\ -1, & k \equiv 3(\text{mod}4) \end{cases}$$

$$\cos\left(\frac{k\pi}{2}\right) = \begin{cases} 0, & \text{odd} \\ 1, & k \equiv 0(\text{mod}4) \\ -1, & k \equiv 2(\text{mod}4) \end{cases}$$



3.1.3 Exponential Fourier Series:

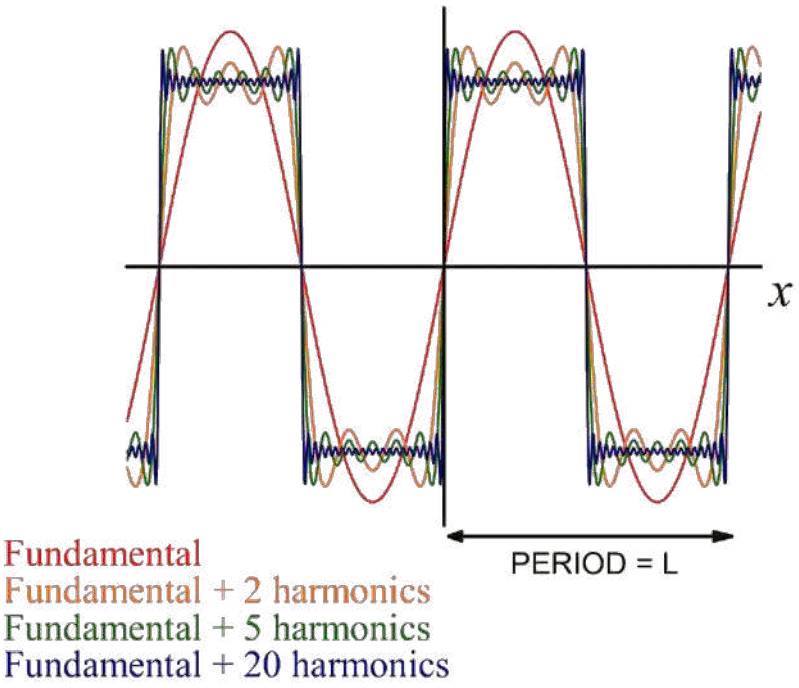
- The general form of EFS:

$$x(t) = \sum_{k=-\infty}^{\infty} C_k \cdot e^{jk\omega_0 t}$$

- Where C_k is the kth EFS Coefficients.
- C_k and ω_0 will be different from signal to another.
- So, all what we need to find the EFS of any signal is to calculate C_k as follow:

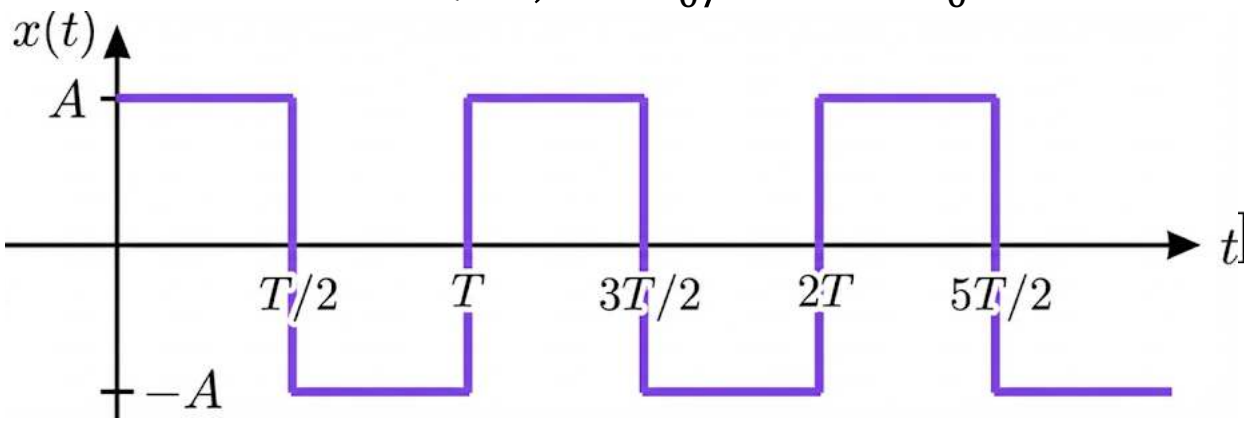
$$C_k = \frac{1}{T_0} \int_{(T)} x(t) \cdot e^{-jk\omega_0 t}\, dt$$

- Remember that $\int_{(T)}$ is an integration over one period, like $(0, T)$ & $\left(-\frac{T}{2}, \frac{T}{2}\right)$ … etc:



Q1: Express this periodic signal using EFS.

$$x(t) = \begin{cases} A, & 0 < t < T_0/2 \\ -A, & T_0/2 < t < T_0 \end{cases}$$



First, we find C_k:

$$C_k = \frac{1}{T_0}\int_0^{T_0} x(t)\cdot e^{-jk\omega_0 t}\,dt$$

$$= \frac{1}{T_0}\left(\int_0^{\frac{T_0}{2}} A\cdot e^{-jk\omega_0 t}\,dt + \int_{\frac{T_0}{2}}^{T_0} -A\cdot e^{-jk\omega_0 t}\,dt\right)$$

$$= \frac{1}{T_0}\left(\frac{A}{-jk\omega_0} e^{-jk\omega_0 t}\Big|_0^{\frac{T_0}{2}} + \frac{-A}{-jk\omega_0} e^{-jk\omega_0 t}\Big|_{\frac{T_0}{2}}^{T_0}\right)$$

$$= \frac{A}{-jkT_0\omega_0}\left(e^{-jk\omega_0 t}\Big|_0^{\frac{T_0}{2}} - e^{-jk\omega_0 t}\Big|_{\frac{T_0}{2}}^{T_0}\right)$$

$$= \frac{A}{-jkT_0\omega_0}\left(e^{-jk\omega_0\left(\frac{T_0}{2}\right)} - 1 - e^{-jk\omega_0(T_0)} + e^{-jk\omega_0\left(\frac{T_0}{2}\right)}\right)$$

$$= \frac{A}{-jkT_0\omega_0}\left(2e^{-jk\omega_0\left(\frac{T_0}{2}\right)} - e^{jk\omega_0 T_0} - 1\right)$$

$$= \frac{A}{-jk(2\pi)}\left(2e^{-jk\frac{2\pi}{2}} - e^{jk(2\pi)} - 1\right)$$

$$= \frac{A}{-jk2\pi}\left(+2\left(\cos(k\pi) + j\sin(k\pi)\right)\right.$$
$$\left. - \left(\cos(k2\pi) + j\sin(k2\pi)\right) - 1\right)$$

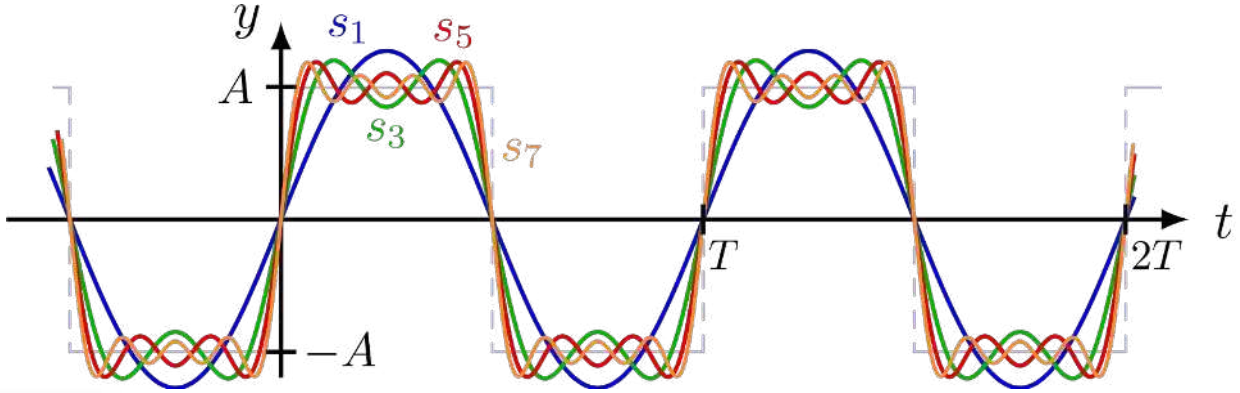
$$= \frac{A}{-jk2\pi}(2\cos(k\pi) - 2)$$

$$= \begin{cases} \frac{A}{-jk2\pi}(2(-1) - 2), & \textit{when } k \textit{ is odd} \\ \frac{A}{-jk2\pi}(2(1) - 2), & \textit{when } k \textit{ is even} \end{cases}$$

$$C_k = \begin{cases} \frac{2A}{jk\pi}, & \textit{when } k \textit{ is odd} \\ 0, & \textit{when } k \textit{ is even} \end{cases}$$

Now the EFS for $x(t)$ is:

$$x(t) = \sum_{\substack{k=-\infty, \\ k \text{ is odd}}}^{\infty} \frac{2A}{jk\pi} e^{jk\omega_0 t} = \sum_{k=-\infty}^{\infty} \frac{2A}{j(2k+1)\pi} e^{j(2k+1)\omega_0 t}$$



Q2: Express this periodic signal using EFS.

$$x(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_0)$$

$$C_k = \frac{1}{T_0} \int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} x(t) \cdot e^{-jk\omega_0 t}\, dt$$

$$= \frac{1}{T_0} \int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} \sum_{n=-\infty}^{\infty} \delta(t - nT_0) \cdot e^{-jk\omega_0 t}\, dt$$

$$= \frac{1}{T_0} \sum_{n=-\infty}^{\infty} \int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} e^{-jk\omega_0 t} \delta(t - nT_0)\, dt$$

We know that if there a delta inside an integration, we substitute the zero of the deltas if the shifting was between the terminals of the integration; so:

$$\delta(t - nT_0) \to t = nT_0, \qquad -\frac{T_0}{2} \le nT_0 \le \frac{T_0}{2} \to -\frac{1}{2} \le n \le \frac{1}{2}$$

Then, over the above period, $x(t)$ is defined when $n = 0$:

$$\frac{1}{T_0} \sum_{n=-\infty}^{\infty} \int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} e^{-jk\omega_0 t} \delta(t - nT_0)\, dt = \frac{1}{T_0} e^{-jk\omega_0 (nT_0)}\Big|_{n=0}$$

$$= \frac{1}{T_0} e^{0} \to C_k = \frac{1}{T_0}$$

$$x(t) = \sum_{k=-\infty}^{\infty} \frac{1}{T_0} e^{jk\omega_0 t}$$

3.1.4 Trigonometric Fourier Series:

- The general form of TFS:

$$x(t) = \frac{a_0}{2} + \sum_{k=1}^{\infty} a_k \cdot \cos k\omega_0 t + \sum_{k=1}^{\infty} b_k \cdot \sin k\omega_0 t$$

- Where a_0 is the DC constant, a_k and b_k is the TFS coefficients for cosine and sine summations.
- You can calculate them as follow:

$$a_0 = C_0 = \frac{2}{T_0}\int_{(T)} x(t)\, dt$$

$$a_k = 2C_{kR} = \frac{2}{T_0}\int_{(T)} x(t) \cdot \cos k\omega_0 t\ dt$$

$$b_k = -2C_{kI} = \frac{2}{T_0}\int_{(T)} x(t) \cdot \sin k\omega_0 t\ dt$$

- If $x(t)$ is odd:

$$a_0\ \&\ a_k = 0, \qquad b_k\ \textit{can be calculated as above}$$

- If $x(t)$ is even:

$$b_k = 0, \qquad a_0\ \&\ a_k\ \textit{can be calculated as above}$$

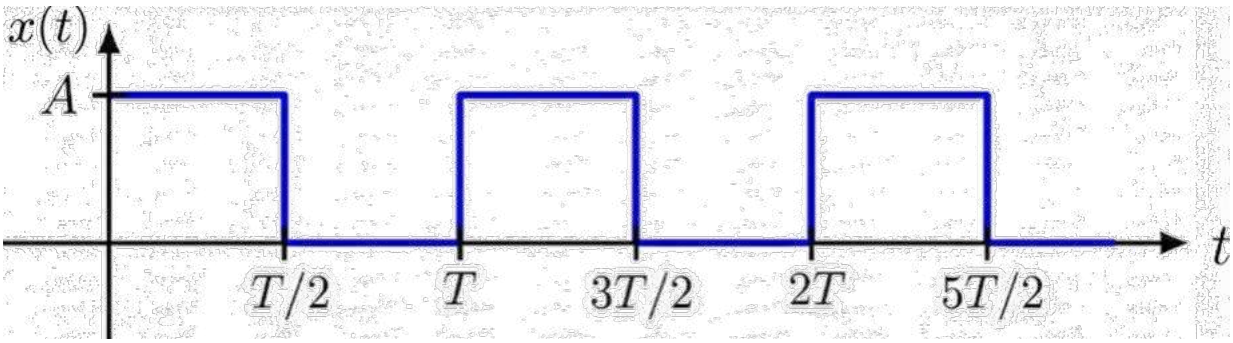
- If $x(t)$ has odd part and even part, calculate them all.

TFS Properties:

1. a_k & b_k are real numbers and are found for positive index values $k = 1,2,3, \ldots$
2. Plotting of a_k & b_k against $k\omega_0$ is a one-sided spectrum.
3. There is no plot for the phase angle spectrum. But there are two plots: one for a_k and the other for b_k.
4. a_0 will be plotted with the a_k spectrum.
5. The following relationships are applied for transformation from one form of FS to another.

Q3: Express this periodic signal using TFS.

$$x(t) = \begin{cases} A, & 0 < t < T_0/2 \\ 0, & T_0/2 < t < T_0 \end{cases}$$



we can conclude that $x(t)$ is neither even or odd , so we find a_0, a_k and b_k:

$$a_0 = \frac{2}{T_0}\int_0^{T_0} x(t)\,dt = \frac{2}{T_0}\int_0^{\frac{T_0}{2}} A\,dt = \frac{2}{T_0}\left(A\frac{T_0}{2}\right) = A$$

$$a_0 = A$$

$$a_k = \frac{2}{T_0}\int_0^{T_0} x(t)\cos(k\omega_0 t)\,dt = \frac{2A}{T_0}\int_0^{T_0/2} \cos(k\omega_0 t)\,dt + 0$$

$$= \frac{2A}{T_0}\cdot\frac{\sin(k\omega_0 t)}{k\omega_0}\Big|_0^{T_0/2} = \frac{2A}{T_0}\cdot\frac{\sin(k\omega_0 T_0/2)}{k\omega_0}$$

But $\frac{k\omega_0 T_0}{2} = k\pi \rightarrow \sin(k\pi) = 0$, so:

$$a_k = 0$$

$$b_k = \frac{2}{T_0}\int_0^{T_0} x(t)\sin(k\omega_0 t)\,dt = \frac{2A}{T_0}\int_0^{T_0/2} \sin(k\omega_0 t)\,dt$$

$$= \frac{2A}{T_0}\cdot\frac{1-\cos(k\omega_0 t)}{k\omega_0}\Big|_0^{T_0/2} = \frac{2A}{T_0}\cdot\frac{1-\cos(k\pi)}{k\omega_0}$$

Use $\cos(k\pi) = (-1)^k$ and $\omega_0 = \frac{2\pi}{T_0}$:

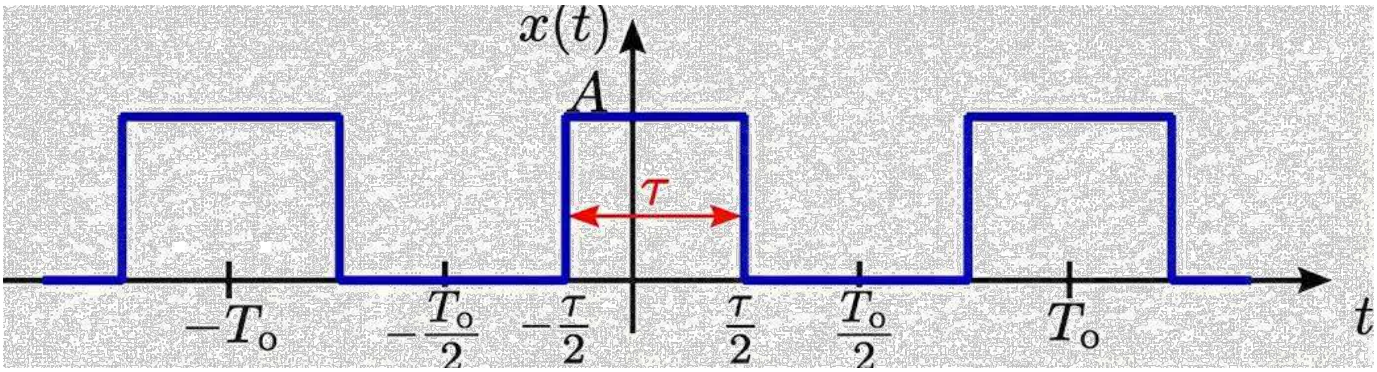
$$\frac{2A}{T_0}\cdot\frac{1-(-1)^k}{k(2\pi/T_0)} = \frac{A(1-(-1)^k)}{k\pi}$$

$$b_k = \begin{cases} 0, & k \text{ is even} \\ \frac{2A}{k\pi}, & k \text{ is odd} \end{cases}$$

$$x(t) = \frac{A}{2} + \sum_{\substack{k=1 \\ k \text{ odd}}}^{\infty} \frac{2A}{k\pi}\sin(k\omega_0 t)$$

Q4: Express this periodic signal using TFS.

$$x(t) = A \cdot rect\left(\frac{t}{\tau}\right) = \begin{cases} A, & -\tau/2 < t < \tau/2 \\ 0, & other\ wise\ over\ T_0 \end{cases}$$



First, we can conclude b_k is zero; because $x(t)$ is even, so we find a_0 and a_k:

$$a_0 = \frac{1}{T_0}\int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} x(t)\,dt = \frac{1}{T_0}\int_{-\frac{\tau}{2}}^{\frac{\tau}{2}} A\,dt = \frac{A\tau}{T_0}$$

$$a_k = \frac{2}{T_0}\int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} x(t)\cdot\cos k\omega_0 t\ dt = \frac{2}{T_0}\int_{-\frac{\tau}{2}}^{\frac{\tau}{2}} A\cdot\cos k\omega_0 t\ dt$$

$$= \frac{2A}{kT_0\omega_0}\sin k\omega_0 t\,\Big|_{-\frac{\tau}{2}}^{\frac{\tau}{2}}$$

$$= \frac{A}{k\pi}\left(\sin\left(k\omega_0\frac{\tau}{2}\right) - \sin\left(k\omega_0\left(-\frac{\tau}{2}\right)\right)\right)$$

$$a_k = \frac{2A}{k\pi}\left(\sin\left(k\omega_0\frac{\tau}{2}\right)\right)$$

$$x(t) = \frac{A\tau}{T_0} + \sum_{k=1}^{\infty}\frac{2A}{k\pi}\left(\sin\left(k\omega_0\frac{\tau}{2}\right)\right)\cdot\cos k\omega_0 t$$

3.1.5 Spectrum Drawing:

- The **spectrum** is a frequency-domain representation of a signal. It shows:
  - ∴ **Which frequencies exist** in the signal
  - ∴ **How strong each frequency is**
  - ∴ **The phase shift** of each frequency
- So instead of looking at $x(t)$in time, we look at its frequency content.

▶ **EFS Spectrum:**

$$x(t) = \sum_{k=-\infty}^{\infty} C_k e^{jk\omega_0 t}$$

- Where:
  - ∴ $k = 0, \pm 1, \pm 2, \ldots$
  - ∴ Frequencies exist only at:
  - ∴ $\omega = k\omega_0$
- Meaning of k
  - ∴ $k = 0$: $C_0 \rightarrow$ DC component (average value)
  - ∴ $k = 1$: $C_1 \rightarrow$ 1st harmonic (fundamental)
  - ∴ $k = n$: $C_n \rightarrow n^{th}$harmonic
- The vertical lines at $\omega = k\omega_0$are called **spectral lines.**

Complex form of C_k:

$$C_k = C_{kR} + jC_{kI}, \qquad C_k = |C_k| \angle\theta_k = |C_k| e^{j\theta_k}$$

- For real signals:
  - ∴ Magnitude is even: $|C_k| = |C_{-k}|$
  - ∴ Phase is odd: $\theta_k = -\theta_{-k}$
- EFS spectrum is drawn as **two graphs**:
  - ∴ **Magnitude spectrum**: $|C_k|$ at $\omega = k\omega_0$
  - ∴ **Phase spectrum**: θ_k at $\omega = k\omega_0$

TFS Spectrum:

$$x(t) = \frac{a_0}{2} + \sum_{k=1}^{\infty} a_k \cos(k\omega_0 t) + \sum_{k=1}^{\infty} b_k \sin(k\omega_0 t)$$

- Frequencies exist only at: $\omega = k\omega_0, k = 1,2,3,\ldots$
- In TFS, each harmonic k has:
  - ∴ a cosine part a_k
  - ∴ a sine part b_k
  - ∴ So the spectrum tells how much cosine and sine exist at each harmonic.
- TFS can also be written in amplitude-phase form

$$x(t) = \frac{a_0}{2} + \sum_{k=1}^{\infty} A_k \cos(k\omega_0 t + \phi_k)$$

- Where:
  - ∴ $A_k = \sqrt{a_k^2 + b_k^2}$
  - ∴ $\phi_k = \tan^{-1}(\frac{-b_k}{a_k})$
- TFS Spectrum plots (2 common ways)

A)

- ∴ A_k vs $k\omega_0$ (Amplitude)
- ∴ ϕ_k vs $k\omega_0$ (Phase)

B)

- ∴ a_k vs $k\omega_0$
- ∴ b_k vs $k\omega_0$

Main difference (EFS vs TFS Spectrum)

Type	Frequencies shown	Coefficients
EFS	$k = 0, \pm1, \pm2, \cdots$ (positive & negative)	C_k (complex)
TFS	$k = 0,1,2, \cdots$ (positive only)	a_k, b_k (real)

Q5: Draw the EFS spectrum for the following expanded signal:

$$x(t) = 4e^{j\frac{\pi}{3}}e^{-j7\omega_0 t} - 5e^{-j3\omega_0 t} + 2 + 5e^{j3\omega_0 t} - 4e^{j\frac{\pi}{3}}e^{j7\omega_0 t}$$

Using $C_k = |C_k|\angle\theta_k = |C_k|e^{j\theta_k}$, We can conclude that:

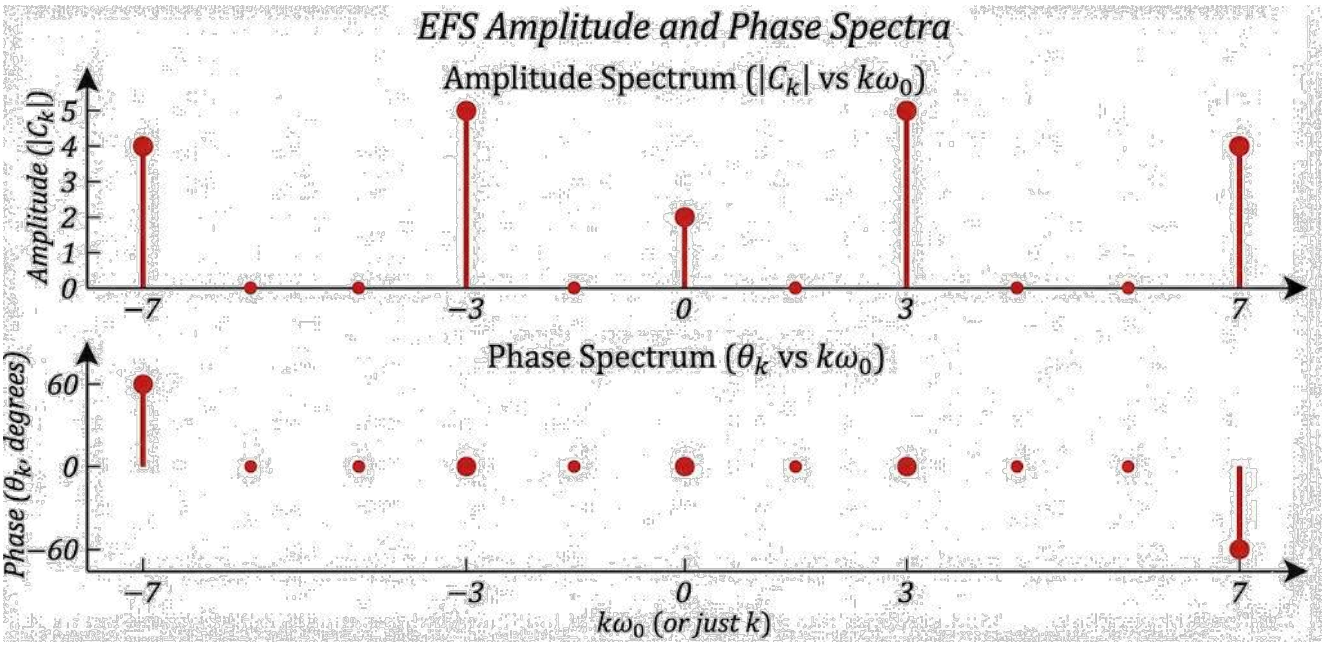
$4e^{j\frac{\pi}{3}}e^{-j7\omega_0 t} \rightarrow k = -7 \rightarrow |C_{-7}| = 4 \rightarrow \theta_{-7} = \frac{\pi}{3}$

$-5e^{-j3\omega_0 t} \rightarrow k = -3 \rightarrow |C_{-3}| = 5 \rightarrow \theta_{-3} = 0$

$2 \rightarrow k = 0 \rightarrow |C_0| = 2 \rightarrow \theta_0 = 0$

$5e^{j3\omega_0 t} \rightarrow k = 3 \rightarrow |C_3| = 5 \rightarrow \theta_3 = 0$

$-4e^{j\frac{\pi}{3}}e^{j7\omega_0 t} \rightarrow k = 7 \rightarrow |C_7| = 4 \rightarrow \theta_7 = -\frac{\pi}{3}$

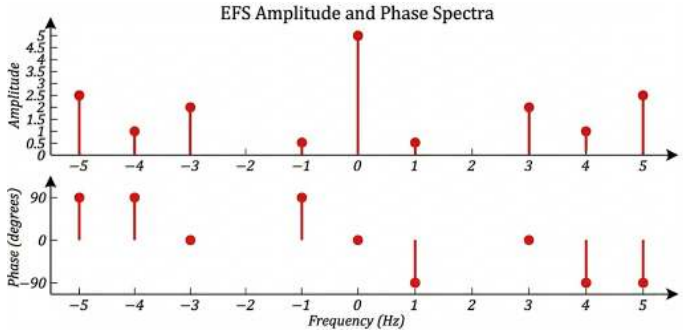


Q6: Draw EFS and TFS spectrum for:

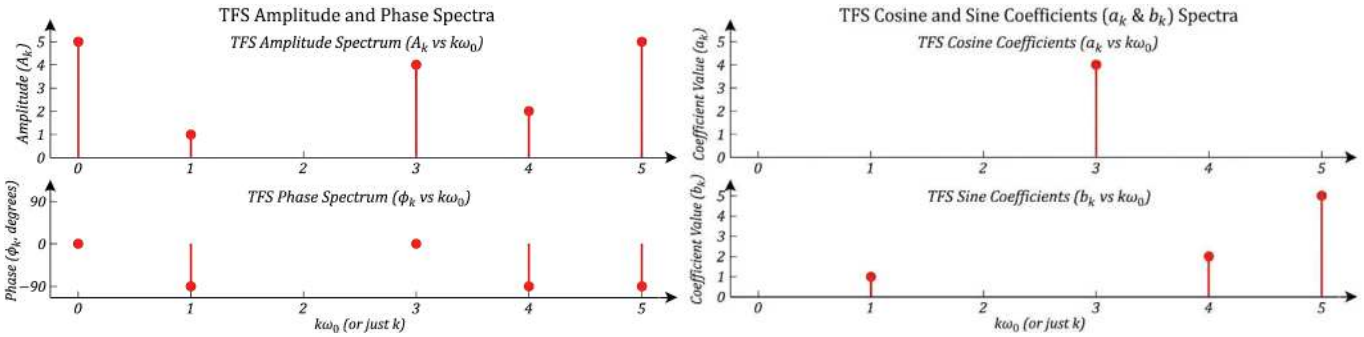
$$x(t) = 5 + \sin 2\pi t + 4\cos 6\pi t + 2\sin 8\pi t + 5\sin 10\pi t$$

C_k			f	$\lvert C_k \rvert$	θ_0
5	$\longrightarrow$		0	5	0
$\sin 2\pi t$	$\rightarrow$	$\dfrac{e^{j2\pi t} - e^{-j2\pi t}}{2j}$	1	$\dfrac{1}{2}$	$\pm 90°$
$4\cos 6\pi t$	$\rightarrow$	$4\dfrac{e^{j6\pi t} + e^{-j6\pi t}}{2}$	3	2	0
$2\sin 8\pi t$	$\rightarrow$	$2\dfrac{e^{j8\pi t} - e^{-j8\pi t}}{2j}$	4	1	$\pm 90°$
$5\sin 10\pi t$	$\rightarrow$	$5\dfrac{e^{j10\pi t} - e^{-j10\pi t}}{2j}$	5	$\dfrac{5}{2}$	$\pm 90°$

$\therefore$ EFS Spectrum:



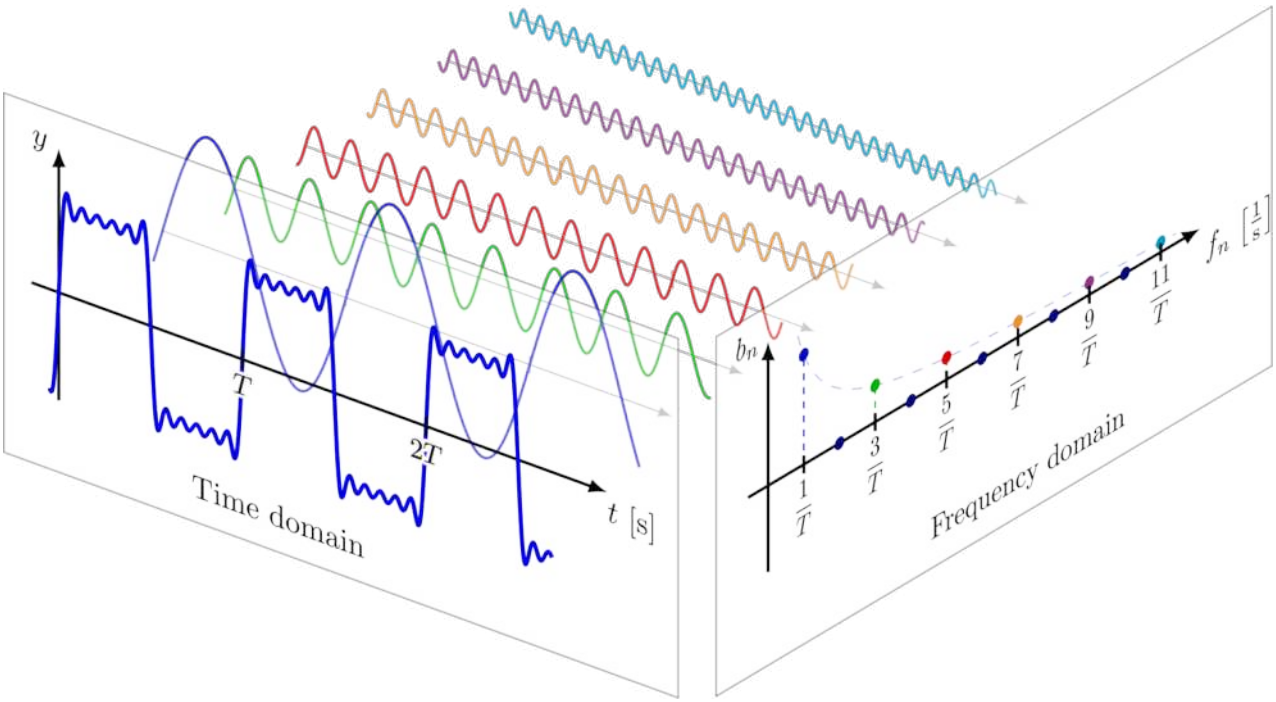
$\therefore$ TFS Spectrum:



3.2 Fourier Transform

3.2.1 Introduction:

- **Fourier Transform** is a mathematical transformation employed to transform signals between time domain and frequency domain, which has many applications in physics and engineering.
- It is reversible, being able to transform from either domain to the other. The term itself refers to both the transform operation and to the function it produces.
- The Fourier Transform (F.T) can be extended to non-periodic waveforms such as a single pulse or a single transient by making $T \rightarrow \infty$.



3.2.2 Fourier Transform Formula:

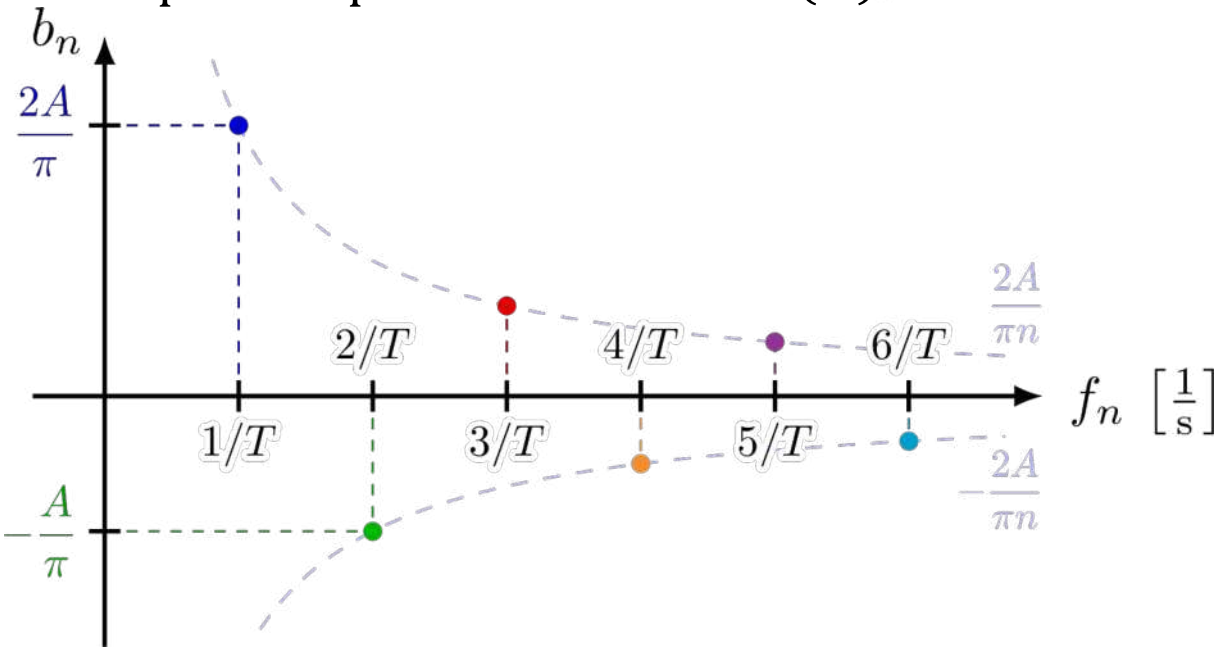
- To transform $x(t)$ from time to frequency domain:

$$X(\omega) = \mathcal{F}\{x(t)\} = \int_{-\infty}^{\infty} x(t) \cdot e^{-j\omega t}\, dt$$

- To transform $X(\omega)$ from frequency to time domain
- (The Inverse F.T):

$$x(t) = \mathcal{F}^{-1}\{X(\omega)\} = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\omega) \cdot e^{j\omega t}\, d\omega$$

- The most-common pairs $x(t)$ and $X(\omega)$ are often given in tables for easy reference.
- F.T has the useful property that many relationships and operations over the original $x(t)$ correspond to simpler relationships and operations over its $X(\omega)$.



Q1: Find F.T for:

A) $x(t) = e^{-at}u(t),\ a > 0$

$$X(\omega) = \mathcal{F}\{x(t)\} = \int_{-\infty}^{\infty} x(t) \cdot e^{-j\omega t}\, dt$$

$$= \int_{-\infty}^{\infty} e^{-at}u(t) \cdot e^{-j\omega t}\, dt = \int_{0}^{\infty} e^{-at} \cdot e^{-j\omega t}\, dt$$

$$= \int_{0}^{\infty} e^{-t(a+j\omega)}\, dt = -\frac{1}{a+j\omega}\left(e^{-t(a+j\omega)}\right)\Big|_0^{\infty}$$

$$= \frac{1}{a+j\omega}\left(e^{0} - e^{-\infty}\right) = \frac{1}{a+j\omega}$$

B) $x(t) = e^{at}u(-t),\ a > 0$

$$X(\omega) = \mathcal{F}\{x(t)\} = \int_{-\infty}^{\infty} x(t) \cdot e^{-j\omega t}\, dt$$

$$= \int_{-\infty}^{\infty} e^{at}u(-t) \cdot e^{-j\omega t}\, dt = \int_{-\infty}^{0} e^{at} \cdot e^{-j\omega t}\, dt$$

$$= \int_{-\infty}^{0} e^{t(a-j\omega)}\, dt = \frac{1}{a-j\omega}\left(e^{-t(a+j\omega)}\right)\Big|_{-\infty}^{0}$$

$$= \frac{1}{a-j\omega}\left(e^{0} - e^{-\infty}\right) = \frac{1}{a-j\omega}$$

C) $x(t) = e^{-a|t|},\ a > 0$

$$X(\omega) = \int_{-\infty}^{\infty} e^{-a|t|} \cdot e^{-j\omega t}\, dt$$

$$= \int_{-\infty}^{0} e^{t(a-j\omega)}\, dt + \int_{0}^{\infty} e^{-t(a+j\omega)}\, dt$$

$$= \mathcal{F}\{e^{at}u(-t)\} + \mathcal{F}\{e^{-at}u(t)\} = \frac{1}{a+j\omega} + \frac{1}{a-j\omega}$$

$$= \frac{2a}{a^2+\omega^2}$$

D) $x(t) = rect\left(\frac{t}{\tau}\right) = \begin{cases} 1, & -\frac{\tau}{2} < t < \frac{\tau}{2} \\ 0, & otherwise \end{cases}$

$$X(\omega) = \int_{-\infty}^{\infty} rect\left(\frac{t}{\tau}\right) \cdot e^{-j\omega t}\, dt = \int_{-\frac{\tau}{2}}^{\frac{\tau}{2}} e^{-j\omega t}\, dt$$

$$= \frac{1}{-j\omega} e^{-j\omega t}\Big|_{-\frac{\tau}{2}}^{\frac{\tau}{2}} = \frac{1}{j\omega}\left(e^{j\omega\frac{\tau}{2}} - e^{-j\omega\frac{\tau}{2}}\right) = \frac{\sin\left(\omega\frac{\tau}{2}\right)}{\frac{\omega}{2}}$$

$$= \frac{\pi\tau}{\pi\tau} \frac{\sin\left(\left(\frac{\pi}{\pi}\right)\frac{\omega\tau}{2}\right)}{\frac{\omega}{2}} = \frac{\tau \sin\left(\frac{\pi\omega\tau}{2\pi}\right)}{\frac{\pi\omega\tau}{2\pi}}$$

$$= \tau\, \mathrm{sinc}\left(\frac{\omega\tau}{2\pi}\right)$$

E) $x(t) = tri\left(\frac{t}{\tau}\right) = \begin{cases} 1 - \frac{|t|}{\tau}, & |t| \le \tau \\ 0, & otherwise \end{cases}$

if $x(t)$ is even, you can use Euler's identity to facilitate the solution by replacing $e^{-j\omega t}$ by $\cos \omega t$ (sine function is zero in even functions), multiply the function by 2 and reset the terminals of the integration from 0 to $\frac{\tau}{2}$:

$$X(\omega) = \int_{-\infty}^{\infty} tri\left(\frac{t}{\tau}\right) \cdot e^{-j\omega t}\, dt = \int_{-\tau}^{\tau} \left(1 - \frac{|t|}{\tau}\right) \cdot e^{-j\omega t}\, dt$$

$$= 2\int_{0}^{\tau} \left(1 - \frac{t}{\tau}\right) \cdot \cos(\omega t)\, dt$$

$$= 2\int_{0}^{\tau} \cos(\omega t)\, dt - \frac{2}{\tau}\int_{0}^{\tau} t \cdot \cos(\omega t)\, dt$$

$$= 2 \cdot \frac{\sin \omega\tau}{\omega} - \frac{2}{\tau}\left(\frac{\tau \sin \omega\tau}{\omega} - \frac{1 - \cos \omega\tau}{\omega^2}\right)$$

$$= 2\frac{1 - \cos \omega\tau}{\tau\omega^2} = \frac{4\sin^2\left(\frac{\omega\tau}{2}\right)}{\tau\omega^2} = \tau\, \mathrm{sinc}^2\left(\frac{\omega\tau}{2\pi}\right)$$

Important notes:

- We concluded that the F.T for the rectangular function is:

$$rect\left(\frac{t}{\tau}\right) = \tau \,\text{sinc}\left(\frac{\omega\tau}{2\pi}\right) \Big|_{when\ \tau\ is\ the\ width\ for\ rect.}$$

- But in different books, τ is defined as the half width of the rectangle, don't get confused between them.
- If τ is defined as the half width of the rectangle, its F.T is:

$$rect\left(\frac{t}{\tau}\right) = 2\tau \,\text{sinc}\left(\frac{\omega\tau}{\pi}\right) \Big|_{when\ \tau\ is\ half\ the\ width\ for\ rect.}$$

- We concluded that the F.T for the triangular function is:

$$tri\left(\frac{t}{\tau}\right) = \tau \,\text{sinc}^2\left(\frac{\omega\tau}{2\pi}\right) \Big|_{when\ \tau\ is\ half\ the\ base\ for\ tri.}$$

- But in different books, τ is defined as the base of the triangle, don't get confused between them.
- If τ is defined as the base of the rectangle, its F.T is:

$$tri\left(\frac{t}{\tau}\right) = \frac{\tau}{2}\,\text{sinc}^2\left(\frac{\omega\tau}{4\pi}\right) \Big|_{when\ \tau\ is\ the\ base\ for\ tri.}$$

- for easy remembering, just replace $\tau \rightarrow \begin{cases} 2\tau \\ \frac{\tau}{2} \end{cases}$ as needed.

Important rules:

$$x(t) = \delta(t) \quad \rightarrow \quad X(\omega) = 1$$

$$x(t) = 1 \quad \rightarrow \quad X(\omega) = 2\pi\delta(\omega)$$

$$x(t) = u(t) \quad \rightarrow \quad X(\omega) = \frac{1}{j\omega} + \pi\delta(\omega)$$

$$x(t)e^{j\omega_0 t} \quad \rightarrow \quad X(\omega) = 2\pi\delta(\omega - \omega_0)$$

Q2: Find F.T for:

A) $x(t) = \cos(at)$

$$X(\omega) = \mathcal{F}\left\{\frac{e^{jat} + e^{-jat}}{2}\right\} = \mathcal{F}\left\{\frac{e^{jat}}{2}\right\} + \mathcal{F}\left\{\frac{e^{-jat}}{2}\right\}$$

$$= \pi\delta(\omega - a) + \pi\delta(\omega + a)$$

$$= \pi\big(\delta(\omega + a) + \delta(\omega - a)\big)$$

B) $x(t) = \sin(at)$

$$X(\omega) = \mathcal{F}\left\{\frac{e^{jat} - e^{-jat}}{2j}\right\} = \mathcal{F}\left\{\frac{e^{jat}}{2j}\right\} - \mathcal{F}\left\{\frac{e^{-jat}}{2j}\right\}$$

$$= \frac{\pi}{j}\delta(\omega - a) - \frac{\pi}{j}\delta(\omega + a)$$

$$= j\pi\big(\delta(\omega + a) - \delta(\omega - a)\big)$$

3.2.3 f and ω domain:

- You can find F.T for $x(t)$ in terms of f instead of ω by 2 ways:

1. Using integral rules:

$$X(f)=\int_{-\infty}^{\infty} x(t)\cdot e^{-j2\pi ft}\,dt\ ,x(t)=\int_{-\infty}^{\infty} X(f)\cdot e^{j2\pi ft}\,df$$

2. If you have $X(\omega)$, you can substitute ω by $2\pi f$:

$$\omega=2\pi f,\qquad f=\frac{\omega}{2\pi}$$

So, we can say:

$$X(f)=X\left(\frac{\omega}{2\pi}\right),\qquad X(\omega)=X(2\pi f)$$

Q3: Find $X(f)$ for:

A) $x(t)=e^{-2t}u(t)$

$$x(t)\Leftrightarrow X(\omega)\rightarrow e^{-2t}\Leftrightarrow\frac{1}{2+j\omega}$$

$$X(\omega)=X(2\pi f)=\frac{1}{2+j2\pi f}$$

B) $x(t)=\cos at$

$$x(t)\Leftrightarrow X(\omega)\rightarrow\cos at\Leftrightarrow\pi(\delta(\omega+a)+\delta(\omega-a))$$

$$X(\omega)=X(2\pi f)=\pi\left(\delta(2\pi f+a)+\delta(2\pi f-a)\right)$$

$$=\frac{1}{2}\left(\delta\left(f+\frac{a}{2\pi}\right)+\delta\left(f-\frac{a}{2\pi}\right)\right)$$

3.2.4 Fourier Transform Properties:

1. Linearity:

$$ax_1(t) + bx_2(t) \Leftrightarrow aX_1(\omega) + bX_2(\omega)$$

Q1: Find $\mathcal{F}\{sgn(t)\}$.

$$X(\omega) = \mathcal{F}\{u(t) - u(-t)\} = \mathcal{F}\{u(t)\} - \mathcal{F}\{u(-t)\}$$
$$= \frac{1}{j\omega} + \pi\delta(\omega) - \frac{1}{j(-\omega)} - \pi\delta(-\omega)$$
$$= \frac{1}{j\omega} + \frac{1}{j\omega} + \pi\delta(\omega) - \pi\delta(\omega) = \frac{2}{j\omega}$$

2. Time Scaling:

$$x(at) \Leftrightarrow \frac{1}{|a|} X\left(\frac{\omega}{a}\right)$$

Q2: If $x(t) = e^{-t}u(t)$, Find $\mathcal{F}\{e^{t}u(-t)\} = \mathcal{F}\{x(-t)\}$ using Time scaling property.

$$e^{-t}u(t) \Leftrightarrow \frac{1}{1 + j\omega}$$

$$e^{t}u(-t) \Leftrightarrow \frac{1}{|-1|} \frac{1}{1 + j\left(\frac{\omega}{-1}\right)} = \frac{1}{1 - j\omega}$$

3. Duality:

$$X(t) \Leftrightarrow 2\pi x(-\omega)$$

- This property allows to find F.T for a function that know that it was transformed to ω domain, but it in t domain, so you can use this property instead of finding it using integration.

Q3: Find $\mathcal{F}\left\{\tau \operatorname{sinc}\left(\frac{\tau t}{2\pi}\right)\right\}$.

We know that $\mathcal{F}^{-1}\left\{\tau \operatorname{sinc}\left(\frac{\tau\omega}{2\pi}\right)\right\} = rect\left(\frac{t}{\tau}\right)$ but to find $\mathcal{F}\left\{\tau \operatorname{sinc}\left(\frac{\tau t}{2\pi}\right)\right\}$ use this propery:

$$\tau \operatorname{sinc}\left(\frac{\tau\omega}{2\pi}\right) \Leftrightarrow rect\left(\frac{t}{\tau}\right)$$

$$\tau \operatorname{sinc}\left(\frac{\tau t}{2\pi}\right) \Leftrightarrow 2\pi\, rect\left(\frac{-\omega}{\tau}\right) = 2\pi\, rect\left(\frac{\omega}{\tau}\right)$$

Q4: Find $\mathcal{F}\left\{\frac{4}{4+t^2}\right\}$.

$$\frac{2a}{a^2+\omega^2} \Leftrightarrow e^{-a|t|} \rightarrow a = 2$$

$$\frac{2a}{a^2+t^2} \Leftrightarrow 2\pi\, e^{-a|-\omega|} = 2\pi\, e^{-2|\omega|}$$

Q5: Find $\mathcal{F}\{\delta(t+1)+\delta(t-1)\}$.

$$\pi\left(\delta(\omega+\omega_0)+\delta(\omega-\omega_0)\right) \Leftrightarrow \cos\omega_0 t \rightarrow \omega_0 = 1$$

$$\pi\left(\delta(t+1)+\delta(t-1)\right) \Leftrightarrow 2\pi\cos(-\omega)$$

$$\left(\delta(t+1)+\delta(t-1)\right) \Leftrightarrow 2\cos\omega$$

4. Time Shifting:

$$x(t-t_0) \Leftrightarrow e^{-j\omega t_0}X(\omega)$$

Q6: Find $\mathcal{F}\{e^{-2(t-3)}u(t)\}$.

We know that $\mathcal{F}\{e^{-at}u(t)\} = \frac{1}{a+j\omega}$ but to find $\mathcal{F}\{e^{-2(t-3)}u(t)\}$ use this propery:

$$e^{-at}u(t) \Leftrightarrow \frac{1}{a+j\omega} \rightarrow a=2, t_0=3$$

$$e^{-a(t-t_0)}u(t) \Leftrightarrow e^{-j\omega t_0}\frac{e^{at_0}}{a+j\omega} = e^{-j3\omega}\frac{e^6}{2+j\omega}$$

Q7: Find $\mathcal{F}\left\{4\,rect\left(\frac{t+2}{\tau}\right)\right\}$.

$$rect\left(\frac{t}{\tau}\right) \Leftrightarrow \tau\,\text{sinc}\left(\frac{\omega\tau}{2\pi}\right) \rightarrow t_0 = -2$$

$$4\,rect\left(\frac{t}{\tau}\right) \Leftrightarrow 4e^{-j\omega t_0}\,\tau\,\text{sinc}\left(\frac{\omega\tau}{2\pi}\right)$$

$$= 4e^{j2\omega}\tau\,\text{sinc}\left(\frac{\omega\tau}{2\pi}\right)$$

Q8: Find $\mathcal{F}\{2u(t-1)-\delta(t-1)\}$.

$$u(t) \Leftrightarrow \frac{1}{j\omega} + \pi\delta(\omega) \;\&\; \delta(t) \Leftrightarrow 1$$

$$2u(t-1)-\delta(t-1) \Leftrightarrow e^{-j\omega}\left(\frac{2}{j\omega} + 2\pi\delta(\omega) - 1\right)$$

5. "ω" Shifting:

$$e^{j\omega_0 t}x(t) \Leftrightarrow X(\omega-\omega_0)$$

Q9: Find $\mathcal{F}\{e^{j2t}e^{-t}u(t)\}$.

We know that $\mathcal{F}\{e^{-at}u(t)\}=\frac{1}{a+j\omega}$ but to find $\mathcal{F}\{e^{j2t}e^{-at}u(t)\}$ use this propery:

$$e^{-at}u(t) \Leftrightarrow \frac{1}{a+j\omega} \rightarrow a=1, \omega_0=2$$

$$e^{j\omega_0 t}e^{-at}u(t) \Leftrightarrow \frac{1}{a+j(\omega-\omega_0)} = \frac{1}{1+j(\omega-2)}$$

Q10: Find $\mathcal{F}\{e^{-j2t}\sin(t)\}$.

$$\sin at \Leftrightarrow j\pi\big(\delta(\omega+a)-\delta(\omega-a)\big)$$

$$\rightarrow a=1, \quad \omega_0=-2,$$

$$e^{j\omega_0 t}\sin a(t) \Leftrightarrow j\pi\Big(\delta(\omega-\omega_0+a)-\delta(\omega-\omega_0-a)\Big)$$

$$= j\pi\big(\delta(\omega+3)-\delta(\omega+1)\big)$$

6. Differentiation in Time Domain:

$$x'(t) \Leftrightarrow j\omega\, X(\omega)$$

$$x^{(n)}(t) \Leftrightarrow (j\omega)^n X(\omega)$$

Q11: If $x(t) = e^{-t}u(t)$, Find $\mathcal{F}\{x''(t)\}$

$$x(t) = e^{-t}u(t) \Leftrightarrow X(\omega) = \frac{1}{1+j\omega}$$

$$x''(t) \Leftrightarrow (j\omega)^2 X(\omega) = -\omega^2 \cdot \frac{1}{1+j\omega}$$

Q12: Find $\mathcal{F}\{(t^2 - 3t)e^{-t}u(t)\}$.

Using linearity and differentiation properties:

$$\mathcal{F}\{(t^2 - 3t)e^{-t}u(t)\} \Leftrightarrow \mathcal{F}\{t^2 e^{-t}u(t) - 3te^{-t}u(t)\}$$

$$\mathcal{F}\{t^2 e^{-t}u(t)\} + \mathcal{F}\{-3t\, e^{-t}u(t)\} \Leftrightarrow \frac{2}{(1+j\omega)^3} + \frac{-3}{(1+j\omega)^2}$$

7. Symmetry:

$$X(-\omega) \Leftrightarrow X^*(\omega)$$

$$|X(\omega)| = |X(-\omega)|$$

$$\theta(\omega) = -\theta(-\omega)$$

8. Multiplication by t:

$$(jt)^n x(t) \Leftrightarrow X^{(n)}(\omega)$$

$$t^n x(t) \Leftrightarrow j^n X^{(n)}(\omega)$$

Q13: Find $\mathcal{F}\{t^2 e^{-t} u(t)\}$

$$x(t) = e^{-t}u(t) \Leftrightarrow X(\omega) = \frac{1}{1+j\omega}$$

$$t^2 x(t) \Leftrightarrow j^2 X''(\omega) = -1\left(-\frac{j}{(1+j\omega)^2}\right)' = j\frac{-2j}{(1+j\omega)^3}$$

$$= \frac{2}{(1+j\omega)^3}$$

Q14: Find $\mathcal{F}\{t^3 e^{-t} u(t)\}$

$$\mathcal{F}\{t^3 e^{-t} u(t)\} = \frac{6}{(1+j\omega)^4}$$

9. Energy:

$$E_{t-Domain} = E_{\omega-Domain}$$

$$\int_{-\infty}^{\infty} |x(t)|^2 \, dt \Leftrightarrow \frac{1}{2\pi}\int_{-\infty}^{\infty} |X(\omega)|^2 \, d\omega$$

10. Integration in Time Domain:

$$\int_0^t x(\tau)\,d\tau \Leftrightarrow \frac{1}{j\omega}X(\omega) + \pi X(0)\delta(\omega)$$

Q15: If $x(t) = e^{-t}u(t)$, Find $\mathcal{F}\left\{\int_0^t x(\tau)\,d\tau\right\}$

$$x(t) = e^{-t}u(t) \Leftrightarrow X(\omega) = \frac{1}{1+j\omega}$$

$$\mathcal{F}\left\{\int_0^t x(\tau)\,d\tau\right\} \Leftrightarrow \frac{1}{j\omega}X(\omega) + \pi X(0)\delta(\omega)$$

$$= \frac{1}{j\omega - \omega^2} + \pi\delta(\omega)$$

Q16: If $x(t) = (t-1)e^{-t}\,u(t)$, Find $\mathcal{F}\left\{\int_0^t x(\tau)\,d\tau\right\}$

$$\mathcal{F}\{(t-1)e^{-t}\,u(t)\} = \left(\frac{1}{(1+j\omega)^2}\right) - \left(\frac{1}{1+j\omega}\right)$$

$$\mathcal{F}\left\{\int_0^t (t-1)e^{-t}\,u(t)dt\right\}$$

$$= \left(\frac{1}{j\omega}\right)\left[\left(\frac{1}{(1+j\omega)^2}\right) - \left(\frac{1}{1+j\omega}\right)\right]$$

11. Convolution:

$$x(t)^{*}h(t) \Leftrightarrow X(\omega)\cdot H(\omega)$$

$$y(t) \Leftrightarrow Y(\omega)$$

Q17: If $x(t)=e^{-t}u(t)$ & $h(t)=e^{t}\,u(-t)$, Find $\mathcal{F}\{x(t)^{*}h(t)\}$

$$\mathcal{F}\{x(t)^{*}h(t)\}=X(\omega)\cdot H(\omega)=\frac{1}{1+j\omega}\cdot\frac{1}{1-j\omega}=\frac{1}{1+\omega^{2}}$$

Q18: If $x(t)=h(t)=rect\left(\frac{t}{\tau}\right)|_{\tau \text{ is the width}}$, Find $y(t)$

$$Y(\omega)=X(\omega)\cdot H(\omega)=\tau\,\text{sinc}\left(\frac{\omega\tau}{2\pi}\right)\cdot\tau\,\text{sinc}\left(\frac{\omega\tau}{2\pi}\right)$$

$$=\tau^{2}\,\text{sinc}^{2}\left(\frac{\omega\tau}{2\pi}\right)$$

$$y(t)=\mathcal{F}^{-1}\{Y(\omega)\}=\mathcal{F}^{-1}\left\{\tau\cdot\tau\,\text{sinc}^{2}\left(\frac{\omega\tau}{2\pi}\right)\right\}$$

$$=\tau\cdot tri\left(\frac{t}{\tau}\right)|_{\tau \text{ is half the base}}$$

3.2.5 F.T Reference Table:

$x(t)$	$X(\omega)$
a	$2\pi a\delta(\omega)$
$\delta(t)$	1
$u(t)$	$\frac{1}{j\omega}+\pi\delta(\omega)$
$sgn(t)$	$\frac{2}{j\omega}$
$rect(t/\tau)$	$\tau\,\mathrm{sinc}\left(\frac{\omega\tau}{2\pi}\right)$
$tri\,(t/\tau)$	$\tau\,\mathrm{sinc}^2\left(\frac{\omega\tau}{2\pi}\right)$
$e^{-\alpha t}\,u(t)$	$\frac{1}{\alpha+j\omega}$
$e^{\alpha t}\,u(-t)$	$\frac{1}{\alpha-j\omega}$
$e^{j\omega_0 t}$	$2\pi\delta(\omega-\omega_0)$
$e^{-a\lvert t\rvert}$	$\frac{2a}{a^2+\omega^2}$
$\sin(at)$	$\pi j[\delta(\omega+a)-\delta(\omega-a)]$
$\cos(at)$	$\pi[\delta(\omega+a)+\delta(\omega-a)]$

3.2.6 F.T Properties Table:

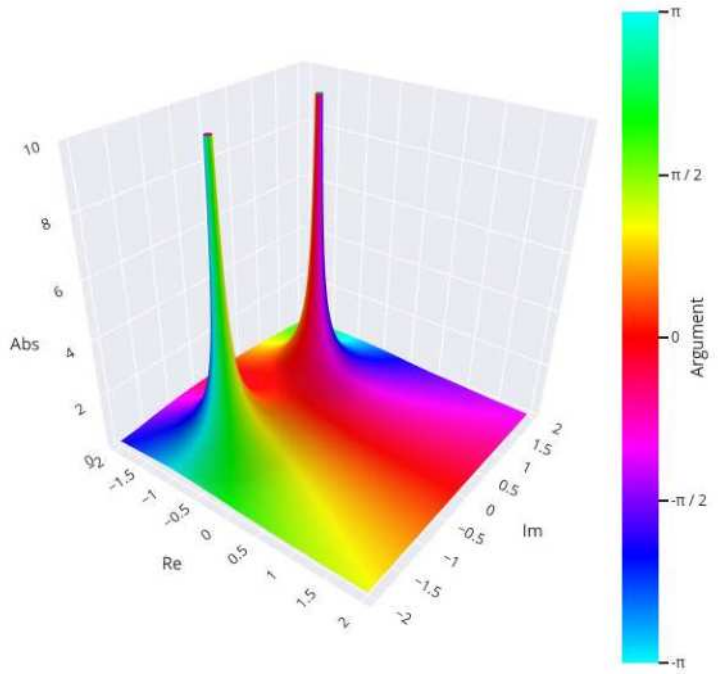
Property	$x(t)$	$X(\omega)$
Shifting	$x(t-a)$	$X(\omega)e^{-j\omega a}$
	$x(t)e^{j\omega_0 t}$	$X(\omega-\omega_0)$
Scaling	$x(at)$	$\frac{1}{\lvert a\rvert}X\left(\frac{\omega}{a}\right)$
Duality	$X(t)$	$2\pi x(-\omega)$
Differentiation	$x'(t)$	$(j\omega)X(\omega)$
	$x^{(n)}(t)$	$(j\omega)^n X(\omega)$
	$tx(t)$	$jX'(\omega)$
Multiplying by t	$(t)^n x(t)$	$j^n X^{(n)}(\omega)$
	$(-jt)^n x(t)$	$X^{(n)}(\omega)$
integration	$\int_{-\infty}^{t} x(\tau)\,d\tau$	$\frac{X(\omega)}{j\omega}+\pi X(0)\delta(\omega)$
Convolution	$x(t)^* h(t)$	$X(\omega)\cdot H(\omega)$
Energy	$\int_{-\infty}^{\infty}\lvert x(t)\rvert^2\,dt$	$\frac{1}{2\pi}\int_{-\infty}^{\infty}\lvert X(\omega)\rvert^2\,d\omega$

4 Laplace Transform

4.1 Laplace Transform

4.1.1 Introduction:

- **Laplace Transform** is a mathematical tool used in physics and engineering to simplify complex problems. It converts a function $x(t)$, which depends on time t (where $t \geq 0$), into another function $X(s)$, which depends on complex variable s.
- This transform is like the Fourier Transform but focuses on capturing the function's overall behavior rather than just its frequencies. It is commonly used to solve differential and integral equations, making it especially useful for analyzing systems with initial conditions.
- Laplace Transform is also a linear operator (follows two properties Additivity & Homogeneity).



4.1.2 Laplace Transform:

- The Laplace Transforms has 2 types:

1. Unilateral Laplace Transform:

$$X(s) = \mathcal{L}\{x(t)\} = \int_0^{\infty} x(t) \cdot e^{-st}\, dt$$

2. Bilateral Laplace Transform:

$$X(s) = \mathcal{L}\{x(t)\} = \int_{-\infty}^{\infty} x(t) \cdot e^{-st}\, dt$$

- Both result in a function of s (complex variable):

$$s = \sigma + j\omega$$

- Where:

  ∴ σ is the real part.

  ∴ ω is the imaginary part.

- The Inverse Laplace transform:

$$x(t) = \mathcal{L}^{-1}\{X(s)\} = \frac{1}{2\pi j} \int_{\sigma - j\omega}^{\sigma + j\omega} X(s) \cdot e^{+st}\, dt$$

- The most-common pairs of $x(t)$ and $X(s)$ are often given in tables for easy reference.
- Laplace Transform makes calculations easier because many difficult operations on $x(t)$ become simpler when transformed into $X(s)$. This helps in solving equations more efficiently.

4.1.3 L.T Reference Table

$x(t)$	$X(s)$	$x(t)$	$X(s)$
$\delta(t)$	1	$e^{-at}\,u(t)$	$\frac{1}{a+s}$
a	$\frac{a}{s}$	$e^{at}\,u(-t)$	$\frac{1}{a-s}$
$u(t)$	$\frac{1}{s}$	$e^{-a\lvert t\rvert}$	$\frac{2a}{a^2-s^2}$
$sgn(t)$	$\frac{2}{s}$	$rect\left(\frac{t}{\tau}\right)$	$\frac{\sinh\left(\frac{s\tau}{2}\right)}{\frac{s}{2}}$
$t^n u(t)$	$\frac{n!}{s^{n+1}}$	$tri\left(\frac{t}{\tau}\right)$	$\frac{\sinh^2\left(\frac{s\tau}{2}\right)}{\left(\frac{s}{2}\right)^2}$
$\sin(at)$	$\frac{a}{a^2+s^2}$	$\sinh(at)$	$\frac{a}{a^2-s^2}$
$\cos(at)$	$\frac{s}{a^2+s^2}$	$\cosh(at)$	$\frac{s}{a^2-s^2}$

4.1.4 Region of Convergence:

What determines if $X(s)$ exists?

- The Laplace transform **does not exist for all values of** s.
- It exists only for the values of s that make the integral converge (give a finite answer).
- All values of $s$that make the integral converge form a region in the s-plane called the:

ROC (Region of Convergence)

- So:

$X(s)$ exists if and only if s is inside the ROC

Why does the ROC depend mainly on σ?

- Since:

$$e^{-st} = e^{-(\sigma+j\omega)t} = e^{-\sigma t} \cdot e^{-j\omega t}$$

- The term $e^{-\sigma t}$controls **growth/decay**, so it affects convergence.
- The term $e^{-j\omega t}$ is just an oscillation (magnitude = 1), so it does **not** affect convergence.
- That's why the ROC depends only on:

$$\sigma = \mathrm{Re}\{s\}$$

ROC and Fourier Transform

- The Fourier Transform is obtained from the Laplace transform by setting:

$$\sigma = 0 \Rightarrow s = j\omega$$

- So the Fourier Transform corresponds to the $j\omega$-**axis** in the s-plane.

When does the Fourier Transform exist?

- A signal $x(t)$ is Fourier-transformable if the ROC of $X(s)$ includes the $j\omega$-axis.

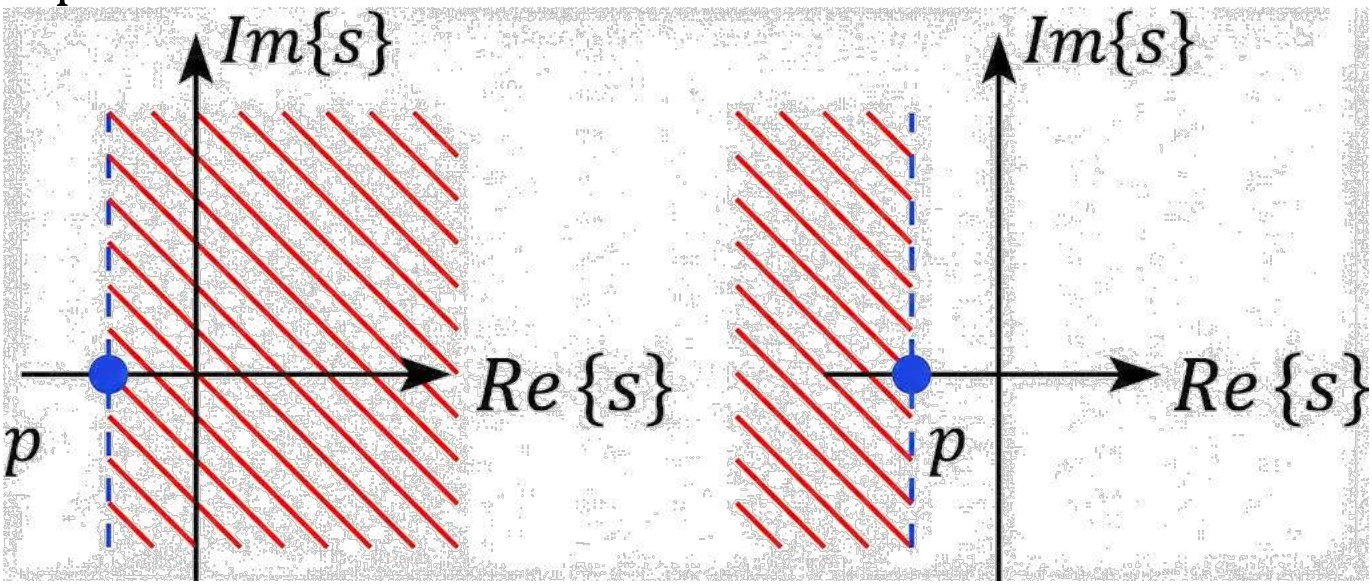
If ROC includes $s = j\omega$, then Fourier Transform exists

How to get $X(\omega)$ from $X(s)$

- If the $j\omega$-axis is inside the ROC, then:

$$X(\omega) = X(s)\,|_{s=j\omega}$$

- Meaning: replace $s$with $j\omega$ in the Laplace transform expression.



Q1: Find L.T for:

A) $x(t) = e^{-at}u(t)$

$$X(s) = \mathcal{L}\{x(t)\} = \int_{-\infty}^{\infty} x(t) \cdot e^{-st}\, dt = \int_{-\infty}^{\infty} e^{-at}u(t) \cdot e^{-st}\, dt$$

$$= \int_{0}^{\infty} e^{-at} \cdot e^{-st}\, dt = \int_{0}^{\infty} e^{-t(s+a)}\, dt$$

$$= \frac{-1}{s+a} e^{-t(s+a)}\Big|_0^{\infty}$$

$$= \frac{-1}{a+s}\left(e^{-\infty(s+a)} - e^{0}\right) \frac{-1}{a+s}(0-1) = \frac{1}{a+s}$$

Note that when we say $e^{-\infty(s+a)}$ and we assumed it 0, thus, $s+a$ must be positive to achieve $e^{-\infty} = 0$, so:

$$a + s > 0, \qquad s > -a$$

So; $X(s)$ with its ROC:

$$X(s) = \frac{1}{a+s}, \qquad Re\{s\} > -a$$

B) $x(t) = e^{at}u(-t)$

$$X(s) = \mathcal{L}\{x(t)\} = \int_{-\infty}^{\infty} x(t) \cdot e^{-st}\, dt = \int_{-\infty}^{0} e^{at} \cdot e^{-st}\, dt$$

$$= \int_{-\infty}^{0} e^{t(a-s)}\, dt = \frac{1}{a-s} e^{t(a-s)}\Big|_{-\infty}^{0}$$

$$= \frac{1}{a-s}\left(e^{0} - e^{-\infty(a-s)}\right) \frac{1}{a-s}(1-0) = \frac{1}{a-s},$$

$$a - s > 0, \qquad s < a;$$

$$X(s) = \frac{1}{a-s}, \qquad Re\{s\} < a$$

4.1.5 ROC properties:

1 . ROC never contains poles:

- A **pole** means $X(s)$ becomes infinite.
- But ROC is the region where the Laplace integral **converges** (finite).
- So poles can **never** be inside the ROC.

2 . Finite-duration signal → ROC is almost everything

- If the signal exists only for a limited time (like one pulse):

∴ The Laplace integral will converge for almost all s

∴ So the ROC is **the whole s-plane** (except sometimes special points like 0 or ∞).

3 . Right-sided signal → ROC is to the RIGHT

- Right-sided means:

$$x(t) = 0 \text{ for } t < t_1$$

- So the signal exists mainly for $t \geq t_1$.
- To make the integral converge, we need the exponential term $e^{-\sigma t}$ to **decay** as $t \to +\infty$.
- That happens when σ is large enough.
- So the ROC is:

$$Re(s) > \sigma_{max}$$

- Meaning:

∴ ROC is a **right half-plane**

∴ It starts to the right of the **rightmost pole**

4. Left-sided signal → ROC is to the LEFT

- Left-sided means:

$$x(t) = 0 \text{ for } t > t_2$$

- So the signal exists mainly for $t \leq t_2$.
- Now the integral must converge as $t \to -\infty$.

That requires the exponential term to decay in the negative direction.

- So the ROC is:

$$Re(s) < \sigma_{min}$$

- Meaning:
  - ∴ ROC is a **left half-plane**
  - ∴ It is to the left of the **leftmost pole**

5. Two-sided signal → ROC is a STRIP

- Two-sided means the signal exists on both sides:
  - ∴ for large positive t
  - ∴ for large negative t
- We need convergence in **both directions**, so that creates:

$$\sigma_1 < Re(s) < \sigma_2$$

- Meaning:
  - ∴ ROC is a **vertical strip**
  - ∴ Between two poles

4.1.7 Unilateral & Bilateral Laplace Transform:

▶ Unilateral Laplace Transform:

- The **unilateral Laplace transform** is very useful when working with **causal signals**, which means:

$$x(t) = 0 \text{ for } t < 0$$

- So the signal only exists for $t \geq 0$, which matches many real systems (circuits, control systems, etc.).
- Why the Bilateral Laplace may fail

▶ Bilateral Laplace transform:

$$X(s) = \int_{-\infty}^{\infty} x(t)e^{-st}\,dt$$

- This integral can be split into two parts:

$$X(s) = \int_{-\infty}^{0} x(t)e^{-st}\,dt + \int_{0}^{\infty} x(t)e^{-st}\,dt$$

- Sometimes:

∴ The first integral converges only when $\text{Re}\{s\} < 0$

∴ The second integral converges only when $\text{Re}\{s\} > 0$

- These conditions cannot happen at the same time, so no ROC exists, and the bilateral Laplace transform does not converge

- In many applications, we only care about $t \geq 0$.

So we use the **unilateral Laplace transform**, defined as:

$$X(s) = \int_0^{\infty} x(t)e^{-st}\,dt$$

- Writing a signal as causal + noncausal parts
- Even if a signal is not fully causal, we can split it into two parts:

∴ A part that exists for $t \geq 0$(causal part)

∴ A part that exists for $t < 0$(noncausal part)

- So:

$$x(t) = x_+(t)u(t) + x_-(t)u(-t)$$

- Where:

∴ $x_+(t)$describes the signal for $t \geq 0$

∴ $x_-(t)$describes the signal for $t < 0$

▶ **Important note**

- This splitting works for signals that do **not** contain impulses like $\delta(t)$exactly at $t = 0$.

4.1.8 Laplace Transform Properties:

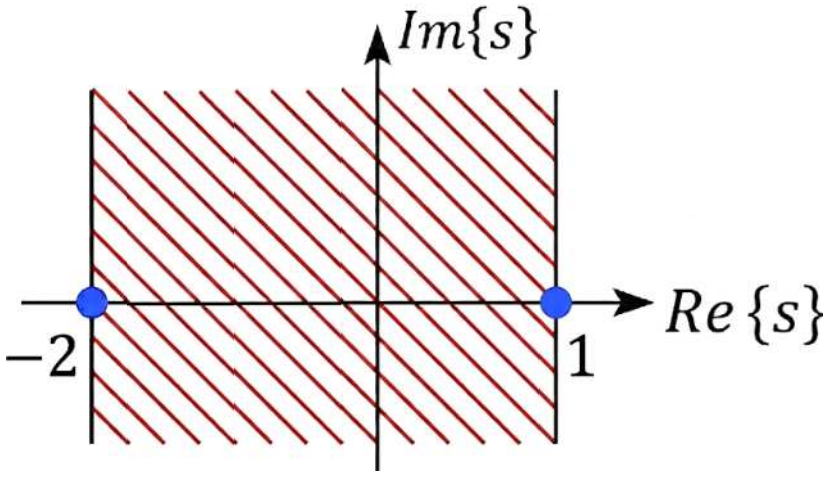
1. Linearity:

$$ax_1(t) + bx_2(t) \Leftrightarrow aX_1(s) + bX_2(s)$$

Q1: Find $\mathcal{L}\{x(t)\}$ for $x(t) = 3e^{-2t}u(t) + 4e^{t}u(-t)$

$$X(s) = \mathcal{L}\{x(t)\} = \mathcal{L}\{3e^{-2t}u(t) + 4e^{t}u(-t)\}$$
$$= 3\mathcal{L}\{e^{-2t}u(t)\} + 4\mathcal{L}\{e^{t}u(-t)\}$$
$$= 3 \cdot \frac{1}{2+s} + 4 \cdot \frac{1}{1-s} = \frac{3}{2+s} + \frac{4}{1-s},$$
$$\uparrow \qquad \uparrow$$
$$-2 < Re\{s\}, \quad Re\{s\} < 1$$

$$X(s) = \frac{3}{2+s} + \frac{4}{1-s}, \qquad -2 < Re\{s\} < 1$$



2. Time Scaling:

$$x(at) \Leftrightarrow \frac{1}{|a|}X\left(\frac{s}{a}\right)$$

3. Time Shifting:

$$x(t-t_0) \Leftrightarrow e^{-st_0}X(s)$$

Q2: Find $\mathcal{L}\{e^{-2(t-3)}u(t)\}$.

$$e^{-at}u(t) \Leftrightarrow \frac{1}{a+s} \rightarrow a=2, t_0=3$$

$$e^{-a(t-t_0)}u(t) \Leftrightarrow e^{-st_0}\frac{1}{a+s} = e^{-3s}\frac{1}{2+s}$$

Q3: Find $\mathcal{L}\{2u(t-1)-\delta(t-1)\}$.

$$u(t) \Leftrightarrow \frac{1}{s} \;\&\; \delta(t) \Leftrightarrow 1$$

$$2u(t-1)-\delta(t-1) \Leftrightarrow e^{-s}(\frac{2}{s}-1)$$

4. "s" Shifting:

$$e^{s_0t}x(t) \Leftrightarrow X(s-s_0)$$

Q4: Find $\mathcal{L}\{\sin 2t\, e^{-t}u(t)\}$.

$$\sin at\, u(t) \Leftrightarrow \frac{a}{a^2+s^2} \rightarrow a=2, s_0=-1$$

$$e^{s_0t}\sin at\, u(t) \Leftrightarrow \frac{a}{a^2+(s-s_0)^2} = \frac{2}{4+(s+1)^2}$$

5. Differentiation in Time Domain:

$$x'(t) \Leftrightarrow sX(s) - x(0)$$

$$x^{(n)}(t) \Leftrightarrow s^n X(s) - s^{n-1}x(0) - s^{n-2}x'(0) - \cdots - x^{(n-1)}(0)$$

Q5: If $x(t) = t^2 u(t)$, Find $\mathcal{L}\{x''(t)\}$

$$x(t) = t^2 u(t) \Leftrightarrow X(s) = \frac{2}{s^3}$$

$$x''(t) \Leftrightarrow s^2 X(s) - s^1 x(0) - x'(0) = \frac{2}{s}$$

6. Multiplication by t:

$$t \cdot x(t) \Leftrightarrow -X'(s)$$

$$t^n \cdot x(t) \Leftrightarrow (-1)^n X^{(n)}(s)$$

Q6: Find $\mathcal{L}\{t^2 e^{-t} u(t)\}$

$$x(t) = e^{-t} u(t) \Leftrightarrow X(s) = \frac{1}{1+s}$$

$$t^2 x(t) \Leftrightarrow X''(s) = \left(\frac{-1}{(1+s)^2}\right)' = \frac{2(1+s)}{(1+s)^4} = \frac{2}{(1+s)^3}$$

7. Integration in Time Domain:

$$\int_0^t x(\tau)\,d\tau \Leftrightarrow \frac{1}{s}X(s)$$

Q7: Find $\mathcal{L}\left\{\int_0^t \sin 2\tau\,d\tau\right\}$

$$x(\tau) = \sin 2\tau \Leftrightarrow X(s) = \frac{2}{s^2 + 2^2}$$

$$\int_0^t x(\tau)\,d\tau \Leftrightarrow \frac{X(s)}{s} = \frac{2}{s(s^2+4)} = \frac{2}{s^3 + 4s}$$

8. Divided by t:

$$\frac{x(t)}{t} \Leftrightarrow \int_s^\infty X(s)\,ds$$

Q8: Find $\mathcal{L}\{\text{sa}(t)\}$

$$x(t) = \sin t \Leftrightarrow X(s) = \frac{1}{s^2+1}$$

$$\frac{x(t)}{t} \Leftrightarrow \int_s^\infty X(s)\,ds = \int_s^\infty \frac{1}{s^2+1}\,ds = \tan^{-1} s\,|_s^\infty$$

$$= \tan^{-1}\infty - \tan^{-1} s$$

$$= \frac{\pi}{2} - \tan^{-1} s$$

9. Modulation:

$$x(t)\cos(\omega_0 t) \Leftrightarrow \frac{1}{2}\Big(X(s+j\omega_0)+X(s-j\omega_0)\Big)$$

$$x(t)\sin(\omega_0 t) \Leftrightarrow \frac{1}{2j}\Big(X(s-j\omega_0)-X(s+j\omega_0)\Big)$$

Q9: Find $\mathcal{L}\{x(t)\cos(\omega_0 t)\}$ if $x(t) = e^{-at}u(t)$.

$$x(t)\cos(\omega_0 t) \Leftrightarrow \frac{1}{2}\Big(X(s+j\omega_0)+X(s-j\omega_0)\Big)$$

$$= \frac{1}{2}\left[\left(\frac{1}{(s+j\omega_0)+a}\right)+\left(\frac{1}{(s-j\omega_0)+a}\right)\right]$$

Q10: Find $\mathcal{L}\{x(t)\cos(\omega_0 t)\}$ if $x(t) = te^{-at}u(t)$.

We know that:

$$X(s) = \frac{1}{(s+a)^2}$$

Using modulation property:

$$x(t)\cos(\omega_0 t) \Leftrightarrow \frac{1}{2}\Big(X(s+j\omega_0)+X(s-j\omega_0)\Big)$$

$$= \frac{1}{2}\left[\left(\frac{1}{\big((s+j\omega_0)+a\big)^2}\right)+\left(\frac{1}{\big((s-j\omega_0)+a\big)^2}\right)\right]$$

10. Convolution:

$$x(t)^{*}h(t) \Leftrightarrow X(s)\cdot H(s)$$

$$y(t) \Leftrightarrow Y(s)$$

Q11: If $x(t)=e^{-t}u(t)$ & $h(t)=e^{t}\,u(-t)$, Find $Y(s)$.

$$\mathcal{F}\{Y(s)\}=X(s)\cdot H(s)=\frac{1}{1+s}\cdot\frac{1}{1-s}=\frac{1}{1-s^{2}}$$

Q12: If $x(t)=h(t)=rect(t)$, Find $y(t)$.

$$Y(s)=X(s)\cdot H(s)=\frac{\sinh\left(\frac{s}{2}\right)}{\frac{s}{2}}\cdot\frac{\sinh\left(\frac{s}{2}\right)}{\frac{s}{2}}=\frac{\sinh^{2}\left(\frac{s}{2}\right)}{\left(\frac{s}{2}\right)^{2}}$$

$$y(t)=\mathcal{L}^{-1}\{Y(s)\}=\mathcal{L}^{-1}\left\{\frac{\sinh^{2}\left(\frac{s}{2}\right)}{\left(\frac{s}{2}\right)^{2}}\right\}=tri(t)$$

11. Initial Value Theorem:

$$\lim_{t\to 0^+} x(t) = \lim_{s\to\infty} s\cdot X(s)$$

$$\lim_{t\to 0^+} x^{(n)}(t) = \lim_{s\to\infty} s^{n+1}\cdot X(s)$$

Q13: Find $\lim_{t\to 0^+} x(t)$ if $X(s)=\frac{1}{s+2}$.

$$\lim_{t\to 0^+} x(t) = \lim_{s\to\infty} s\cdot\left[\frac{1}{s+2}\right] = 1$$

Q14: Find $\lim_{t\to 0^+} x'(t)$ if $X(s)=\frac{12}{s^2+6s+9}$.

$$\lim_{t\to 0^+} x'(t) = \lim_{s\to\infty} s^2\cdot\left[\frac{12}{s^2+6s+9}\right] = 12$$

12. Final Value Theorem:

$$\lim_{t\to\infty} x(t) = \lim_{s\to 0} s\cdot X(s)$$

$$\lim_{t\to\infty} x^{(n)}(t) = \lim_{s\to 0} s^{n+1}\cdot X(s)$$

Q15: Find $\lim_{t\to\infty} x(t)$ if $X(s)=\frac{1}{s+2}$.

$$\lim_{t\to\infty} x(t) = \lim_{s\to 0} s\cdot\left[\frac{1}{s+2}\right] = 0$$

Q16: Find $\lim_{t\to\infty} x'(t)$ if $X(s)=\frac{12}{s^2+6s+9}$.

$$\lim_{t\to\infty} x'(t) = \lim_{s\to 0} s^2\cdot\left[\frac{12}{s^2+6s+9}\right] = 0$$

4.1.9 Inverse Laplace Transform:

- It is an easy object, just remember that:

$$x(t) = \mathcal{L}^{-1}\{X(s)\}$$

Q1: Find $\mathcal{L}^{-1}\{X(s)\}$ for:

A) $X(s) = \frac{2}{3+s}$

$$x(t) = \mathcal{L}^{-1}\left\{\frac{2}{3+s}\right\} = 2e^{-3t}u(t)$$

B) $X(s) = \frac{6}{(s-1)^4}$

$$x(t) = \mathcal{L}^{-1}\left\{\frac{3!}{(s-1)^4}\right\} = e^{t}t^{3}u(t)$$

C) $X(s) = e^{-4s}\frac{s}{s^2-9}$

$$x(t) = \mathcal{L}^{-1}\left\{e^{-4s}\frac{s}{s^2-3^2}\right\} = \cosh\big(3(t-4)\big)$$

D) $X(s) = \frac{s-1}{(s-3)^2+16}$

$$x(t) = \mathcal{L}^{-1}\left\{\frac{s-1-3+3}{(s-3)^2+16}\right\} = \mathcal{L}^{-1}\left\{\frac{s-3+2}{(s-3)^2+16}\right\}$$

$$\mathcal{L}^{-1}\left\{\frac{s-3}{(s-3)^2+4^2}\right\} + \mathcal{L}^{-1}\left\{\frac{4}{4}\frac{2}{(s-3)^2+4^2}\right\}$$

$$= e^{3t}\cos(4t) + \frac{1}{2}e^{3t}\sin(4t)$$

E) $X(s) = \frac{s+2}{s^5}$

$$x(t) = \mathcal{L}^{-1}\left\{\frac{s}{s^5} + \frac{2}{s^5}\right\} = \mathcal{L}^{-1}\left\{\frac{3!}{3!}\cdot\frac{1}{s^4}\right\} + \mathcal{L}^{-1}\left\{\frac{4!}{4!}\cdot\frac{2}{s^5}\right\}$$

$$= \frac{t^3}{6} + \frac{t^4}{12}$$

F) $X(s) = 2\frac{s+1}{s(s+2)}$

In this case, we use the partial fractions to make it as sum of two functions then take I.L.T for these functions:

$$\frac{2s+2}{s(s+2)} = \frac{A}{s} + \frac{B}{s+2} \rightarrow 2s+2 = A(s+2) + Bs$$

$$s = 0 \rightarrow 2 = 2A \rightarrow A = 1$$
$$s = -2 \rightarrow -2 = -2B \rightarrow B = 1$$

$$\frac{s+1}{s(s+2)} = \frac{1}{s} + \frac{1}{s+2}$$

$$\mathcal{L}^{-1}\left\{\frac{2s+2}{s(s+2)}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{s}\right\} + \mathcal{L}^{-1}\left\{\frac{1}{s+2}\right\} = \left(1 + e^{-2t}\right)u(t)$$

G) $X(s) = \frac{s}{s^2+4s+3}$

$$\frac{s}{(s+1)(s+3)} = \frac{A}{s+1} + \frac{B}{s+3} \rightarrow s = A(s+3) + B(s+1)$$

$$s = -1 \rightarrow -1 = 2A \rightarrow A = -\frac{1}{2}$$
$$s = -3 \rightarrow -3 = -2B \rightarrow B = \frac{3}{2}$$

$$\frac{s}{(s+1)(s+3)} = -\frac{1}{2}\cdot\frac{1}{s+1} + \frac{3}{2}\cdot\frac{1}{s+3}$$

$$\mathcal{L}^{-1}\left\{\frac{s}{s^2+4s+3}\right\} = -\frac{1}{2}\mathcal{L}^{-1}\left\{\frac{1}{1+s}\right\} + \frac{3}{2}\mathcal{L}^{-1}\left\{\frac{1}{3+s}\right\}$$
$$= \left(-\frac{1}{2}e^{-t} + \frac{3}{2}e^{-3t}\right)u(t)$$

H) $X(s) = \frac{s+1}{s^2+2s+2}$

In this case, since the denominator can't be factored, so we use square completing:

$$\mathcal{L}^{-1}\left\{\frac{s+1}{s^2+2s+2}\right\} = \mathcal{L}^{-1}\left\{\frac{s+1}{s^2+2s+2-1+1}\right\} = \mathcal{L}^{-1}\left\{\frac{s+1}{(s^2+2s+1)+1}\right\}$$
$$= \mathcal{L}^{-1}\left\{\frac{s+1}{(s+1)^2+1}\right\} = e^{-t}\cos t$$

4.1.10 Laplace Transform Applications:

- The L.T has so many applications in engineering subjects, like differential equations that we could use L.T to solving the differential equations by the property:

$$x^{(n)}(t) \Leftrightarrow s^n X(s) - s^{n-1}x(0) - s^{n-2}x'(0) - \cdots - x^{(n-1)}(0)$$

Q2: Solve this differential equation:

$$y''(t) - y'(t) + 2y(t) = 0$$

If $y(0) = 0$ and $y'(0) = 1$.

$$y''(t) \Leftrightarrow s^2Y(s) - s\,y(0) + y'(0)$$
$$y'(t) \Leftrightarrow sY(s) - y(0)$$
$$y(t) \Leftrightarrow Y(s)$$

$$y''(t) + 3y'(t) + 2y(t) = 0:$$

$$s^2Y(s) - s\,y(0) - y'(0) + 3\left(sY(s) - y(0)\right) + 2Y(s) = 0,$$
$$s^2Y(s) - 0 - 1 + 3sY(s) - 0 + 2Y(s) = 0,$$
$$s^2Y(s) + 3sY(s) + 2Y(s) = 1,$$
$$Y(s)\left(s^2 + 3s + 2\right) = 1, \qquad Y(s) = \frac{1}{(s^2 + 3s + 2)}$$
$$y(t) = \mathcal{L}^{-1}\{Y(s)\} = \mathcal{L}^{-1}\left\{\frac{1}{s^2 + 3s + 2}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{s+2} - \frac{1}{s+1}\right\}$$
$$y(t) = e^{-2t} + e^{-t}$$

- Also, there are so many applications in control systems.

Q3: The input $x(t) = A \cdot u(t)$ is applied to an automatic positioning-control system whose transfer function is:

$$H(s) = \frac{c}{s(s+b)+c}$$

Find the final value of the output $y(t)$.

$$Y(s) = X(s) \cdot H(s) = \frac{A}{s} \cdot \frac{c}{s(s+b)+c}$$

$$\lim_{t\to\infty} y(t) = \lim_{s\to 0} sY(s) = \lim_{s\to 0} s \cdot \frac{A}{s} \cdot \frac{c}{s(s+b)+c}$$

$$= \lim_{s\to 0} \frac{Ac}{s(s+c)+c} = \frac{Ac}{c} = A$$



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5 References:

Books:

- Signals and systems 2nd edition
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